

Monetary Policy and Firm Innovation*

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Abstract

This paper studies how monetary policy shapes firm-level innovation, using linked Compustat, USPTO, and CRSP data with local projection methods. The R&D response to monetary policy operates almost entirely through the intensive margin: expansionary shocks raise R&D and knowledge capital among firms already engaged in innovation, while other firms remain largely unresponsive. On the output side, expansionary shocks lower the aggregate number of patents but raise average citations and, over the medium run, patent importance and the share of patents in the upper tail of the importance distribution—consistent with firms reallocating R&D from incremental patenting toward higher-impact projects as financing conditions ease. A two-stage analysis indicates that this reallocation operates through the R&D response itself rather than through a direct effect of the shock on patenting. The effects are also asymmetric in sign: contractionary shocks reduce both the quantity and quality of innovation, and this reallocation-toward-quality channel is cleanest among economically large shocks. These results suggest that monetary policy shapes not just the scale but the direction of firms’ innovative activity, and that policymakers could pair monetary policy with targeted innovation support to sustain long-run technological progress.

Keywords: Monetary Policy; Firm-Level Innovation; R&D Investments; Patent Quality; Knowledge Capital; Technological Progress; **JEL Classification Numbers:** E52, O31, O32, O34;

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1 Introduction

Monetary policy is traditionally viewed as a tool for managing short-run economic fluctuations, primarily through its effects on output, inflation, and employment, with conventional wisdom holding that its influence on long-run growth is minimal because markets return to equilibrium once temporary disturbances subside. Recent literature increasingly challenges this view, arguing that monetary policy can have persistent effects on long-run economic outcomes through its influence on investment and innovation decisions (Ma and Zimmermann, 2023).

Innovation is inherently risky, entails long gestation periods, and depends heavily on external financing (Hall and Lerner, 2010)—characteristics that make firms’ innovation decisions particularly sensitive to changes in monetary conditions. By lowering borrowing costs and relaxing financial constraints, expansionary monetary policy can induce firms to undertake more ambitious innovation projects; contractionary policy, by tightening financial conditions, may instead compel firms to curtail R&D spending or shift toward short-term, incremental innovation at the expense of long-run technological progress.

Despite growing attention to this topic, there are several key gaps in the current literature. First, much of the existing research operates at the aggregate level and does not examine how monetary policy affects firms’ internal R&D decisions—in particular, whether it operates by drawing new firms into innovation or by changing how much already-active innovators invest. Second, the link between monetary policy and innovation has rarely been studied in terms of both inputs and outputs together: while some studies examine R&D spending, few investigate how this spending translates into measurable innovation outcomes such as patent quality and quantity, and fewer still test whether R&D is the actual channel connecting the two. Third, existing studies typically estimate the average effect of a monetary policy shock and rarely ask whether this effect is symmetric across the sign of the shock or scales predictably with its size.

This paper addresses these gaps by jointly analyzing the inputs (R&D investment) and outputs (patent quantity and quality) of innovation, and by identifying the mechanism connecting the two. Specifically, I examine how firms adjust R&D spending in response to monetary policy shocks, how these adjustments translate into innovation outcomes—the quantity, quality, and importance of patents—and whether the resulting effects depend on the sign and magnitude of the shock. Addressing these questions provides a firm-level account of how monetary policy shapes both innovation investment and its downstream contribution to technological progress.

I combine firm-level data on R&D spending and innovation outcomes with local projection methods and Probit/Tobit models to answer these questions. First, I examine how monetary pol-

icy shocks affect firms' R&D spending along both the intensive margin—how much active firms invest—and the extensive margin—whether inactive firms begin to invest. The results show that the observed growth in R&D investment is driven almost entirely by firms already engaged in innovation: monetary policy shocks have little effect on inducing inactive firms to begin R&D, whereas active firms respond substantially by increasing investment. The intensive-margin response is an order of magnitude larger than the extensive-margin response and precisely estimated, while the extensive-margin response cannot be distinguished from zero at any horizon. This indicates that expansionary monetary policy primarily benefits firms already positioned to exploit favorable financing conditions, and points to sharply different responses among active and inactive firms to policy shocks.

Second, I measure the direct effect of monetary policy on innovation outcomes, including patent counts, citations, market value, and importance. Expansionary monetary policy has opposing short- and long-run effects: patent counts decline in the short run, as firms shift resources away from immediate, lower-value projects, while patent quality and importance rise over the medium run, as firms come to produce fewer but more impactful patents. Per-patent market value, by contrast, shows no reliable movement in either direction, so the decline in aggregate patent value that accompanies this reallocation is a quantity effect rather than a deterioration in the value of the individual inventions firms produce. This pattern is consistent with firms substituting toward research of greater long-run value at the cost of a temporary decline in output quantity.

Third, I check whether the effects of monetary policy on innovation documented above operate through changes in R&D spending, as opposed to simply reflecting the direct effect of the shock. Using a two-stage regression framework, I find that changes in R&D spending play a central role in explaining how monetary policy affects innovation. In this analysis, I focus specifically on the intensive margin, examining how the R&D investment of firms already engaged in innovation translates into patenting outcomes. Expansionary monetary policy drives sustained growth in patent citations and in aggregate patent value, reflecting improvements in the quality and importance of innovation. Traced through this R&D channel specifically, the total number of patents shows no statistically distinguishable change, in contrast to the direct decline documented above, while firms shift their focus toward producing fewer but more impactful innovations. High-impact patents grow significantly, reflecting a strategic reallocation of resources toward projects with greater economic and technological significance. R&D spending thus acts as a central channel linking monetary policy to innovation, promoting higher-quality and more valuable outcomes.

Finally, I examine whether these effects are symmetric across the sign and magnitude of the

shock. Comparing expansionary and contractionary shocks directly, I find that the two are not mirror images of each other: expansionary shocks ease financial constraints and shift firms toward long-term, high-value projects, temporarily lowering patent output but raising its quality and importance, while contractionary shocks raise patent counts and reduce the quality and impact of innovation, reversing the composition of output rather than simply scaling it down. Splitting shocks by magnitude instead of sign shows that big and small shocks differ in more than just estimation precision: for several outcomes, including the investment rate and knowledge capital growth, the point estimates for big and small shocks carry opposite signs, so the shock-size split is not simply a rescaled version of a common underlying response.

To disentangle these two dimensions, I interact shock sign and magnitude directly. The sign asymmetry documented above holds within both magnitude groups, but its precision and dynamics differ sharply between them: the reallocation from patent quantity toward quality is identified most cleanly among economically large shocks, whereas the response to small shocks—particularly small contractionary shocks—is considerably noisier and, in some cases, outsized relative to the corresponding large-shock response. These results indicate that shock sign, rather than magnitude alone, is the primary source of asymmetry in how monetary policy affects firm innovation, and that the transmission mechanism documented throughout the paper is most sharply identified among economically significant policy moves.

Literature Review This paper contributes to five strands of literature. The first three strands connect to the monetary policy literature: monetary policy and innovation/intangible investments, monetary policy and heterogeneous firms, and monetary policy and long-run growth, respectively. The last two strands connect to the innovation literature: innovation quantity and quality, and innovation inputs and outputs, respectively.

Monetary Policy and Firm Innovation/Intangible Investments The effects of monetary policy on innovation or intangible investments are quite under-studied. [Döttling and Ratnovski \(2023\)](#) showed that intangible investment is less sensitive to monetary policy than tangible investment. [Annicchiarico and Pelloni \(2021\)](#) argued that innovation-led growth requires counter-cyclical policy. [Moran and Queralto \(2018\)](#); [Zhang et al. \(2020\)](#); [Ma and Zimmermann \(2023\)](#); [Elfsbacka-Schmöller et al. \(2024\)](#) indicate that monetary policy is a key determinant of innovation investment, with accommodative policy boosting R&D and tighter policy undermining innovation spending. Building on these insights, this paper provides novel firm-level evidence on the relationship between monetary policy and innovation investment.

Monetary Policy and Heterogeneous Firms An emerging literature studies how firm heterogeneity shapes the investment channel of monetary policy, mostly for physical investment. Starting with [Ottonello and Winberry \(2020\)](#), this literature shows that several dimensions of firm

heterogeneity matter for monetary policy transmission, including firm leverage (Ottonello and Winberry, 2020; Auer et al., 2021; Lakdawala et al., 2021), balance sheet liquidity (Jeenas, 2018, 2023), firm age and size (Cloyne et al., 2023; Durante et al., 2022; Casiraghi et al., 2021), and corporate debt maturity (Deng and Fang, 2022; Jungherr et al., 2024). For the innovation channel, however, I find that these traditional dimensions of firm heterogeneity matter less than a different distinction: whether a firm is already engaged in R&D. The bulk of the innovation response to monetary policy operates through the intensive margin—the scale of R&D investment among active innovators—rather than through firms entering or exiting innovation activity altogether.¹

Monetary Policy and Long-run Growth In addition, a growing body of research explores how monetary policy shapes long-run growth. Endogenous growth theory (Aghion, 1990; Grossman and Helpman, 1991; Romer, 1990; Jones, 1999) identifies innovation as the driving force behind technological progress and long-run growth. Recent research shows that monetary policy influences innovation incentives, can have persistent real effects, and affects corporate innovation and total factor productivity (Moran and Queralto, 2018; Jordà et al., 2020; Grimm et al., 2022). Moreover, New Keynesian models with endogenous innovation indicate that recessionary shocks can have permanent adverse effects on technology growth (Anzoategui et al., 2019; Ikeda and Kurozumi, 2019; Bianchi et al., 2019). In this paper, I contribute by identifying the firm-level mechanisms through which monetary policy influences long-term growth.

Innovation Quantity and Quality Another important topic in this area is the quality and impact of innovation, especially patents. Trajtenberg (1990) pioneered using patent citations as proxies for economic impact. As refined by (Hall, 2000; Kogan et al., 2017; Kelly et al., 2018) using stock market reactions, citation weights, and quality metrics, market valuations and related indicators better reflect patent value than simple counts. Complementing these findings, (Lanjouw and Schankerman, 2004; Kuhn and Thompson, 2019) show that incorporating refined quality and scope indicators improves patent valuation². In this paper, I extend these insights by investigating the causal links between monetary policy and both the quality and quantity of firm-level innovation. A further contribution of this paper to that literature is to show that the three families of quality measures do not move together in response to a monetary policy shock: citation-based and text-based measures both rise, while market-value-based measures show no reliable per-patent response, so conclusions about innovation quality depend materially on which proxy is used.

¹This difference between the innovation and physical investment channels may reflect their different reliance on the extensive margin. As shown in Fang (2020) and Gnewuch and Zhang (2025), the extensive margin is central to the physical investment channel of monetary policy.

²As demonstrated by Moser (2005, 2013, 2016); Khan (2013), patent laws play a crucial role in determining both the quantity and direction of innovation.

Innovation Inputs and Outputs Finally, patents are central to innovation incentives and outcomes. However, the innovation process is often hindered by mounting educational burdens, R&D spillovers, and systemic inefficiencies (Griliches, 1991, 1998; Jones, 2009; Hall and Harhoff, 2012). Public policy can diffuse breakthroughs (Goolsbee, 1998), yet short-term incentives may deter long-horizon research. While patent invalidation may spur follow-on innovation, overly strong patents can stifle subsequent R&D (Williams, 2013; Budish et al., 2015; Galasso and Schankerman, 2015). Creative destruction and taxes on incumbents reallocate resources to more innovative firms (Aghion and Jaravel, 2015). Both incremental advances and major breakthroughs fuel economic growth (Bresnahan and Trajtenberg, 1995; Akcigit and Kerr, 2018). Building on these insights, this paper examines how shifts in financial conditions—mediated by monetary policy—affect patenting behavior and shape innovation incentives and outcomes.

Layout. The rest of the paper proceeds as follows. Section 2 describes the data, the construction of the knowledge capital stock, and the monetary policy shock series. Section 3 outlines the empirical strategy and reports the baseline results for innovation inputs, innovation outputs, and the R&D channel connecting the two. Section 4 discusses the asymmetric effects of monetary policy by shock sign, by shock magnitude, and by their interaction. Section 5 concludes. A separate appendix reports additional descriptive evidence, supplementary derivations for the static and dynamic frameworks, and the full set of robustness checks referenced throughout the text.

2 Data

To quantify firms’ accumulated R&D investments, I develop a knowledge capital database. Unlike tangible assets, which are directly reflected on companies’ balance sheets, knowledge capital is intangible and evolves dynamically through firms’ innovation processes, which makes it difficult to measure and analyze. To address this, I build on the methodology of Peters and Taylor (2017), constructing knowledge capital from R&D expenditures and depreciation. This approach recognizes that knowledge capital accumulates over time through investment and utilization in innovation activities, making its measurement essential for understanding how firms drive long-term growth and technological progress.

2.1 Data Sources and Sample Selection

Measuring knowledge capital is challenging because R&D expenditures are not consistently reported or accounted for in financial statements. Unlike tangible assets, which are capitalized and systematically depreciated, R&D is often treated as an expense. This leads to irregular reporting

across firms and time periods. For example, some firms disclose R&D expenditures only in their annual financial statements, while others provide inconsistent or incomplete data. Additionally, R&D investments vary greatly across industries. Innovation-heavy sectors, such as technology or biotechnology, invest heavily in R&D, while other industries may spend very little. These differences make it difficult to create a consistent and reliable measure of knowledge capital.

To overcome these challenges, I build the dataset from five main sources that together capture both the inputs and outputs of innovation. The CRSP-Compustat Merged (CCM) database, covering 1990Q1 to 2020Q4, lets me track firms' financial information consistently over time and measure R&D investment accurately. Compustat quarterly data, spanning 1960Q1 to 2020Q4, provide the detailed R&D expenditure figures needed to analyze firms' innovation investments. For firms reporting R&D only annually, I evenly distribute their annual spending across quarters to maintain consistency in the time series, and I exclude firms that have never reported R&D expenditures so as to focus on firms actively engaged in innovation. This last restriction matters for how the extensive margin should be interpreted below: because firms that never report R&D are not in the sample, the extensive margin I estimate is the decision of a firm with some history of research to remain above or fall below a minimum threshold of research intensity, rather than the decision of a non-innovating firm to begin innovating for the first time.

To measure the outputs of innovation, I use patent filings from the United States Patent and Trademark Office (USPTO) covering 1990 to 2017. These filings report the number and quality of patents granted to firms, letting me assess how effectively R&D investment translates into innovation. I also draw on the patent database of [Kogan et al. \(2017\)](#), which links patent activity to economic outcomes such as resource allocation and growth over the same 1990–2017 period and includes measures of patent quality, such as citation counts and market values, to evaluate the impact of innovations. In addition, I use the text-based importance measure of [Kelly et al. \(2018\)](#), who focus on breakthrough innovations during the same period; their work helps capture the significance of high-quality patents in contributing to knowledge capital. Finally, I supplement the dataset with data from Datastream, which adds financial and economic information that provides a broader view of firms' innovation processes across different economic environments.

Combining these data sources lets me capture both the scale of firms' R&D investments and the results of those investments, such as patents and their quality. All financial data are adjusted to real 1990 U.S. dollars using the Consumer Price Index (CPI) to account for inflation. This approach lets me study the relationship between R&D spending, knowledge capital growth, and innovation outcomes in a way that reflects firms' activities more accurately.

Throughout the paper I exclude the Great Recession from the estimation sample. High-frequency identification of monetary policy shocks becomes unreliable once the federal funds

rate approaches its effective lower bound, because the policy rate no longer moves in response to Federal Open Market Committee announcements and the interest-rate surprises measured around those announcements instead reflect forward guidance and balance-sheet policy rather than conventional changes in the policy stance. The baseline shock series of [Ottonello and Winberry \(2020\)](#) ends in December 2007 for the same reason, so retaining these quarters would add outcome observations without adding identifying variation in the shock. Dropping them follows standard practice in the firm-level monetary policy literature, and Appendix [D.3](#) reports the baseline specification re-estimated on alternative samples to confirm that the results do not depend on this choice. Appendix [D.3](#) also shows that the baseline results are not an artifact of the broader sample construction described above: they hold when I restrict attention to the top patenters, to the drug and semiconductor industries, and to NBER recession quarters outside the Great Recession.

2.2 Constructing the Knowledge Capital Stock

To address the challenges of measuring knowledge capital, I build upon the methodologies introduced by [Peters and Taylor \(2017\)](#). Knowledge capital is a dynamic and intangible asset that evolves through continuous R&D investments while simultaneously depreciating over time. Unlike tangible assets, knowledge capital is not directly reflected on balance sheets, making its measurement inherently complex. To address this complexity, I adopt the perpetual inventory method, which provides a structured and transparent framework for capturing the dual processes of accumulation and depreciation inherent in knowledge capital.

The perpetual inventory method systematically tracks the evolution of knowledge capital by recognizing its dynamic nature. It accounts for how firms build their stock of knowledge through ongoing R&D investments while simultaneously reflecting the diminishing value of existing knowledge due to technological obsolescence, market competition, and changing economic conditions. The knowledge capital stock for a firm at a given time is calculated as the previous period’s stock multiplied by one minus the depreciation rate, plus the R&D expenditures of the current period. Specifically, if the prior period’s knowledge capital stock is $KC_{i,t-1}$, the current R&D expenditures are $R\&D_{i,t}$, and the depreciation rate is δ , the formula becomes $KC_{i,t} = (1 - \delta) \times KC_{i,t-1} + R\&D_{i,t}$. I set the depreciation rate at 15%, reflecting the rapid pace of obsolescence in innovation-driven industries.

Two key measures derived from this framework are the investment rate of knowledge capital and the growth rate of knowledge capital. The investment rate, defined as the ratio of a firm’s R&D expenditures in the current period to its knowledge capital stock in the previous period,

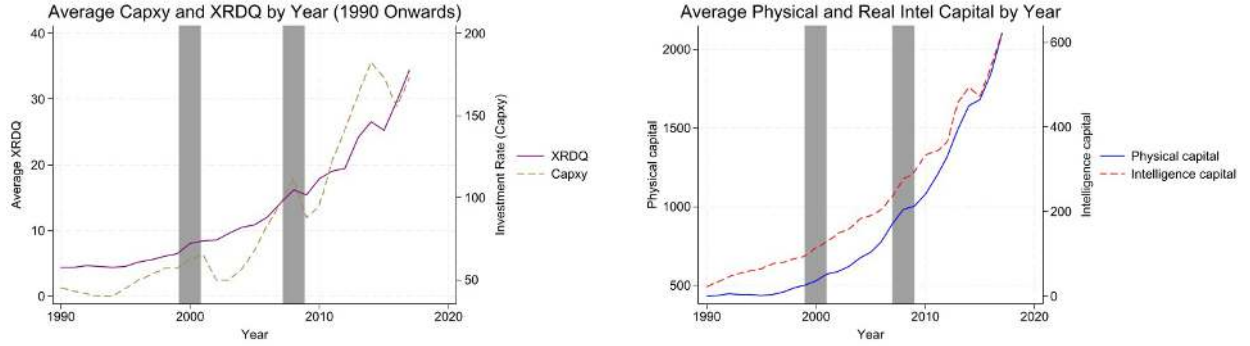
reflects a firm's decision-making process regarding R&D investments. For firm i at time t , this is expressed as $ikc_{i,t} = \frac{R\&D_{i,t}}{KC_{i,t-1}}$. A higher investment rate indicates a deliberate effort to allocate resources toward expanding the firm's knowledge base and maintaining its competitive advantage. This measure provides insights into how firms prioritize and manage their innovation activities in response to economic conditions and market dynamics. Additionally, the growth rate of knowledge capital tracks the proportional change in a firm's knowledge capital stock over time. It is calculated as the difference between the logarithm of the knowledge capital stock in the current period and the logarithm of the stock in the previous period. For firm i at time t , this is given by $\Delta \log KC_{i,t} = \log KC_{i,t} - \log KC_{i,t-1}$. This measure reflects the cumulative impact of a firm's R&D investments and its ability to sustain and grow its innovation-driven assets. While the investment rate focuses on the firm's active R&D decisions, the growth rate captures the long-term outcomes of these decisions and the overall progression of knowledge accumulation, offering a detailed understanding of how firms strategically manage their innovation resources over time.

The use of this method provides a clear and flexible way to measure knowledge capital. By looking at both the investment rate and the growth rate of knowledge capital, I can better understand how firms plan their innovation strategies, balancing short-term spending with long-term knowledge building. This approach helps explain the connection between R&D investments, changes in knowledge capital, and innovation results, while also considering the impact of changing economic and technological conditions. The dataset spans from 1960 to 2020, ensuring a comprehensive analysis across different time periods without significant gaps in the data, thereby capturing the evolution of firm-level innovation behaviors over decades.

Figure 1 illustrates trends in capital expenditures and intellectual capital over time. Panel (a) plots the average R&D expenditures (XRDQ, solid line, left axis) against the average capital expenditure investment rate (CAPX, dashed line, right axis) from 1990 onward, with the shaded gray bands marking the 2001 and 2008–2009 recessions. R&D expenditures rise smoothly and almost without interruption throughout the sample, while capital expenditures dip visibly around each recession before accelerating more sharply after 2010, overtaking R&D spending around 2013–2014—consistent with R&D commitments being stickier and less cyclically sensitive than physical investment.

Panel (b) shows the corresponding stocks: physical capital (solid line, left axis) and knowledge capital (dashed line, right axis), both constructed via the perpetual inventory method. Physical capital grows unevenly through the 1990s, with visible slowdowns around each recession, before growing markedly faster from 2010 onward. Knowledge capital, by contrast, grows steadily and almost monotonically throughout, reflecting its slow depreciation rate (15% per year) and consequent insensitivity to short-run fluctuations in current R&D flows. This contrast—procyclical

Figure 1: Trends in Physical and Knowledge Capital Expenditures



(a) Average CAPX and R&D Expenditures Over Time **(b) Average Physical and Knowledge Capital Over Time**

Notes: This figure illustrates trends in physical capital and R&D expenditures for US firms over time. The solid line represents physical capital expenditures, while the dashed line represents expenditures on XRDQ, which include R&D expenditures and their total. Panel (a) shows average capital expenditures (CAPX) and R&D expenditures over time, with gray shaded areas indicating periods of economic recessions. Panel (b) shows trends in average physical capital and knowledge capital, with both variables plotted over the same time frame. The curves reflect observed changes in expenditures and capital accumulation during the sample period.

physical investment against a comparatively smooth, persistent innovation stock—motivates the local projection approach used throughout the paper to trace out the dynamic response of knowledge capital to monetary policy shocks.

2.3 Patent Data Application

I focus on the outcomes of companies' innovation activities, particularly their patent results, which serve as a key indicator of innovation output. To capture the economic and technological impact of patents, I utilize data from prior studies as [Kogan et al. \(2017\)](#) and [Kelly et al. \(2018\)](#). [Kogan et al. \(2017\)](#) provide assessments of the market value of patents and their influence on firms' financial performance, demonstrating that patents are not only technological milestones but also valuable economic assets that strengthen firms' competitive positions. Meanwhile, [Kelly et al. \(2018\)](#) evaluate the technological novelty and broader influence of patents, offering insights into the originality of innovations and their impact across industries. Together, these indicators enable a more precise evaluation of firms' research achievements and help establish clear links between patenting activity and the accumulation of knowledge capital.

Table 1 illustrates the heterogeneity in patent outcomes across firms. The mean values for key variables, such as total patent number, citations, and market value, are notably skewed, as indicated by large disparities between mean and median values. For instance, the median total

patent number and citations are zero, indicating that a substantial proportion of firms hold no patents or receive no citations. However, the upper percentiles (P75 and P95) demonstrate that firms with patents can generate exceptionally high value and influence. This pattern suggests that while most patents have limited or no measurable impact, a small subset significantly drives innovation output, market valuation, and overall influence. The dataset includes 201,059 observations from publicly traded firms with available data, which reinforces the picture of patent outcomes concentrated within a small group of highly innovative firms.

Table 1: Summary Statistics of Patent Variables

	Mean	P50 (Median)	P75	P95	SD	Max
Total patent number	4.287	0	1	14	34.285	2467
Total patent citation	85.760	0	6	292	651.825	39249
Average patent citation	8.635	0	3	41.750	35.475	1944
Total patent market value	60.602	0	0.726	135.826	530.268	29579
Average patent market value	2.938	0	0.525	12.554	15.968	1151
Total patent innovation	3.047	0	0	9.214	25.867	1874
Average patent innovation	0.274	0	0	1.445	0.565	12
Observations	201059					

Notes: This table presents summary statistics for the patent-related variables used in this analysis. The variables are defined as follows: Total patent number is the total number of patents; Total patent citation is the total number of citations received by patents; Average patent citation is the average number of citations per patent; Total patent market value is the total real value of patents following Kogan et al. (2017); Average patent market value is the average real value per patent; Total patent innovation and Average patent innovation are measures of patent importance as defined by Kelly et al. (2018). All variables are calculated over the sample period and are winsorized at the 1% and 99% levels to mitigate the impact of outliers.

2.4 Monetary Policy Shocks

I incorporate smoothed monetary policy shock data from [Ottonello and Winberry \(2020\)](#), covering January 1990 to December 2007. Their dataset identifies 164 shocks with a mean near zero and a standard deviation of approximately 9 basis points. To merge these shocks with the firm-level data, I adopt their time-aggregation method, constructing a moving average of shocks weighted by the number of days in the quarter after the shock occurs. This approach better captures firms' exposure to monetary policy changes over time. For comparison, I also apply their alternative method of summing shocks within each quarter to test the robustness of these findings. As a further robustness check, Appendix [A.3](#) re-estimates all baseline results using the information-robust shock of [Jarociński and Karadi \(2020\)](#), which extends the sample through 2019 and purges the central-bank information component embedded in high-frequency interest-rate surprises.

Table 2 shows that the summary statistics for high-frequency and smoothed shocks are consistent, with both measures exhibiting similar moments across mean, standard deviation, and extreme values. These results indicate that different aggregation methods preserve the key characteristics of the shocks, confirming the suitability of the data for analyzing the effects of monetary policy on firms' innovation outcomes.

Table 2: Summary Statistics of Monetary Policy Shocks

	High Frequency	Smoothed	Sum
Mean	-0.0185	-0.0429	-0.0421
Median	0	-0.0127	-0.00509
S.D.	0.0855	0.108	0.124
Min	-0.463	-0.480	-0.479
Max	0.152	0.233	0.261
Observations	164	71	72

Notes: This table reports the summary statistics of monetary policy shocks. The "High Frequency" column represents the high-frequency identified shocks, "Smoothed" refers to the smoothed shocks obtained from a structural model, and "Sum" indicates the cumulative monetary policy shocks over the sample period. The monetary policy shocks are derived from [Ottonello and Winberry \(2020\)](#), providing insights into the unexpected changes in monetary policy that are not anticipated by market participants. These shocks are crucial for understanding the causal impact of monetary policy on firm behavior.

2.5 Other Variables

Furthermore, I collect operational data from Compustat, including earnings before interest and taxes (EBIT), sales revenue, total assets, and other financial indicators. These data let me control for factors such as firms' operational conditions, profitability, and size, allowing for a more precise analysis of the impact of knowledge capital on firm performance. Appendix D.3 also checks that the baseline estimates are not sensitive to the particular set of firm-level controls used throughout the paper.

Table 3 presents summary statistics for research firms, defined as those reporting R&D expenditures, covering the period from 1991 to 2016. The table provides key financial and operational metrics, including the knowledge capital stock (KC), the growth rate of knowledge capital ($\Delta \log KC$), capital expenditure relative to R&D (Capx/R&D), physical capital, cash holdings, leverage, and other firm characteristics such as age, size, and debt growth. These statistics, based on a total of 201,059 observations, offer insights into the financial conditions and investment behaviors of research-oriented firms, further supporting the empirical analysis. Appendix A reports addi-

tional descriptive evidence, including the annual evolution of patenting activity and a complete breakdown of firm-level financial variables by research status.

Table 3: Summary Statistics of Research Firms

	Mean	P50 (Median)	P25	P75	SD	Min	Max
KC	7.36	5.34	0.12	8.6	11.37	0	100
$\Delta \log KC$	0.04	0.02	-0.01	0.05	0.11	-0.05	6.4
Capx / R&D	11.83	1.19	0.34	4.28	153.45	-28.74	23476
Physical Capital	628.61	15.26	2.27	122.32	4961.34	0	255556
Cash	0.21	0.12	0.03	0.32	0.22	-0.07	1
Distance to Default	6.08	5.02	2.48	8.48	5.12	-4.25	172.52
Leverage	0.23	0.16	0.02	0.32	0.38	0	9.75
Age	16.52	13.75	8	22.75	11.17	0.75	56.5
Total Asset	2082.1	97.76	19.75	624.63	11582.97	0.01	444097
Size	5	4.84	3.26	6.65	2.4	-5.07	13
Dividend Paid	0.09	0	0	0	0.29	0	1
Cashflow	0	0.02	-0.01	0.03	0.18	-46.9	4.78
Debt Growth	1.04	-0.01	-0.08	0.04	105.71	-1	27779.56
Observations	201,059						

Note: Summary statistics are reported for research firms only, defined as those that report R&D expenditures. The sample spans from 1991 to 2016 and includes all research firms from Compustat with available data. The variables are defined as follows: *KC* is the knowledge capital stock constructed via the perpetual inventory method described above; $\Delta \log KC$ is the growth rate of knowledge capital; *CAPX/R&D* is capital expenditure divided by R&D expenditure; *Physical Capital* is net property, plant, and equipment; *Cash* is cash holdings divided by total assets; *Distance to Default* is a measure of the firm's default risk following [Gilchrist and Zakrajšek \(2012\)](#); *Leverage* is total debt divided by total assets; *Age* is the number of years since the firm's first appearance in Compustat; *Total Assets* is the book value of assets; *Size* is the natural logarithm of total assets; *Dividend Paid* is a dummy variable equal to 1 if the firm paid dividends in the fiscal year; *Cash Flow* is operating income before depreciation scaled by total assets; *Debt Growth* is the annual growth rate of total debt.

3 Empirical Strategy and Baseline Results

This section sets out the empirical strategy for investigating the impact of monetary policy shocks (MPS) on firms' R&D expenditures and innovation outcomes, and reports the corresponding baseline estimates. Specifically, I seek to understand whether innovation is driven by changes in R&D investment induced by monetary policy. This approach helps unravel the dynamic mechanisms through which monetary policy influences both the inputs and outputs of firm-level innovation activities.

First, I investigate the impact of MPS on firms' investment in innovation. By analyzing the relationship between MPS and R&D expenditures at the firm level, I aim to understand how mone-

tary policy shapes firms' decisions to allocate resources toward innovation activities. Specifically, I assess whether MPS influence firms' propensity to invest in R&D and the extent to which they adjust their innovation-related expenditures. To achieve this, I employ both Probit and Tobit models to examine the effects of MPS on firms' innovation investment decisions.

Next, I assess the direct impact of MPS on innovation outcomes. Using measures such as patent counts, patent citations, or other innovation indicators, I evaluate whether changes in monetary policy have a significant effect on the innovative output of firms. This analysis helps determine if monetary policy affects innovation independently of its influence on R&D spending. It provides insights into the extent to which monetary policy directly drives innovation outputs beyond its impact on the financial constraints that influence R&D decisions.

Finally, I investigate the relationship between the fitted values of R&D expenditures from the first step and innovation outcomes using a two-stage regression model. This approach lets me identify the extent to which the effect of monetary policy on innovation operates through its impact on R&D investment. By isolating the mediating role of R&D expenditures, this method provides a clear understanding of how monetary policy-induced changes in R&D spending influence innovation outcomes.

In the subsequent subsections, I provide detailed descriptions of the econometric models, variable definitions, and estimation methods used in each step of the analysis. This structured framework ensures a comprehensive understanding of how monetary policy shapes firms' innovation behaviors and outputs. Appendix B provides supplementary derivations and specification details for the static and dynamic empirical frameworks introduced below.

3.1 Impact of MPS on Innovation Inputs

In this section, I investigate the effects of MPS on firms' R&D investment and the accumulation of knowledge capital. Employing local projection methods along with Probit and Tobit models, I study the responsiveness of firm-level R&D expenditures to changes in monetary policy over time. This analysis differentiates between the extensive margin, which assesses whether firms remain engaged in innovation activities at a minimum level of intensity, and the intensive margin, which evaluates the magnitude of R&D investment among firms already engaged in such activities. This dual approach lets me examine not only the probability of firms sustaining innovation efforts in response to MPS but also the adjustments in R&D expenditure levels among existing innovators under varying monetary conditions.

3.1.1 Local Projection Model

I first employ a local projection model (Òscar Jordà, 2005) to capture how R&D expenditures respond to monetary policy shocks over time. The model is specified as follows:

$$y_{j,t+h} = \alpha_{j,h} + \beta_h \varepsilon_t + \Gamma'_1 Z_{j,t-1} + \Gamma'_2 Y_{t-1} + e_{j,t,h} \quad (1)$$

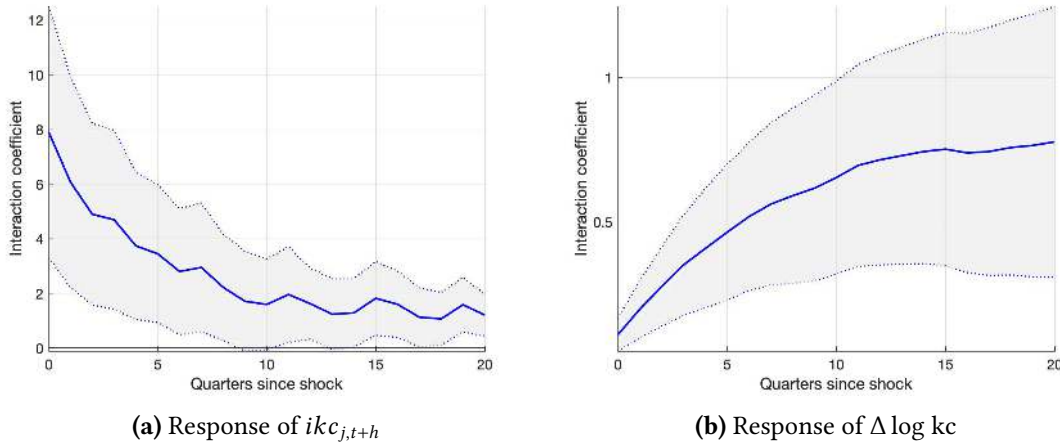
Here, $y_{j,t+h}$ represents the firm-level outcome variable of interest for firm j at time $t + h$, where h denotes the forecast horizon. The key coefficient of interest, β_h , captures the cumulative response of the outcome variable to monetary policy shocks (ε_t). In this analysis, y includes measures related to firms' R&D and knowledge capital dynamics, such as the investment rate of knowledge capital ($ikc_{i,t}$) and the growth rate of knowledge capital ($\Delta \log KC_{i,t}$). The investment rate, $ikc_{i,t} = \frac{R\&D_{i,t}}{KC_{i,t-1}}$, reflects how much firms allocate to R&D relative to their existing stock of knowledge capital, while the growth rate of knowledge capital, $\Delta \log KC_{i,t} = \log KC_{i,t} - \log KC_{i,t-1}$, measures the proportional change in knowledge capital, encapsulating its cumulative evolution over time. The firm-specific fixed effects $\alpha_{j,h}$ address time-invariant characteristics specific to each firm. The vector $\Gamma'_1 Z_{j,t-1}$ contains lagged firm-level controls, including leverage, distance to default, total assets, and sales growth, which capture financial and operational conditions specific to each firm. $\Gamma'_2 Y_{t-1}$ incorporates lagged macroeconomic controls such as log CPI, real GDP growth, and the employment ratio, accounting for broader aggregate economic conditions. The error term $e_{j,t,h}$ represents unobserved heterogeneity and stochastic shocks. I control for the levels of these firm characteristics and their interaction with the monetary policy shock (ε_t), ensuring that any observed differences in the response of knowledge capital investment and innovation outcomes are not driven by other observable firm characteristics. These controls enable the analysis to isolate the dynamic impact of monetary policy shocks on firm-level metrics such as R&D investment, knowledge capital accumulation, and innovation outputs over various time horizons.

The results are presented in Figure 2, which illustrates the dynamic responses to a 100 bp expansionary monetary policy shock. Panel (a) shows that the R&D investment rate ($ikc_{j,t+h}$) responds strongly to the shock in the initial quarters, with the peak effect occurring on impact, followed by a gradual decline over time while remaining positive throughout the horizon. A 100 basis point expansionary shock raises the investment rate of knowledge capital (ikc) by roughly 8 percentage points on impact; the response then decays steadily, falling to about 5 percentage points by quarter 2, roughly 2 percentage points by quarter 10, and a little over 1 percentage point by quarter 20.³ Firms already engaged in research therefore front-load their R&D response,

³Throughout the paper, responses of ikc are reported in percentage points of the investment rate, so a coefficient

scaling up sharply when financing conditions ease and unwinding gradually rather than sustaining the elevated investment rate indefinitely. Panel (b) shows the response of the growth rate of knowledge capital ($\Delta \log KC$), revealing a steady upward trend throughout the 20-quarter horizon. Knowledge capital growth demonstrates a persistent and cumulative effect, indicating that the initial boost in R&D investment translates into sustained accumulation of knowledge capital over time. The contrast between the two panels is what one would expect from a slowly depreciating stock accumulating a front-loaded flow: the flow response peaks immediately and decays, while the stock response builds monotonically and does not revert within the horizon. Together, these results show that expansionary monetary policy has both immediate and lasting impacts on firms' innovation activities, first stimulating R&D expenditures in the short term and subsequently contributing to long-term knowledge capital growth. Appendix C.4 shows that these dynamics remain robust when the horizon is extended to $h = 30$ quarters and when the specification is augmented with additional lags of the dependent variable.

Figure 2: Dynamic Responses to a 100 bp Monetary Policy Loosening.



Notes: These lines illustrate the dynamic responses to a 100 bp expansionary monetary policy shock. The solid line represents the estimated β_h interaction coefficient, and the dashed lines represent the 90% error bands. The horizontal axis represents the quarters since the monetary policy shock. Panel (a) shows the dynamic response of the R&D investment rate, while Panel (b) illustrates the growth of knowledge capital.

of 8 corresponds to an increase of 8 percentage points in R&D expenditure scaled by the lagged knowledge capital stock. The 90% band excludes zero over roughly the first eight quarters and includes it thereafter, so the front-loaded portion of the response is the precisely estimated one.

3.1.2 Extensive and Intensive Margins

In this section, I explore both the extensive and intensive margins of firms' responses to MPS. To analyze these dimensions, I employ Tobit and Probit models. The extensive margin captures whether monetary policy shocks influence firms to remain 'active' in innovation activities, while the intensive margin examines how firms already engaged in such activities adjust the intensity of their investment. Specifically, the Probit model is used to investigate the likelihood of firms staying 'active' in innovation under monetary policy shocks, whereas the Tobit model evaluates the adjustments in investment levels by firms that are already engaged in R&D activities.

The Probit model specifies a latent index for investment in knowledge capital as follows:

$$ikc_{j,t+h}^* = \alpha_{j,h} + \beta_h \varepsilon_t + \Gamma'_1 Z_{j,t-1} + \Gamma'_2 Y_{t-1} + e_{j,t,h} \quad (2)$$

The corresponding observed binary outcome, $d_{j,t+h}$, is defined as:

$$d_{j,t+h} = \begin{cases} 1, & \text{if } ikc_{j,t+h} > 1\% \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

In this framework, $ikc_{j,t+h}$ is the observed investment rate of knowledge capital for firm j at time $t+h$, while $ikc_{j,t+h}^*$ is the unobserved latent index that governs whether the firm crosses the activity threshold. Firms are classified as "active" if their investment rate exceeds the 1% threshold; otherwise, they are categorized as "inactive." The binary variable $d_{j,t+h}$ therefore records whether a firm is "active" in knowledge capital investment, and it is distinct from the latent index that generates it. The model includes the same firm-level controls ($Z_{j,t-1}$), macroeconomic controls (Y_{t-1}), and firm fixed effects ($\alpha_{j,h}$) as described earlier.⁴ This classification distinguishes firms that actively engage in innovation from those that do not, based on their observed investment behavior. The Probit model focuses on the extensive margin, capturing whether monetary policy shocks (ε_t) influence firms to become or remain active in innovation. By estimating the key coefficient β_h , I assess the dynamic impact of monetary policy shocks on the probability of firms crossing the activity threshold. This approach shows whether monetary policy stimulates firms to initiate or expand innovation activities, reflecting its broader effects on firm behavior.

⁴Because the Probit specification includes firm fixed effects, $\hat{\beta}_h$ is subject to the incidental parameters problem: with a large number of firms and a comparatively short time dimension, the firm effects are estimated imprecisely and this imprecision transmits to the slope coefficients in a nonlinear model, so the estimates need not be consistent even in large samples. I therefore read the extensive-margin estimates as descriptive rather than as consistent estimates of a marginal effect. The substantive conclusion below—that the extensive margin accounts for little of the aggregate R&D response—rests on the contrast in magnitude and precision with the intensive-margin estimates rather than on the point value of $\hat{\beta}_h$.

The Tobit model focuses on analyzing the behavior of firms that are already “active” in innovation when facing changes in monetary policy. Unlike the Probit model, which examines whether firms cross the threshold to become active (extensive margin), the Tobit model studies the magnitude of investment among those firms whose knowledge capital investment rate exceeds the 1% threshold, capturing the intensive margin of investment. The latent index is specified as:

$$ikc_{j,t+h}^* = \alpha_{j,h} + \beta_h \varepsilon_t + \Gamma_1' Z_{jt-1} + \Gamma_2' Y_{t-1} + e_{j,t,h} \quad (4)$$

and the observed, censored outcome $\widetilde{ikc}_{j,t+h}$ is defined as:

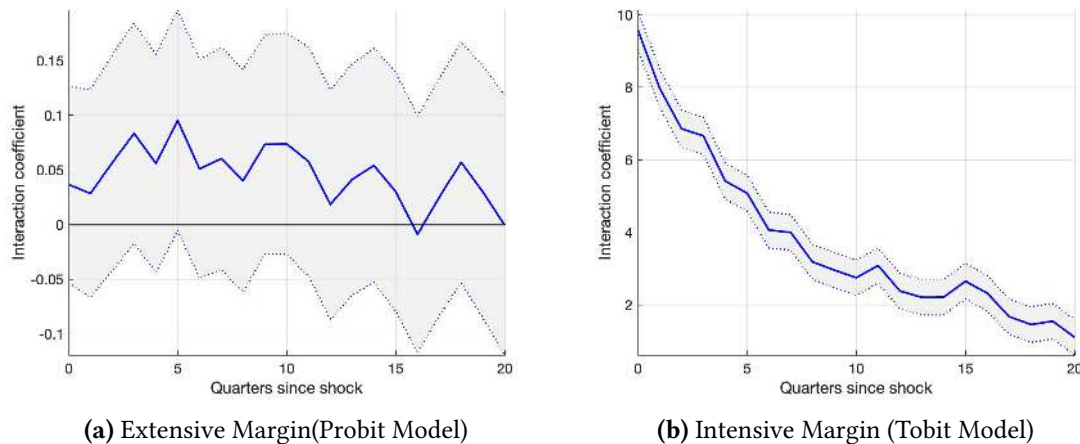
$$\widetilde{ikc}_{j,t+h} = \begin{cases} ikc_{j,t+h}^*, & \text{if } ikc_{j,t+h}^* > 1\% \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

This model accounts for the fact that only firms with an investment rate above the 1% threshold are ‘actively’ engaging in innovation, and focuses on how these firms adjust their level of investment in response to monetary policy shocks (ε_t). By estimating the key coefficient β_h , the Tobit model helps identify how expansionary monetary policy shapes the scale of innovation-related activities among firms already engaged in knowledge capital investment, providing insights into its role in influencing the intensity of such investments.

Figure 3 illustrates dynamic responses to a 100 bp expansionary monetary policy shock using Probit and Tobit models. Panel (a) presents the results of the Probit model, which captures the probability of firms becoming active in innovation. The coefficients are never statistically significant at any horizon: they oscillate between roughly 0.03 and 0.095 over the first ten quarters, and the 90% band contains zero throughout, so the point estimates suggest at most a small and temporary increase in the likelihood of firms sustaining investment activities and cannot be distinguished from no response at all. Panel (b) shows the response of the Tobit model, which measures the level of investment among already ‘active’ firms. The results closely track the pooled response in Figure 2: the coefficient starts near 10 percentage points on impact, falls to roughly 5 percentage points by quarter 4 and about 3 percentage points by quarter 8, and declines to a little over 1 percentage point by quarter 20, with confidence bands narrow enough to exclude zero at every horizon. The contrast between the two panels is the substantive point of this subsection. The intensive-margin response is an order of magnitude larger than the extensive-margin response and precisely estimated over the full horizon, whereas the extensive-margin response is small, erratic, and statistically indistinguishable from zero, which points to a substantial and lasting increase in the intensity of investment for firms already engaged in innovation and essentially no measurable movement at the participation threshold. Appendix C reports additional Probit-

and Tobit-based predictions for citations, patent value, and patent importance that supplement the baseline extensive- and intensive-margin results discussed here.

Figure 3: Intensive and Extensive Margins (Probit and Tobit Models)



Notes: These lines illustrate the dynamic responses to a 100 bp expansionary monetary policy shock. The solid line represents the estimated coefficients, while the dashed lines indicate the 90% confidence intervals. Panel (a) shows the estimated β_h interaction coefficient from the Probit model, which measures the probability of firms becoming active in innovation. Panel (b) shows the response of the Tobit model, which measures the level of investment in knowledge capital among active firms. The horizontal axis represents the quarters since the monetary policy shock, and the shaded area represents the 90% confidence interval.

The results from the Probit and Tobit models reveal that the observed growth in the investment rate of knowledge capital (*ikc*) is predominantly driven by firms already engaged in research and innovation activities. While monetary policy shocks exhibit a limited and imprecisely estimated effect on encouraging inactive firms to initiate innovation, the majority of the increase in innovation investment originates from firms that are already active. This indicates that active firms are more responsive to monetary policy changes, likely due to their greater reliance on funding availability to support ongoing R&D activities. Expansionary monetary policy alleviates financial constraints for these firms, enabling them to scale up their innovation efforts. In contrast, inactive firms remain largely unresponsive, even under expansionary conditions, suggesting that barriers beyond funding—such as expertise gaps, uncertainty about returns, or structural limitations—may prevent them from initiating R&D. Monetary policy changes therefore fall disproportionately on firms already involved in innovation, and this asymmetry between active and inactive firms is the first result that the rest of the paper builds on: because the participation margin barely moves, the innovation response to monetary policy has to be understood as a change in what already-innovating firms choose to do with their research budgets rather than as a change in how many firms are doing research at all.

3.2 Impact of MPS on Innovation Outcomes

In this subsection, I investigate the impact of MPS on innovation outcomes using a local projection model. The dependent variable $p_{j,t+h}$ represents firm-level innovation outcomes at time $t+h$. The model is specified as:

$$p_{j,t+h} = \alpha_{j,h} + \beta_h \varepsilon_t + \Gamma'_1 Z_{j,t-1} + \Gamma'_2 Y_{t-1} + e_{j,t,h} \quad (6)$$

In this framework, $\alpha_{j,h}$ captures firm fixed effects. The key coefficient of interest, β_h , measures the dynamic response of innovation outcomes to monetary policy shocks (ε_t) at time t . The model incorporates the same firm-level controls ($Z_{j,t-1}$) and macroeconomic controls (Y_{t-1}) as described earlier. Firm-level controls include variables such as leverage, total assets, R&D expenditures, and prior innovation activity, while macroeconomic controls account for broader economic conditions, such as lagged GDP growth and other aggregate factors. The error term $e_{j,t,h}$ captures unobserved heterogeneity and idiosyncratic shocks that may affect innovation outcomes. By estimating β_h , this approach evaluates the extent to which monetary policy shocks influence firms' innovation activities, offering insights into how financial and economic conditions shape firm behavior.

The innovation outcome variable $p_{j,t+h}$ encompasses a range of measures to capture firms' innovation performance. Specifically, $p_{j,t+h}$ includes the total number of patents, total patent citations, and the average citation count per patent, which reflect the quantity, quality, and influence of innovation output. To provide a more comprehensive analysis, I also incorporate measures of patent value and the upper percentiles of patent performance (p80, p90, and p95). These metrics let me evaluate both the economic and technological significance of firms' innovation outcomes, offering insights into the distribution, scale, and impact of innovation activity. Because many firm-quarters record zero patents, Appendix D.1 re-estimates the count-based outcomes via Poisson pseudo-maximum-likelihood, which retains these zero observations, and confirms that the baseline standard errors are not sensitive to Driscoll–Kraay or industry-by-quarter clustering.

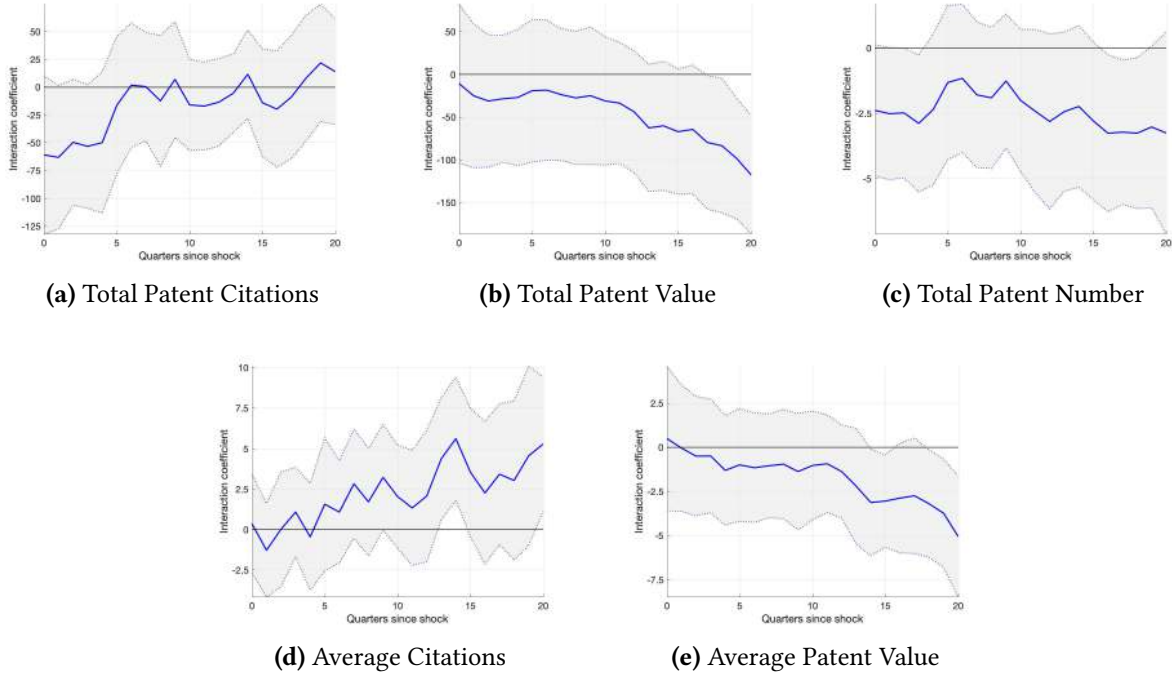
Figure 4 shows that a 100 basis point expansionary monetary policy shock moves the quantity and the per-patent quality of firm-level patent output in opposite directions over a 20-quarter horizon. Panel (a), Panel (b), and Panel (c) illustrate that total patent citations, total patent value, and total patent number all decrease. Total patent citations decline on impact and then recover gradually toward zero over the middle of the horizon. Total patent value is negative at every horizon, drifting from about -10 on impact to roughly -30 by quarter 10 and about -117 by quarter 20, with the 90% band excluding zero only over the final three or four quarters, so the

aggregate value response is large in magnitude but precisely estimated only at long horizons. The total number of patents falls by about two to three patents on impact, recovers partially to roughly -1.2 around quarters 5 and 6, and then declines again to about -3 by the end of the horizon; the effect is distinguishable from zero in the first few quarters and again toward the end, and imprecise in between, so the quantity response is best described as persistently negative with the sharpest declines at the beginning and end of the horizon. In contrast, Panel (d) shows that average citations per patent increase steadily over time, reaching approximately 10 citations by the 15th quarter. This suggests that although fewer patents are produced, the individual impact of each patent becomes higher. Panel (e) shows that the average patent value moves very little: the point estimate begins slightly above zero, drifts down to about -1 over the first ten quarters and about -5 by quarter 20, and its confidence band contains zero at all but the final few horizons, so per-patent commercial value cannot be said to respond reliably in either direction.

Read together, the five panels separate a quantity effect from a per-patent effect, and this distinction matters for how the aggregate declines should be interpreted. Since total patent value is by construction the product of the number of patents and their average value, the decline in Panel (b) is accounted for almost entirely by the decline in the number of patents in Panel (c) rather than by any deterioration in per-patent value, which Panel (e) shows to be statistically flat over almost the whole horizon. Firms produce fewer patents of roughly unchanged average market value, so the sum falls mechanically. The one per-patent margin that does respond, and responds substantially, is citations. Together, these results indicate that monetary policy shocks have heterogeneous effects on firms' patenting outcomes, with aggregate measures of commercial significance and quantity declining while metrics of quality, such as average citations, reveal relative improvements. This suggests that expansionary monetary policy reduces the overall scale of patent output while raising the average technological impact of the inventions firms do patent, and that the fall in aggregate patent value should not be read as evidence that easier financing conditions degrade the economic value of individual innovations.

Figure 5 explores further aspects of patent importance following a 100 basis point expansionary monetary policy, focusing on both the importance and distribution of patents. Panel (a) and Panel (b) show the log sum importance and log mean importance of patents, capturing aggregate and average measures of patent significance, respectively. Log sum importance initially decreases in response to the monetary policy shock but begins to rise steadily over time, ultimately showing a sustained positive trend. Log mean importance shows a gradual positive response, pointing to an increase in the average importance of individual patents following monetary policy shocks. Panel (c), Panel (d), and Panel (e) focus on the upper percentiles of patenting activity, tracing the impact of expansionary monetary policy on patents with varying levels of innovation sig-

Figure 4: Impact of Monetary Policy Shocks on Patenting Outcomes

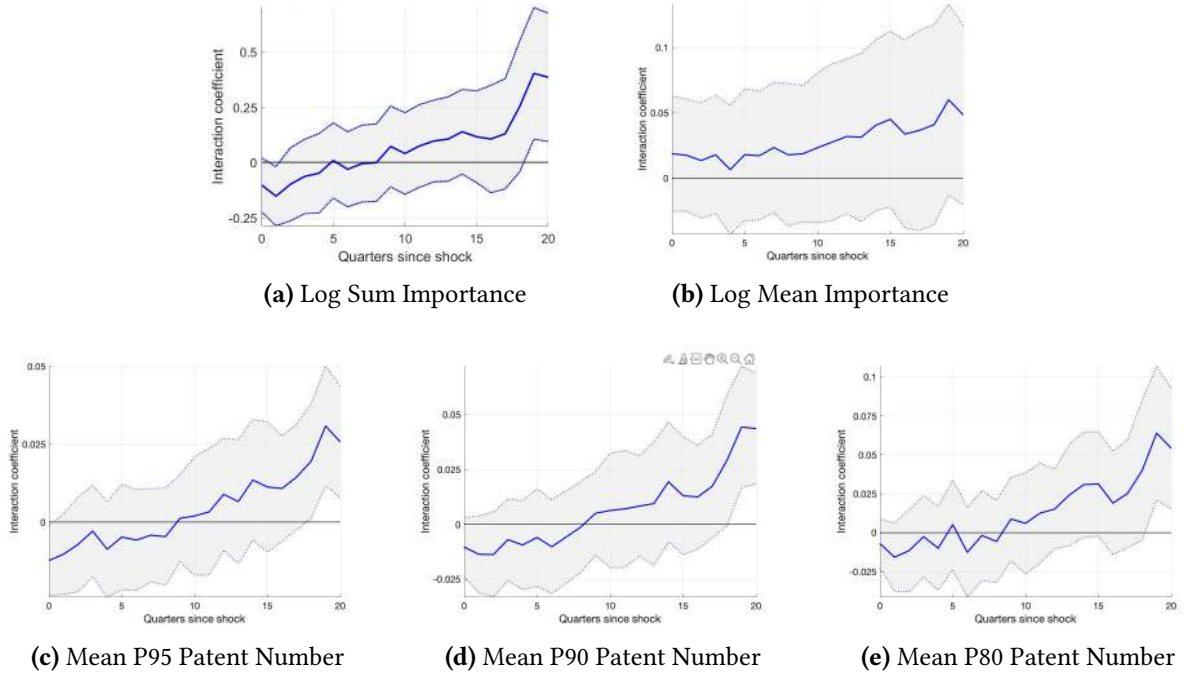


Notes: These lines illustrate the dynamic responses to a 100 bp expansionary monetary policy shock. The solid line represents the estimated coefficients, while the dashed lines indicate the 90% confidence intervals. This figure shows the dynamic effects of monetary policy shocks on aggregate and average patenting outcomes, including total patent citations, total patent value, total patent number, average citations per patent, and average patent value over the quarters since the monetary policy shock. Panels (a)–(c) capture aggregate measures, while Panels (d)–(e) focus on average-level metrics.

nificance. The mean P95 patent number, representing the most innovative patents, shows the strongest and most consistent response, followed by the mean P90 and P80 patent numbers, indicating that while monetary policy shocks positively influence the activity of the most innovative patents, the magnitude of the effect diminishes for patents lower in the innovation distribution. This gradient across percentile cutoffs is informative in its own right: if easier financing simply shifted the entire distribution of patent importance upward, the response would be proportionally similar at each cutoff, whereas the concentration of the effect in the narrowest tier indicates that the additional research effort is directed at the most ambitious projects specifically. This demonstrates that in the short term, both overall quality and top-percentile innovation outputs decline slightly, but in the long term, firms appear to shift their focus towards producing higher-quality patents.

The results suggest contrasting short-term and long-term effects of expansionary monetary

Figure 5: Impact of Monetary Policy Shocks on Patent Importance



Notes: These lines illustrate the dynamic responses to a 100 bp expansionary monetary policy shock. The solid line represents the estimated coefficients, while the dashed lines indicate the 90% confidence intervals. This figure presents the dynamic effects of monetary policy shocks on patent importance. Panels (a)–(b) focus on log sum and mean importance, reflecting the aggregate and average significance of patents. Panels (c)–(e) analyze the upper percentiles of patent activity, showing how the top-performing patents (P95, P90, and P80) respond to monetary policy shocks.

policy on firms' innovation decisions. In the short term, metrics such as total importance, log sum importance, and aggregate patent numbers decrease following expansionary monetary policy, indicating an initial reduction in both output quantity and quality. However, this shift reflects a deliberate reallocation of resources by firms. Rather than focusing on short-term, less significant innovations, firms redirect efforts toward projects with greater long-term value. This prioritization leads to an initial drop in innovation quantity as firms deprioritize immediate, lower-impact research in favor of pursuing projects with lasting significance. Over time, the long-term trends confirm this shift, as metrics like the number of top-performing P95 patents rise steadily, indicating an increase in high-value patents. Similarly, log mean importance and mean log importance improve, reflecting the growing significance of individual patents. These patterns show that firms adjust their innovation choices in response to monetary policy, focusing on producing fewer but more impactful patents, with the caveat noted above that the improvement in per-patent quality appears in citation-based and text-based measures rather than in per-patent market value,

which does not move reliably in either direction. While there is an initial decline in aggregate value and top-percentile outputs, firms later allocate resources to projects that yield sustained, higher-quality outcomes. Expansionary monetary policy initially reduces the scale of innovation projects but leads to a concentration of efforts on more impactful research. The following sections of the paper will further explore how firms' responses to monetary policy changes shape their innovation choices.

3.3 Two-Stage Analysis: From Innovation input to output

In this part of the analysis, I examine whether the observed changes in innovation outcomes are driven by increased R&D investments in response to expansionary monetary policy shocks. To address this, I employ a two-stage approach. In the first stage, I estimate the effect of monetary policy shocks on R&D investment, represented by the latent variable $ikc_{j,t}^*$. In the second stage, I use the fitted values from the first-stage regression to assess their impact on innovation outcomes, denoted by $p_{j,t+h}$.

The first stage employs a Tobit model to examine how MPS influence firms' R&D investments. Since the primary growth and investment in knowledge capital come from firms that are already 'active', I focus on understanding how these firms adjust their investment levels in response to changes in MPS. By addressing the intensive margin, the Tobit model quantifies the scale of investment among these active firms, capturing the impact of monetary policy shocks on their research outcomes. The first-stage regression is specified as follows:

$$ikc_{j,t}^* = \alpha_{j,t} + \gamma \varepsilon_t + \Gamma_1' Z_{j,t-1} + \Gamma_2' Y_{t-1} + e_{j,t}$$

Here, $\alpha_{j,t}$ represents firm fixed effects for firm j , capturing time-invariant firm-specific characteristics, while γ measures the effect of the monetary policy shock ε_t on firm j 's R&D investment $ikc_{j,t}^*$. The terms $\Gamma_1' Z_{j,t-1}$ and $\Gamma_2' Y_{t-1}$ include the same firm-level and aggregate control variables as described earlier.

In the second stage, I investigate whether the fitted values of R&D investment from the first stage account for the changes in innovation outcomes documented above, letting me assess how much of the observed shift in innovation is attributable to monetary policy-induced changes in R&D investment. The second-stage regression is specified as follows:

$$p_{j,t+h} = \alpha_{j,h} + \beta_h \widehat{ikc}_{j,t}^* + \Gamma_1' Z_{j,t-1} + \Gamma_2' Y_{t-1} + e_{j,t+h}$$

Here, $\widehat{ikc^*}_{j,t}$ represents the fitted R&D investment derived from the first-stage Tobit model, which captures the portion of R&D investment explained by monetary policy shocks and firm-level and aggregate controls. The Tobit model focuses on how firms already classified as “active” adjust their investment levels in response to changes in monetary policy, addressing the censored nature of R&D investment data. The key coefficient β_h quantifies the association between fitted R&D investment and innovation outcomes, such as patent value, patent number, and patent citations, measuring how changes in R&D intensity translate into these outcomes. A positive and statistically significant β_h is consistent with increases in R&D investment, driven by expansionary monetary policy, leading to measurable improvements in innovation outcomes.⁵ The firm-specific fixed effects $\alpha_{j,h}$ capture time-invariant characteristics of firm j , while $\Gamma'_1 Z_{j,t-1}$ and $\Gamma'_2 Y_{t-1}$ represent vectors of firm-level and aggregate control variables, consistent with those used in the first stage.

This two-stage framework provides a structured approach to understanding how monetary policy influences innovation outcomes. In the first stage, the Tobit model examines how the shock affects the scale of investment among firms already engaged in knowledge capital investment. The fitted values from this stage are used in the second stage to evaluate the association between R&D investment and patent-related metrics, such as the number, value, and quality of patents. This approach lets me link monetary policy-induced changes in R&D investment to observable innovation outcomes, offering insights into how monetary policy shapes firms’ investment decisions and their resulting contributions to innovation.

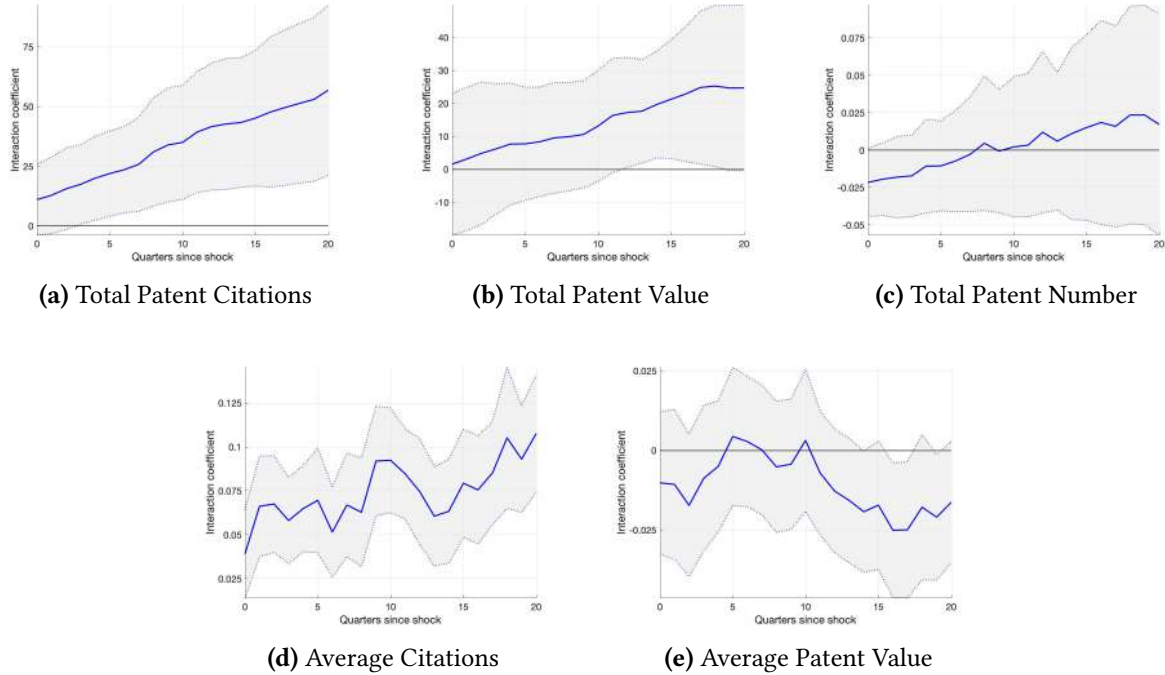
Figures 6 and 7 examine the effects of fitted R&D investments, derived from the first-stage regression, on various patenting outcomes. These figures provide a detailed analysis of how monetary policy-induced R&D investments influence both aggregate and distributional measures of innovation, including patent citations, value, quantity, and importance.

Figure 6 reports the response of patenting outcomes to a 1 percentage point increase in $\widehat{ikc^*}_{j,t}$, the fitted investment rate of knowledge capital from the Tobit model. Panels (a) and (b) show that total patent citations and total patent value both rise over time, underscoring the long-term positive association between R&D investment and the technological and economic significance of innovation. Total patent value increases steadily from close to zero on impact to about 10 by quarter 9 and roughly 25 by the end of the horizon, and its 90% band excludes zero over

⁵Because the first stage and the second stage share the same set of firm-level and aggregate controls, this specification is best read as a descriptive mediation decomposition rather than as an instrumental-variables estimate: there is no exclusion restriction under which $\widehat{ikc^*}_{j,t}$ affects $p_{j,t+h}$ only through R&D, and the second-stage coefficient should not be interpreted as the causal return to R&D. The reported standard errors also do not correct for the fact that $\widehat{ikc^*}_{j,t}$ is a generated regressor, and so should be treated as a lower bound on the true sampling uncertainty. What the exercise does establish is whether the patenting responses documented in Section 3.2 line up in sign and timing with the component of R&D investment that monetary policy moves.

the middle of the horizon, approximately quarters 11 through 18, before widening again at the longest horizons. Panel (c), however, reveals an initial decline followed by a gradual increase in the total number of patents, though the changes are not statistically significant at any horizon. This suggests that while firms maintain a broadly stable scale of patent production, their focus is shifting toward more meaningful and long-term innovation efforts. This trend is partly supported by Panel (d), where average citations per patent display steady and substantial growth, reflecting improvements in the technological influence of individual patents. Panel (e), by contrast, shows no reliable movement in average patent value: the point estimate oscillates within a narrow band around zero for the entire horizon, never departing far from the origin in either direction, and its confidence band always contains zero, so the per-patent commercial value margin is as unresponsive here as it was in the direct specification of Section 3.2.

Figure 6: Impact of Fitted R&D Investment on Patenting Outcomes



Notes: These lines illustrate the effects of a 1 percentage point increase in $\widehat{ikc}_{j,t}^*$, derived from the first-stage regression, on patenting outcomes. The solid line represents the estimated coefficients, while the dashed lines indicate the 90% confidence intervals. Panels (a)–(c) report aggregate outcomes: total patent citations, total patent value, and the total number of patents. Panels (d)–(e) report the corresponding average-level outcomes: average citations per patent and average patent value. These results show how R&D investment changes induced by monetary policy affect both the scale and the per-patent significance of patenting activity.

The sign reversal between Panel (b) here and the corresponding aggregate response in Sec-

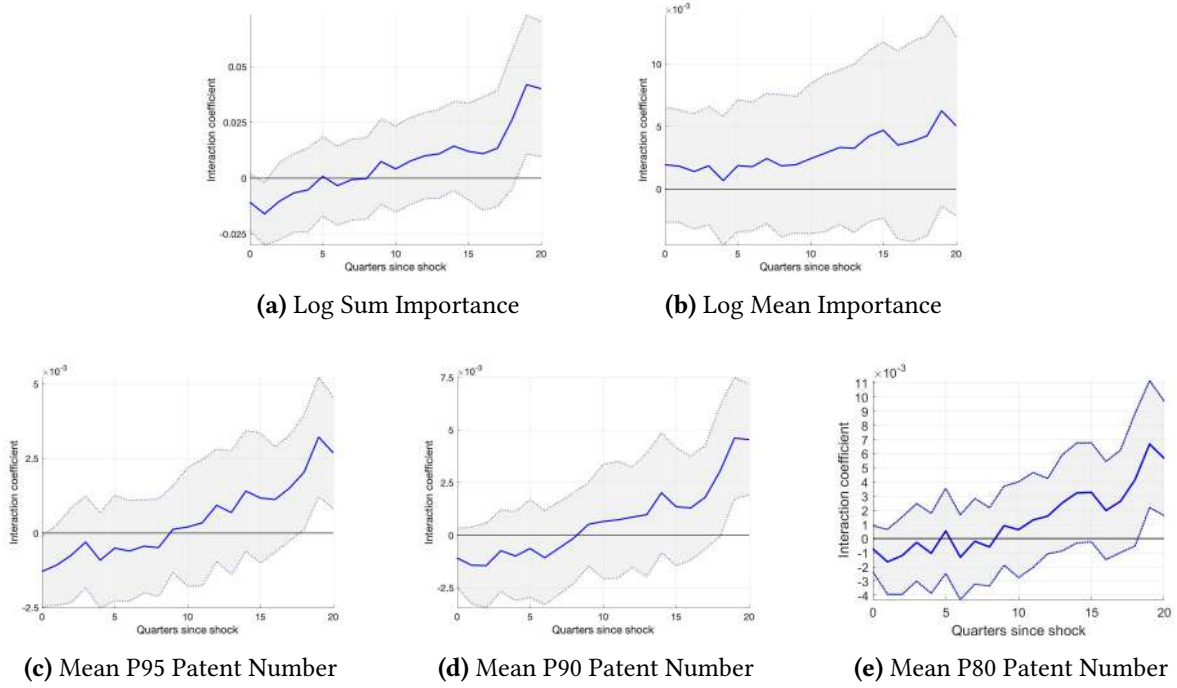
tion 3.2 is the informative comparison, and it deserves to be stated precisely. In the direct specification, total patent value falls; traced through the R&D response of already-active firms, it rises. Because per-patent value is statistically flat in both specifications, neither result can be attributed to a change in the average commercial value of the inventions produced; both are driven by the number of patents. The direct response reflects the fall in patent counts documented above, whereas the R&D channel, which by construction holds the composition of patenting firms fixed at those already active and isolates the component of investment that monetary policy moves, delivers a positive value response over the medium run. The aggregate decline is therefore not evidence that easier financing degrades the value of innovation; it reflects a smaller volume of patents produced by firms that are reallocating toward longer-horizon projects, and once the analysis conditions on the R&D response of those firms, the value of what they produce rises rather than falls.⁶ These findings indicate that “active” firms, following an expansionary monetary policy, are able to reallocate their resources toward producing patents with greater technological significance, and that the aggregate value decline documented earlier operates through the quantity margin rather than through any deterioration in what individual patents are worth.

Figure 7 reports the effects of a 1 percentage point increase in $\widehat{ikc^*}_{j,t}$ on patent importance. Panels (a) and (b) present log sum and log mean importance, capturing aggregate and average measures of patent significance. Log sum importance initially declines and then steadily increases, reflecting a shift in focus toward higher-value innovation over time. Log mean importance shows a consistent upward trend, indicating that the average significance of individual patents improves as firms adjust their innovation strategies. Panels (c), (d), and (e) report the behavior of top-performing patents, showing the mean number of patents in the highest percentiles. These panels reveal an initial decline followed by an increase, with the strongest effects observed for the most innovative patents. This suggests that firms reduce lower-value patenting efforts to concentrate on producing higher-quality patents, aligning their R&D investments with long-term technological and economic value. Taken with Panel (d) of the previous figure, the importance measures and the citation measures therefore agree, which is worth noting because the text-based importance measure of Kelly et al. (2018) is constructed without reference to citation counts and provides a largely independent read on the same reallocation.

The results suggest that monetary policy-induced R&D investments play a pivotal role in driving both the quantity and quality of innovation. Aggregate measures, such as total patent

⁶An alternative reading of the aggregate decline—that expansionary policy draws less innovative firms into R&D and thereby dilutes aggregate value—is inconsistent with two results reported above. First, the extensive-margin estimates in Section 3.1 find no detectable entry response, so there is no measurable inflow of new innovators to do the diluting. Second, a dilution mechanism would show up as a fall in per-patent value, and per-patent value does not respond in either specification.

Figure 7: Impact of Fitted R&D Investment on Patent Importance



Notes: These lines illustrate the dynamic responses to a 1 percentage point increase in $\widehat{ikc}_{j,t}^*$, derived from the first-stage regression, on patent importance metrics. The solid line represents the estimated coefficients, while the dashed lines indicate the 90% confidence intervals. Panels (a)–(b) capture aggregate and average significance through log sum and log mean importance. Panels (c)–(e) focus on the mean number of top-performing patents, including the P95, P90, and P80 percentiles. These results show how R&D investment changes induced by monetary policy affect both the distribution and significance of patents.

citations and total patent value, show sustained growth over time, reflecting the broader positive effects of these investments on economic and technological outcomes. However, the relatively stable total number of patents indicates a strategic focus on enhancing the quality of innovation rather than merely increasing its scale. This reallocation of resources is further evidenced by steady improvements in citation-based and text-based average metrics, which reflect firms' efforts to produce patents of greater significance, while per-patent market value remains flat, so the improvement in quality registers on the technological rather than the commercial margin. Top-performing patents, such as those in the upper percentiles of innovation significance, experience the most pronounced benefits, with substantial increases in their representation over time. Read against the direct estimates of Section 3.2, the comparison also clarifies what the earlier aggregate declines do and do not show: the fall in total patent value there arises from the drop in patent counts, and firms already active in R&D—especially those producing high-impact

innovations—continue to demonstrate improvements in patent quality once the analysis conditions on the R&D response through which monetary policy operates. Collectively, these findings emphasize that monetary policy-driven R&D investments not only enhance the quality of innovation outputs but also foster a strategic transition toward producing higher-value, economically impactful innovations, with the composition of output changing more than its scale.

4 Asymmetric Effects of Monetary Policy

Previous findings indicate that monetary policy plays a dual role in shaping firm-level innovation strategies, with contractionary and expansionary policies exerting distinct and often opposing effects on firms' R&D investment behaviors and innovation outcomes. Contractionary monetary policy, characterized by higher interest rates, raises the cost of borrowing, constrains credit availability, and forces firms to allocate resources more conservatively. As a result, firms facing tighter financial conditions often reduce R&D expenditures or prioritize incremental, lower-risk projects over high-risk, high-reward innovations, thereby reducing the overall scale of innovation activities. One might expect this conservative stance to raise the average quality of what firms continue to produce, on the reasoning that the projects cut first under a tighter budget are the marginal ones; the estimates reported below do not support that expectation, and instead show tightening raising patent counts while lowering the per-patent impact and importance of the resulting output. In contrast, expansionary monetary policy, marked by lower interest rates and increased liquidity, alleviates financial constraints and reduces the cost of capital. This environment incentivizes firms to increase R&D investment and to undertake more ambitious innovation projects. While expansionary policies raise aggregate R&D spending, the extensive-margin results of Section 3.1 show that this additional spending is concentrated among firms already conducting research rather than spread across new entrants, so the relevant question for this section is not how many firms begin innovating but how already-active firms change the composition of the projects they pursue when the sign of the shock flips.

Building on these insights into contractionary and expansionary monetary policies, I now turn to a more detailed examination of how these opposing monetary policies influence firm-level innovation outcomes. Specifically, I aim to disentangle the effects of these asymmetric monetary policies on the extensive margin—whether firms engage in R&D—and the intensive margin, which captures the scale of investment among firms that are already active. Additionally, I assess how changes in R&D investment translate into shifts in innovation outputs, with a focus on patent-related metrics such as value, citations, and the distribution of top-performing patents. By analyzing these nuanced effects, I aim to provide a comprehensive understanding of

the mechanisms through which monetary policy shapes firms' strategic decisions and the broader trajectory of technological progress and long-term economic growth.

4.1 Asymmetry by Shock Sign

I examine asymmetries in the transmission of monetary policy to firm-level innovation along two dimensions: the *sign* of the shock and its *magnitude*. The three specifications used in this section all share the same local projection structure, dependent variables, and set of firm-level and macroeconomic controls used throughout the paper; they differ only in how the monetary policy shock ε_t is split. The dependent variable $y_{j,t+h}$ again includes both R&D investment dynamics (e.g., $ikc_{j,t}$) and innovation outcomes such as patent counts, citations, value, and text-based importance, including the upper percentiles (P80, P90, P95) that capture the distribution of top-performing patents. I begin with asymmetry by sign, before turning to magnitude alone (Section 4.2) and their interaction (Section 4.3).

To capture asymmetry by sign, I split the shock into its expansionary (ε_t^+) and contractionary (ε_t^-) components and estimate

$$y_{j,t+h} = \alpha_{j,h} + \beta_h^+ \varepsilon_t^+ + \beta_h^- \varepsilon_t^- + \Gamma_1' Z_{j,t-1} + \Gamma_2' Y_{t-1} + e_{j,t,h}. \quad (7)$$

The results from Figure 8 reveal the asymmetric effects of a 100 basis point expansionary monetary policy shock (solid line) and a 100 basis point contractionary shock (dashed line) on various firm-level outcomes. Panel (a) shows that the investment rate rises sharply after an expansionary shock, by about 10 percentage points on impact, before decaying to approximately 2.5 percentage points after 10 quarters and remaining near that level through the end of the horizon; the 90% band excludes zero over roughly the first two years, so this is the precisely estimated part of the response. The contractionary side is not the mirror image of it: the point estimate is positive for the first eight or nine quarters, crosses zero around quarter 9 or 10, and then settles between -2.5 and -3 percentage points for the remainder of the horizon, with a band that contains zero at every horizon. The contractionary investment response therefore cannot be signed with confidence, and the sign asymmetry in Panel (a) rests on the expansionary side. Panel (b) turns to knowledge capital dynamics, where expansionary shocks lead to a stable upward trajectory over the 20-quarter horizon, while contractionary shocks result in a decline to a lower level, so the stock measure reproduces the same asymmetry as the flow measure with the direction of the two responses more clearly separated.

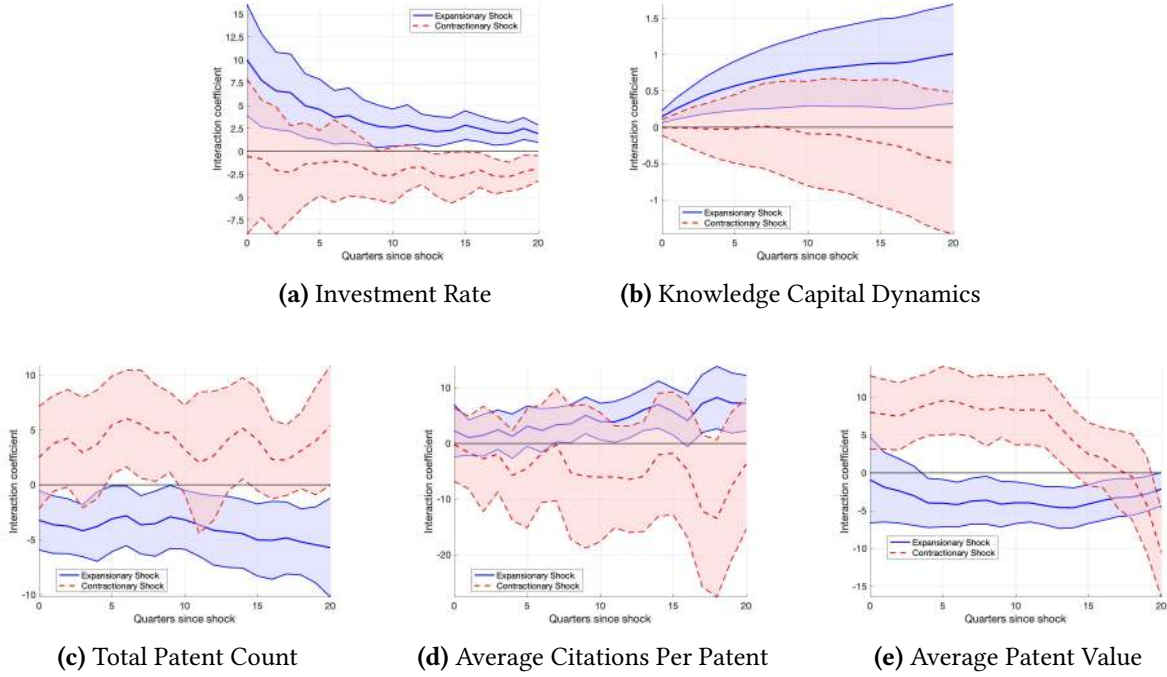
Panels (c) through (e) turn to patenting outcomes. Panel (c) illustrates the effects on total

patent count, with expansionary shocks driving persistent *declines*—from about -3 on impact to roughly -5.7 by quarter 20, with the 90% band below zero over most of the horizon—and contractionary shocks driving increases, with point estimates between roughly $+2.5$ and $+5$ for most of the horizon, interrupted by a temporary dip to about -4 around quarter 11 and estimated considerably less precisely than the expansionary response. This is the sign pattern that the reallocation mechanism implies: when financing conditions ease, firms can afford to substitute away from incremental, quickly-filed patents toward projects with longer development horizons, so measured patent counts fall; when financing conditions tighten, firms are pushed back toward exactly the kind of fast, incremental output that raises counts in the short run. Panel (d) demonstrates that expansionary shocks lead to a consistent increase in average citations per patent, reflecting improvements in innovation quality, while contractionary shocks result in a steady decline over time, so the quantity and quality margins move in opposite directions within each sign. Panel (e) focuses on average patent value, where expansionary shocks cause a gradual decline, but contractionary shocks initially drive a modest increase over 12 quarters, followed by a sharp decline by the 20th quarter. Given that per-patent market value shows no reliable response in the baseline specification either, as documented in Section 3.2, the natural reading of Panel (e) is that market-value-based measures are a noisier proxy for the quality reallocation than citations and text-based importance, rather than that the reallocation runs in the opposite direction on the value margin.

The results from Panels (a)–(e) reveal the distinct and asymmetric effects of monetary policy shocks. Expansionary shocks raise R&D investment and knowledge capital and shift the composition of patent output away from quantity and toward per-patent impact, while contractionary shocks raise patent counts and lower citations and importance, reversing the composition of innovative output rather than simply scaling it down. The two are not mirror images in either magnitude or precision: the expansionary responses are larger and more sharply identified across most outcomes, while the contractionary responses are directionally consistent with the mechanism but estimated with wider bands, particularly for the investment-side outcomes.

The results from Figure 9 reveal the asymmetric effects of a 100 basis point expansionary monetary policy shock (solid line) and a 100 basis point contractionary shock (dashed line) on patent importance and high-performing patents. Panel (a) shows the log sum importance, which initially decreases following an expansionary shock before stabilizing and steadily increasing, while contractionary shocks cause an initial rise followed by a persistent decline. Panel (b) presents the log mean importance, where expansionary shocks result in a consistent upward trend, reflecting enhanced average patent significance, while contractionary shocks lead to a steady decline over time. Panels (c), (d), and (e) focus on top-performing patents, capturing the mean number of

Figure 8: Asymmetric Interaction Coefficients

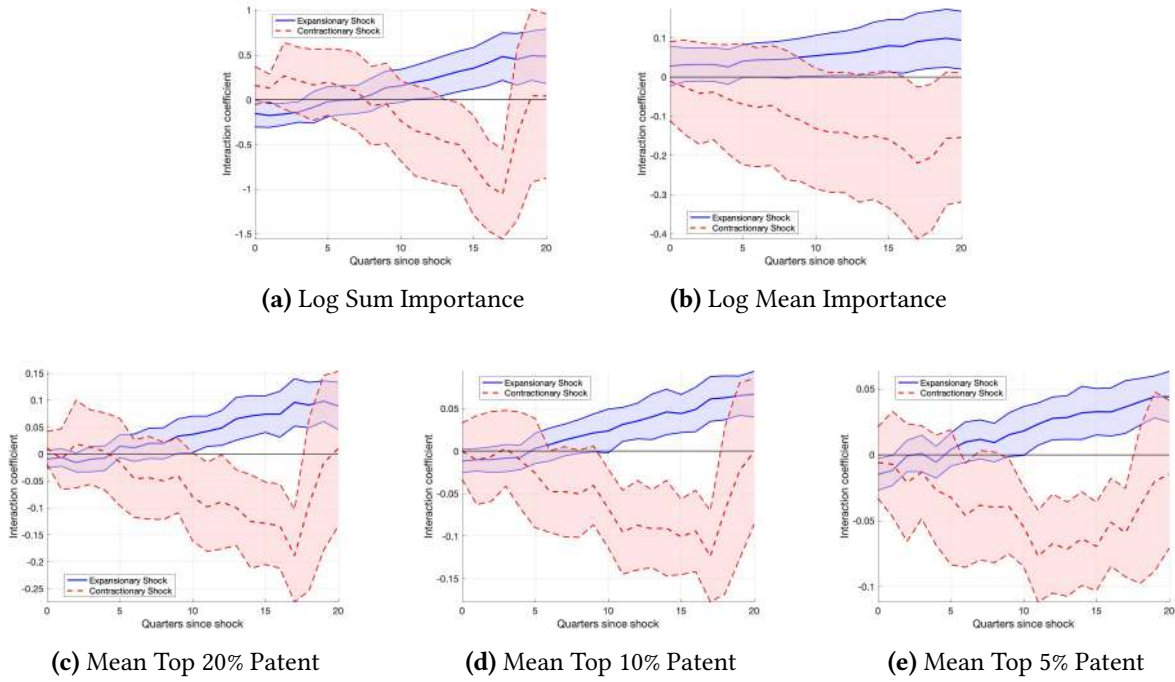


Notes: This figure illustrates the asymmetric effects of a 100 bp monetary policy shock on various firm-level metrics. The solid line represents the effects of an expansionary shock, while the dashed line represents the effects of a contractionary shock. The shaded area indicates the 90% confidence interval. Panels (a)–(b) show the effects on investment rate and knowledge capital dynamics. Panels (c)–(e) present the effects on total patent count, average citations per patent, and average patent value.

patents in the top 20%, 10%, and 5% percentiles. For all three percentiles, expansionary shocks initially result in a decline, followed by a gradual and steady increase, demonstrating the long-term benefits for high-performing patents; in the top 5% category the expansionary response moves from roughly -0.014 on impact to about $+0.045$ by the end of the horizon and its band separates from zero from around quarter 10 onward. Conversely, contractionary shocks lead to consistent reductions across all percentiles, though with wider bands than their expansionary counterparts. The results from Panels (a)–(e) reveal that expansionary policy facilitates a recovery and growth in patent significance and top-performing innovations, while contractionary policy disproportionately harms both the importance and the distribution of high-quality innovations.

The observed asymmetry in the effects of monetary policy shocks can be explained through the differing pathways firms take under expansionary and contractionary conditions. Expansionary monetary policy typically encourages firms to pursue long-term research projects by easing financial constraints and lowering borrowing costs. As firms shift resources toward these more

Figure 9: Asymmetric Metrics: Selected Outcomes



Notes: This figure shows the asymmetric effects of a 100 bp monetary policy shock on patent importance metrics. The solid line represents the effects of an expansionary shock, while the dashed line represents the effects of a contractionary shock. The shaded area indicates the 90% confidence interval. Panels (a)–(b) present log sum and log mean importance, capturing aggregate and average measures of patent significance. Panels (c)–(e) show the mean number of patents in the top 20%, 10%, and 5% percentiles, illustrating the distributional effects of monetary policy shocks on high-performing patents.

ambitious and impactful endeavors, the measured quantity of patents produced declines, and the aggregate importance of the resulting output experiences a temporary decline as well, because the projects being started take time to reach the point of filing. This short-term decline reflects the strategic reallocation of resources, with firms focusing on breakthroughs that require longer development horizons but promise significant long-term benefits. Over time, this approach yields sustained improvements in patent quality, economic significance, and technological impact, particularly among top-tier patents.

In contrast, contractionary monetary policy forces firms to adapt to tighter financial conditions by prioritizing short-term results over long-term innovation. Firms often abandon or scale back projects with extended timelines, redirecting resources toward initiatives that deliver faster outcomes. This shift allows firms to maintain or even increase patent production in the short term, demonstrating their ability to quickly adjust to a constrained economic environment, and it is visible directly in Panel (c) of Figure 8, where the contractionary patent-count response is

positive over most of the horizon. However, this comes at a cost: the overall quality of patents declines, and the production of high-impact, top-tier patents experiences a sharp and sustained drop. This trade-off points to the divergent paths firms take under monetary tightening, with a focus on short-term efficiency and quantity at the expense of long-term innovation quality and economic value.

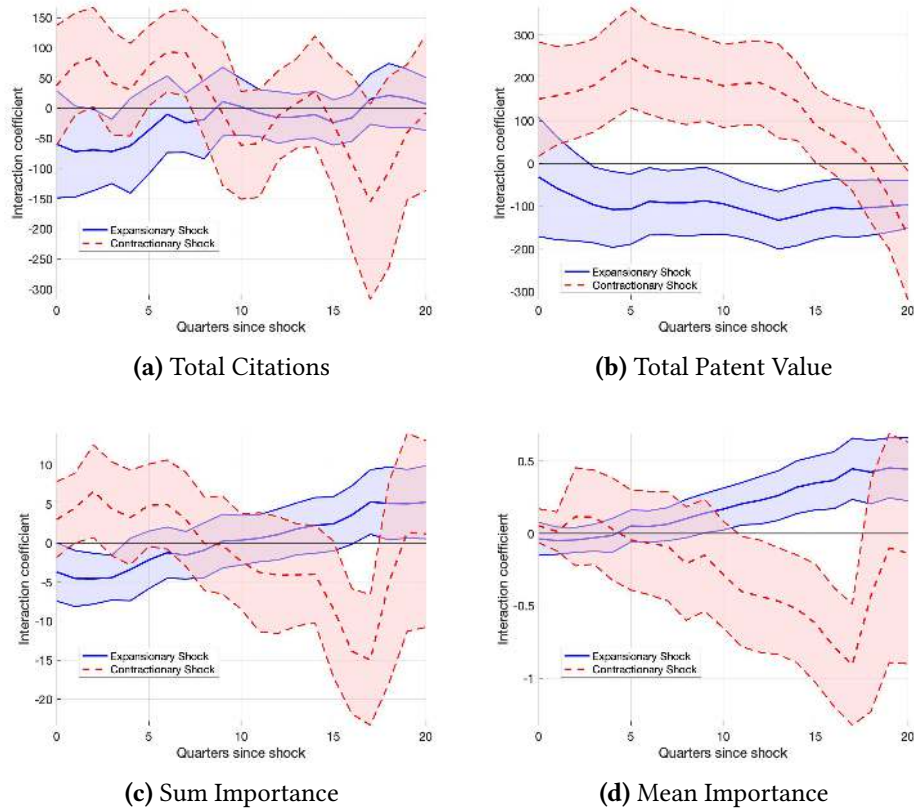
The results from Figures 8 and 9 reveal how expansionary and contractionary policies shape firm-level innovation strategies in fundamentally different ways. While expansionary policies promote long-term technological advancement at the cost of short-term declines in measured output, contractionary policies prioritize immediate adaptability and output quantity but risk undermining the foundational quality and impact of innovation. This divergence points to the need for a nuanced understanding of how monetary policy influences innovation dynamics and the broader economic landscape, and in particular for treating patent counts and per-patent quality measures as distinct outcomes rather than as interchangeable indicators of innovative performance.

The results discussed above focus on a core set of outcomes—investment, knowledge capital, patent counts, average citations, and average patent value—chosen because they most directly speak to the quantity-versus-quality trade-off at the center of this paper. These outcomes are not the only plausible measures of innovation quality, however, and a natural concern is that the asymmetric pattern I document could be sensitive to this particular choice. To address this concern, I re-estimate the same asymmetric local projection specification, equation (7), for a broader set of aggregate outcomes: total citations, total patent value, the sum and the mean of the text-based importance measure of Kelly et al. (2018), and the total number of patents in the top 20%, 10%, and 5% of the importance distribution. Because these outcomes aggregate over the same underlying patenting activity through different weighting schemes—simple counts, citation-weighted counts, market-value weights, and text-based importance weights—agreement across all of them would indicate that the asymmetric reallocation of innovative effort is a robust feature of the data rather than an artifact of any single measurement choice.

The results from Figure 10 report total citations, total patent value, and the sum and mean importance measures. Panels (a) and (b) show that both total citations and total patent value decline persistently following an expansionary shock (solid line) before beginning to recover toward the end of the horizon, while a contractionary shock (dashed line) produces the mirror-image pattern—an initial increase followed by a decline. This pattern is consistent with the mechanism at the heart of this paper: firms deprioritize near-term, lower-value patenting activity when financial conditions ease, redirecting effort toward more ambitious projects that take longer to mature but ultimately generate more citations and higher market value; the opposite reallocation

occurs when financial conditions tighten and firms are pushed toward faster, more incremental output. It is also consistent with the accounting decomposition of Section 3.2, since these aggregate measures inherit the movement in patent counts as well as any movement in per-patent quality. Panels (c) and (d) show that the sum and mean importance measures follow a similar qualitative pattern, reinforcing the view that the reallocation documented for citations and value extends to the text-based importance measure of Kelly et al. (2018) as well, which is constructed independently of citation counts and patent market valuations and therefore provides an additional, largely independent confirmation of the same underlying mechanism.

Figure 10: Additional Asymmetric Interaction Coefficients

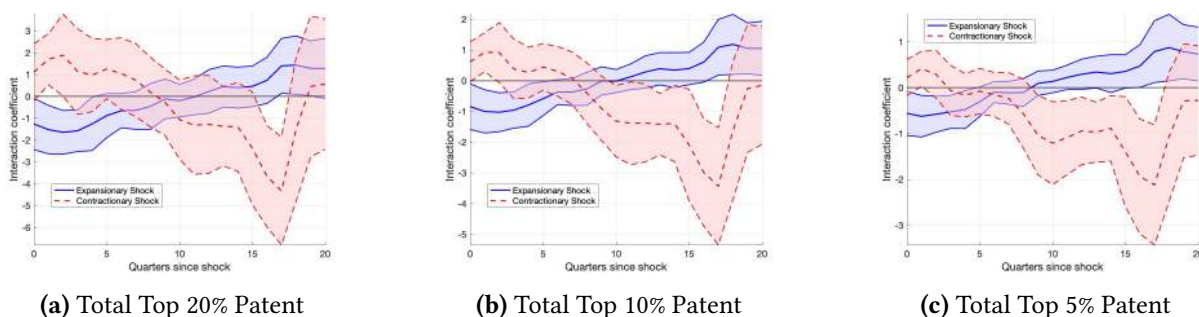


Notes: This figure presents additional asymmetric interaction coefficients capturing the effects of monetary policy shocks. The solid line represents the effects of an expansionary shock, while the dashed line represents the effects of a contractionary shock. The shaded area indicates the 90% confidence interval. Panels (a)–(b) show total citation count and total patent value. Panels (c)–(d) report the sum and mean importance of patents, indicating the overall and average impact of patents in response to monetary policy changes.

The results from Figure 11 turn from average- and sum-based measures to the distribution of high-impact patents, reporting the total number of patents in the top 20%, 10%, and 5% of the importance distribution. Focusing on these upper-tail measures is informative for a distinct reason:

if the reallocation documented above simply shifted the entire distribution of patent importance uniformly, the response would be the same proportional size across all three percentile cutoffs. Instead, across all three percentiles, expansionary shocks produce a temporary decline followed by a sustained increase in the total number of high-impact patents, while contractionary shocks produce the opposite pattern—an initial increase followed by a persistent decline—and, critically, the effect is most pronounced for the narrowest, most exclusive percentile (top 5%) and progressively weaker for the broader top-20% category. In the top 5% category the expansionary response moves from roughly -0.55 on impact to about $+0.87$ by quarter 18, an order of magnitude larger than the corresponding response in the mean-based version of the same outcome, which reflects the fact that the total and mean versions of these percentile measures are scaled differently and should not be compared directly across figures. This gradient suggests that the reallocation of innovative effort documented throughout this section is concentrated among the firms and projects capable of producing the most significant breakthroughs, rather than being spread evenly across the distribution of patenting activity: easier financing conditions appear to free up resources specifically for the most ambitious research projects, not merely for a general uptick in patenting of any quality.

Figure 11: Asymmetric Metrics: Total High-Impact Patents



Notes: This figure illustrates the asymmetric effects of monetary policy shocks on high-impact patents. The solid line represents the effects of an expansionary shock, while the dashed line represents the effects of a contractionary shock. The shaded area indicates the 90% confidence interval. Panels (a) and (b) present the total number of patents in the top 20% and 10% percentiles, while panel (c) shows the total number in the top 5% percentile, showing how monetary policy shocks influence the distribution of the most significant patents.

The results from these additional outcomes confirm that the asymmetric reallocation documented for the core set of results extends to a broader range of measures of innovation quality and impact, and that the effect is, if anything, stronger among the most exclusive tier of breakthrough patents. This robustness across citation-based, value-based, and text-based measures of

importance, and across the full range of the importance distribution, indicates that the mechanism I document—firms trading off near-term output for longer-horizon, higher-impact research as financing conditions change—operates on the substance of firms’ innovation choices, rather than being an artifact of how any single outcome variable happens to be constructed.

4.2 Asymmetry by Shock Magnitude

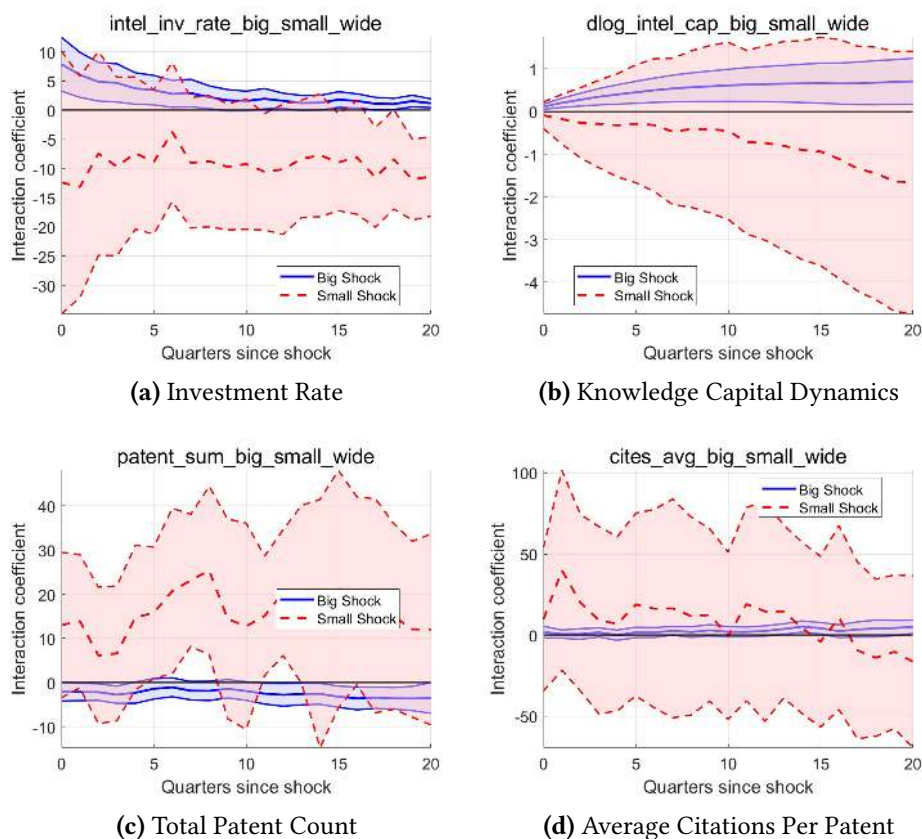
Section 4.1 established that the *sign* of the monetary policy shock is a first-order source of asymmetry in firms’ innovation activity. I now ask whether the *magnitude* of the shock matters as well, on its own. To do so, I instead split the shock into “big” and “small” components—defined respectively as those falling outside and within the middle 50% of the empirical distribution of the shock series—pooling both signs within each magnitude group, and estimate

$$y_{j,t+h} = \alpha_{j,h} + \beta_h^b \varepsilon_t^b + \beta_h^s \varepsilon_t^s + \Gamma_1' Z_{j,t-1} + \Gamma_2' Y_{t-1} + e_{j,t,h}. \quad (8)$$

The results from Figure 12 report the investment rate, knowledge capital growth, total patent counts, and average citations per patent. Panels (a) and (b) present the investment-side outcomes, while panels (c) and (d) turn to the innovation-output side. Two features stand out. First, the big-shock responses track the pooled baseline closely and are tightly estimated, whereas the small-shock responses are barely informative. In Panel (a), the big-shock investment response starts near 8 percentage points on impact and decays to roughly 1 percentage point by quarter 20, with a narrow band that excludes zero over the first two years—essentially reproducing the baseline response of Figure 2. The small-shock response, by contrast, sits near -12 percentage points for the entire horizon, but its 90% band runs from roughly -33 to $+10$, so it is consistent with almost any value one might propose and cannot be signed. Panel (b) repeats the pattern for knowledge capital growth: the big-shock response rises steadily to about $+0.7$ with a band that stays above zero from the middle of the horizon onward, while the small-shock response drifts to about -1.6 inside a band spanning roughly -5 to $+1.5$. This difference in precision is informative rather than merely inconvenient: big shocks, being larger movements in the underlying policy instrument, generate more identifying variation relative to the noise in the local projection residuals, which mechanically improves the precision of $\hat{\beta}_h^b$ relative to $\hat{\beta}_h^s$; small shocks carry comparatively little identifying variation, so the same underlying noise translates into a far more volatile estimated response. Second, for the investment rate and knowledge capital growth the point estimates for big and small shocks are of opposite sign—the big-shock response is positive throughout, while the small-shock response is negative for most of the horizon. Panels (c) and (d) show the same imprecision in the patenting outcomes: total patent counts and average citations per patent

are both close to zero and narrowly banded for big shocks, but oscillate erratically over a much wider range for small shocks, making the small-shock patenting response impossible to sign with confidence. This indicates that the shock-size split is not simply rescaling a common underlying response; instead, big and small shocks appear to pick up qualitatively different components of the innovation response, a possibility I explore further below by interacting magnitude with sign directly.

Figure 12: Asymmetric Metrics: Key Investment and Innovation Indicators

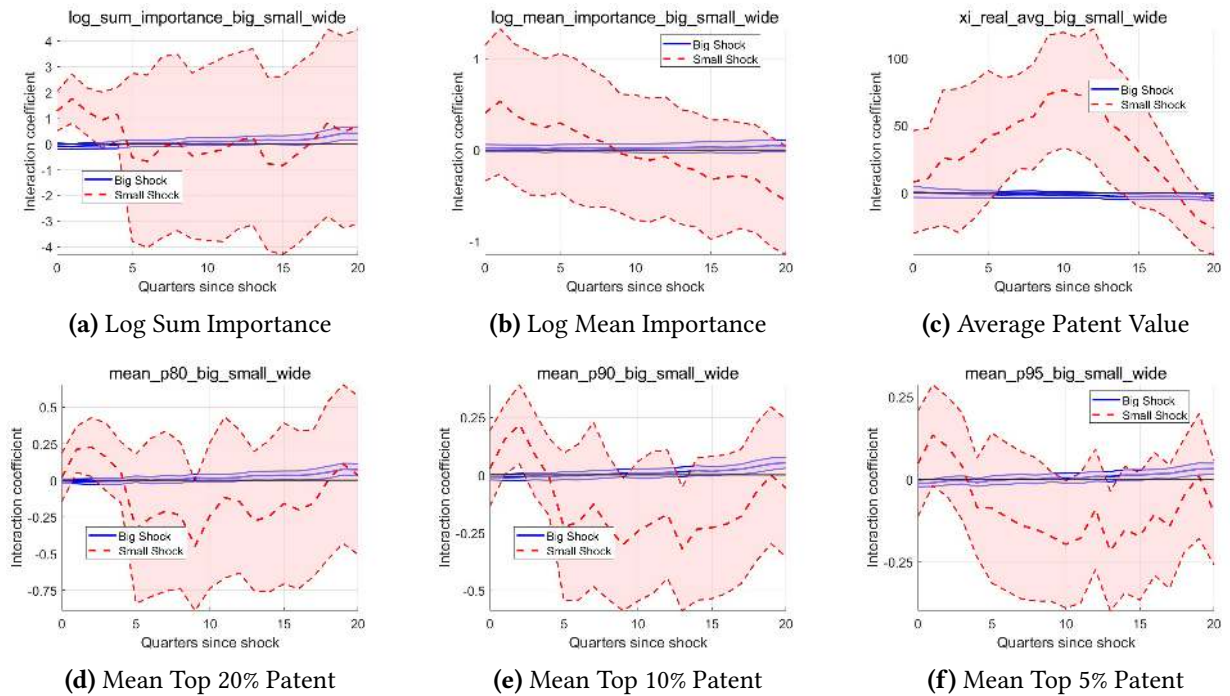


Notes: This figure presents the effects of big versus small monetary policy shocks on investment and innovation. The shaded area indicates the 90% confidence interval. Panels (a)–(b) show investment rate and knowledge capital dynamics. Panels (c)–(d) present total patent count and average citations per patent.

The results from Figure 13 extend this comparison to patent importance and to the distribution of high-impact patents. The same pattern recurs across all six panels: big-shock responses are modest in magnitude and tightly estimated, while small-shock responses are considerably more volatile, with much wider confidence bands and, for log mean importance and average patent value in particular, a visibly negative trend over the medium run that is not present in the corresponding big-shock estimates. Notably, the divergence between big- and small-shock point

estimates does not narrow as one moves from the aggregate importance measures (panels (a)–(c)) to the upper-percentile patent counts (panels (d)–(f)); if anything, the small-shock estimates for the top-percentile categories remain volatile and imprecisely signed throughout the horizon, suggesting that the imprecision documented above is a pervasive feature of the small-shock subsample rather than one confined to a particular type of outcome. As with the core outcomes above, this imprecision makes it difficult to draw sharp conclusions about the size-dependence of the response from the pooled-sign comparison alone. I therefore turn next to a joint decomposition by magnitude and sign, which, by separating expansionary from contractionary shocks within each magnitude group, reveals a sharper and more interpretable pattern than either dimension considered on its own.

Figure 13: Asymmetric Metrics: Patent Importance and Distribution



Notes: This figure presents the effects of big versus small monetary policy shocks on patent importance and distribution. The shaded area indicates the 90% confidence interval. Panels (a)–(b) illustrate log sum importance and log mean importance. Panel (c) shows average patent value, and panels (d)–(f) show the mean number of patents in the top 20%, 10%, and 5% of the importance distribution.

The results from this section establish that shock magnitude is not simply a rescaled version of the sign-based asymmetry documented in Section 4.1: big and small shocks differ not only in the precision with which their effects can be estimated, but in some cases in the sign of the point estimate itself. Because equation (8) pools expansionary and contractionary shocks within

each size bin, however, it cannot distinguish a genuine size-dependence in the response from a composition effect arising from the mix of signs that happens to fall in each bin. This motivates a more direct test of how magnitude and sign interact, to which I turn next.

4.3 Joint Heterogeneity by Sign and Magnitude

Sections 4.1 and 4.2 look at sign-based and magnitude-based asymmetry separately, but this leaves open whether the sign asymmetry is uniform across shock sizes or concentrated in one magnitude group. The magnitude comparison in Section 4.2 pools expansionary and contractionary shocks within each size bin, so it cannot speak to this question on its own. I therefore interact sign and magnitude directly.

I split the shock into four components—large expansionary, large contractionary, small expansionary, and small contractionary—and estimate

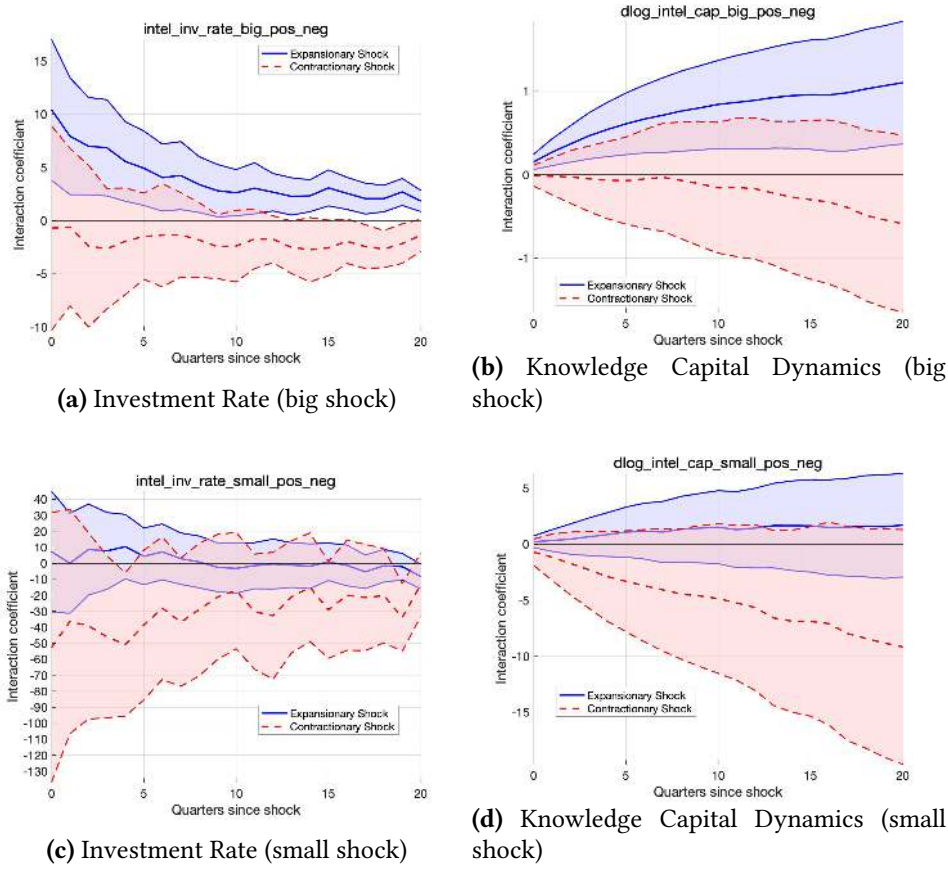
$$y_{j,t+h} = \alpha_{j,h} + \beta_h^{b+} \varepsilon_t^{b+} + \beta_h^{b-} \varepsilon_t^{b-} + \beta_h^{s+} \varepsilon_t^{s+} + \beta_h^{s-} \varepsilon_t^{s-} + \Gamma_1' Z_{j,t-1} + \Gamma_2' Y_{t-1} + e_{j,t,h}, \quad (9)$$

where β_h^{b+} and β_h^{b-} are the effects of large expansionary and large contractionary shocks, and β_h^{s+} and β_h^{s-} are the corresponding effects for small shocks. This four-way split lets me check whether the sign asymmetry from Section 4.1 is driven by large shocks, small shocks, or both.

The results from Figure 14 show sign-based asymmetry within both magnitude groups, but the two look quite different. Among big shocks (panels (a)–(b)), expansionary shocks raise both the investment rate and knowledge capital, while contractionary shocks lower them by a comparable amount—an asymmetry similar in shape to the pooled sample. Among small shocks (panels (c)–(d)), the pattern is more lopsided: contractionary shocks produce a large and volatile negative response, an order of magnitude bigger than the big-shock contractionary response, while expansionary shocks stay close to zero and are much weaker than their big-shock counterpart. So the expansionary–contractionary asymmetry among small shocks is not attenuated; it is driven almost entirely by an outsized, if noisily estimated, response to small contractionary shocks.

The results from Figure 15 extend this comparison to patenting outcomes: total patent counts, average citations, and average patent value, again for big and small shocks separately. Among big shocks (panels (a)–(c)), expansionary shocks lower the raw patent count and raise average citations—the citation response rises from about +4 on impact to roughly +9 by the end of the horizon, and its band excludes zero at most horizons—consistent with the reallocation-toward-quality result from earlier sections. The contractionary response points the other way on both margins, toward more patents and fewer citations, but its confidence band contains zero at nearly

Figure 14: Economic Metrics under Big and Small Monetary Policy Shocks

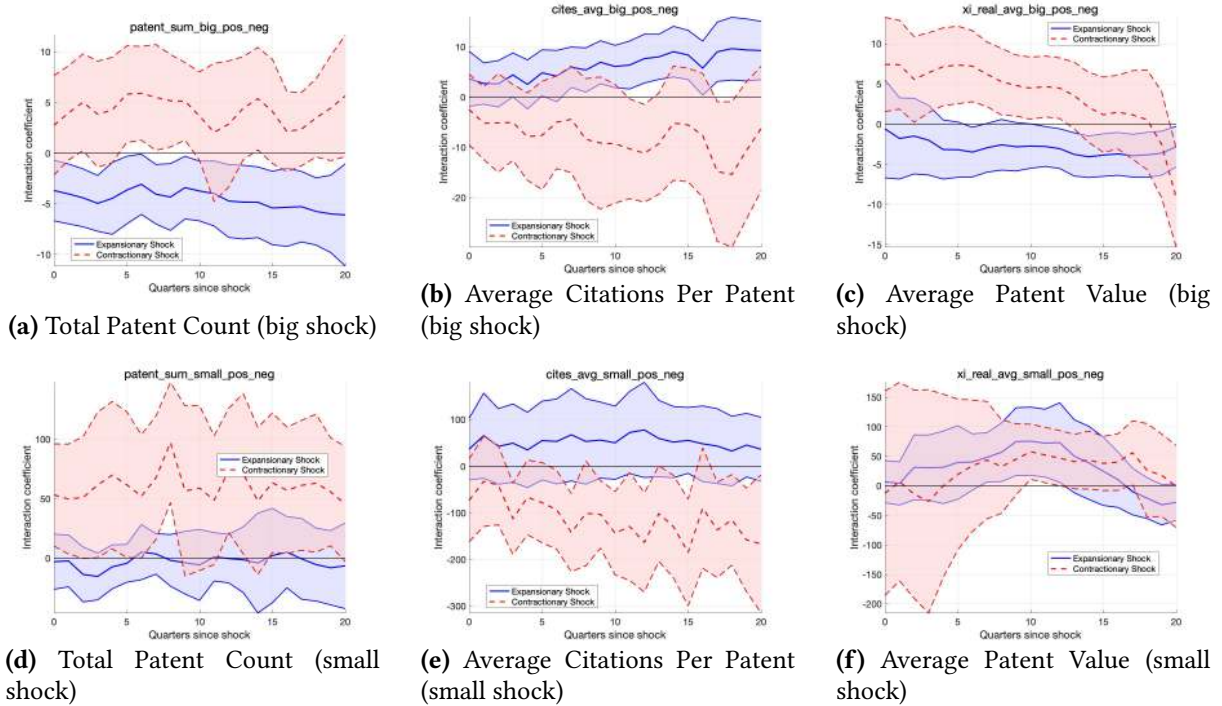


Notes: This figure illustrates the effects of expansionary versus contractionary monetary policy shocks, separately for big and small shocks. Panels (a)–(b) show the effects on investment rate and knowledge capital dynamics for big shocks, while Panels (c)–(d) display the corresponding metrics for small shocks.

every horizon, so the asymmetry within this magnitude group is identified mainly off the expansionary side rather than off a symmetric pair of significant responses. Average patent value (panel (c)) breaks this pattern: for most of the horizon, expansionary shocks are associated with a modest *decline* in average patent value and contractionary shocks with a modest increase, the reverse of the citation result, before the two lines converge near the end of the horizon. Among small shocks (panels (d)–(f)), the picture is more mixed: patent counts and average patent value have overlapping confidence bands for both signs, so no clear asymmetry emerges there, while average citations separate cleanly, positive under expansionary shocks and negative under contractionary shocks throughout. Combined with Figure 14, this points to the reallocation-toward-quality mechanism being clearest among big shocks, with patent value a noisier—and relative to citations, partly conflicting—measure of it, while small shocks give a genuinely mixed picture

rather than a uniformly imprecise one.

Figure 15: Patent Metrics under Big and Small Monetary Policy Shocks

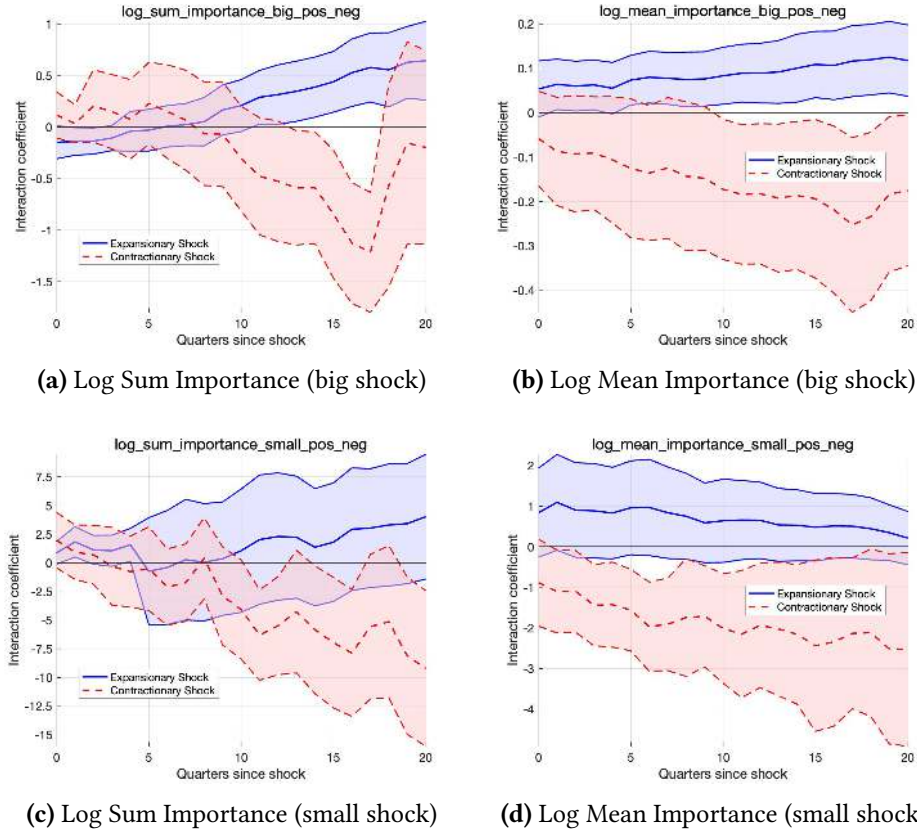


Notes: This figure presents patent-related metrics under expansionary versus contractionary monetary policy shocks, separately for big and small shocks. Panels (a)–(c) show total patent count, average citations, and average patent value for big shocks, while Panels (d)–(f) show the corresponding metrics for small shocks.

The results from Figures 16 and 17 extend the sign comparison to the importance measures and to the distribution of top-percentile patents. Here the sign asymmetry is clear for both big and small shocks: expansionary shocks are associated with higher importance and more top-percentile patents, contractionary shocks with lower importance and fewer top-percentile patents, in each magnitude group. Among big shocks, though, the contractionary response in log sum importance and in the top-percentile patent counts does not decline monotonically—it deepens over the first three quarters of the horizon before partially or fully reversing by quarter 20, which suggests the negative effect of tightening on breakthrough patenting is at least partly transitory rather than permanent. Among small shocks, the point estimates swing over a much wider range with no comparable late-horizon reversal, but the confidence bands widen correspondingly, so this larger swing is better read as suggestive than as clear evidence of a stronger or more persistent effect.

The results from Figures 14 through 17 indicate that shock sign, rather than magnitude, is the

Figure 16: Log Importance Metrics under Big and Small Monetary Policy Shocks

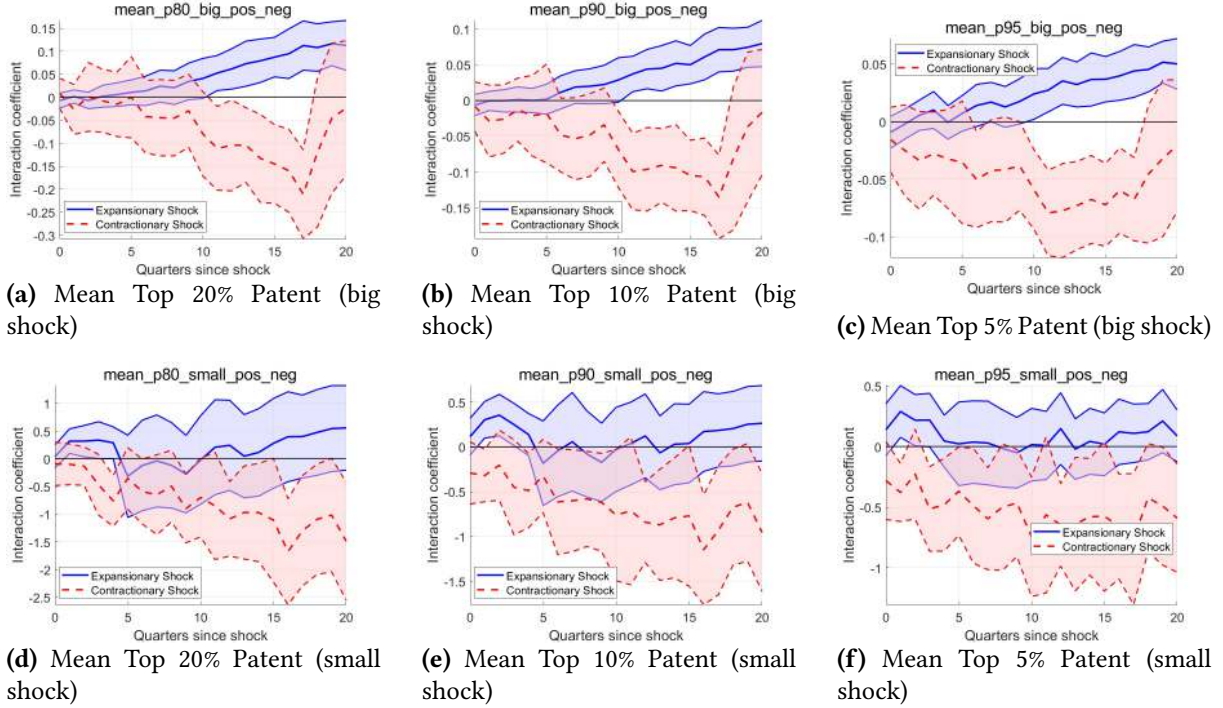


Notes: This figure displays the effects of expansionary versus contractionary monetary policy shocks on patent importance, separately for big and small shocks. Panels (a)–(b) show the log sum and log mean importance for big shocks, while Panels (c)–(d) show the corresponding metrics for small shocks.

primary source of asymmetry in how monetary policy is transmitted to firm-level innovation, including at the extreme upper tail of the importance distribution. The sign asymmetry appears within both magnitude groups rather than being confined to one of them, so this feature of the transmission mechanism holds across the shock-size distribution, even though its dynamics—and how precisely they can be estimated—depend considerably on shock magnitude.

The results from this section indicate that the asymmetric transmission of monetary policy to firm-level innovation operates mainly through the sign of the shock, with magnitude playing a smaller, outcome-specific role: the effects do not scale monotonically with shock size. For investment and knowledge capital, the sign asymmetry is precisely estimated among big shocks but is dominated among small shocks by a noisily estimated, outsized contractionary response. For patenting outcomes, the sign asymmetry is sharp among big shocks and too imprecise among small shocks to support firm conclusions. For the importance-based measures, sign asymmetry

Figure 17: Top Patent Metrics under Big and Small Monetary Policy Shocks



Notes: This figure illustrates the distributional effects of expansionary versus contractionary monetary policy shocks on high-performing patents, separately for big and small shocks. Panels (a)–(c) show metrics for the top 20%, 10%, and 5% patents under big shocks, while Panels (d)–(f) show the corresponding metrics for small shocks.

holds across both magnitude groups, though small-shock estimates carry substantially wider confidence bands. This pattern is consistent with firms treating economically large monetary policy shocks as decision-relevant events that justify a deliberate reallocation of R&D toward long-term, high-value projects; the imprecision of the small-shock estimates leaves open whether firms respond to small shocks with the same deliberateness, or simply do not treat them as consequential enough to warrant a change in innovation strategy.

5 Conclusion

This paper examines the effects of monetary policy shocks on firms' innovation strategies, focusing on both the inputs and outputs of R&D investment. Expansionary monetary policy generates an immediate increase in R&D expenditures among firms already engaged in research, resulting in a rapid accumulation of knowledge capital that persists over time. The initial boost in investment prompts firms to reallocate resources away from short-term, lower-impact projects toward

more ambitious, long-term initiatives, which leads to an early decline in overall patent output but later improves the quality and economic significance of patents. Firms already active in innovation benefit most from the eased financial constraints, while inactive firms remain largely unaffected; the extensive margin is small and statistically indistinguishable from zero at every horizon, so the innovation channel of monetary policy runs through the scale and direction of research among incumbents rather than through participation. The increase in R&D investment drives qualitative improvements in innovation, producing fewer but more impactful patents and marking a strategic shift toward high-value research activities.

Additionally, this analysis shows that expansionary and contractionary monetary shocks have markedly different effects on innovation. Expansionary policies ease financial constraints, boosting R&D investment and ultimately improving patent quality, while contractionary shocks limit spending, raising patent counts in the short run and reducing both patent quality and high-impact innovations. This suggests that an accommodative monetary environment helps firms shift toward long-term, high-value projects, enhancing the economic significance and technological impact of their innovations. The asymmetry operates mainly through the sign of the shock rather than through its size: interacting sign and magnitude shows that the reallocation toward quality is identified most cleanly among economically large policy moves, while the response to small shocks is too imprecisely estimated to sign. A further result worth flagging for future work is that the three families of quality proxies do not agree: citation-based and text-based importance measures both rise following an expansionary shock, whereas per-patent market value shows no reliable response in either direction, so the aggregate decline in patent value documented here operates through the number of patents rather than through their individual worth.

Overall, the evidence points to the importance of monetary policy in shaping the direction and depth of innovative activity beyond its traditional macroeconomic role. For policymakers, these insights suggest that pairing an accommodative policy with targeted support, such as easing credit constraints for smaller innovators, can broaden innovation benefits and sustain the development of transformative inventions.

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Online Appendix to “Monetary Policy and Firm Innovation”*

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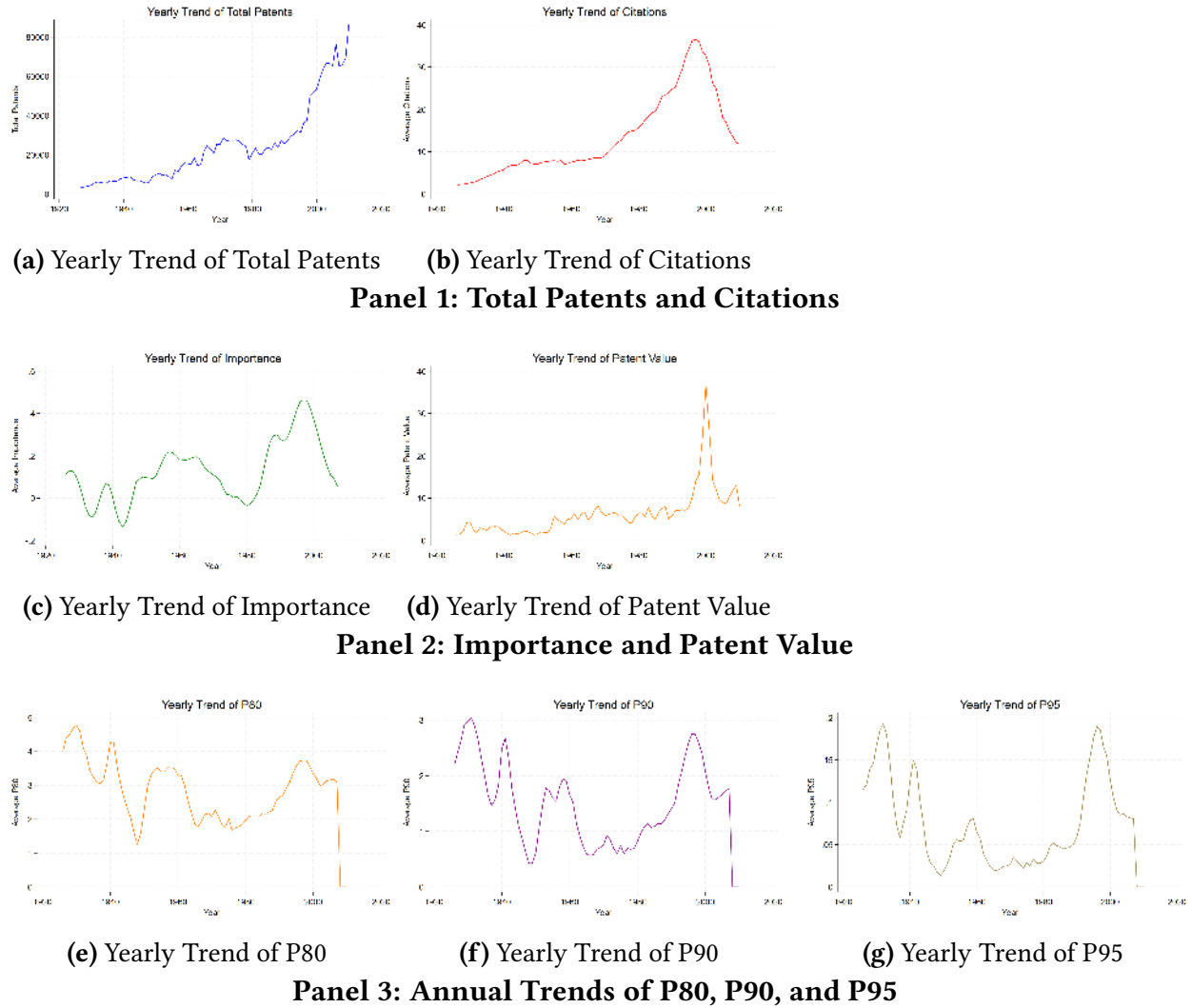
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A Supplements to the Data

A.1 Annual Trends in Patent

Figure A1: Combined Annual Trends of Total Patents, Citations, Importance, Patent Value, and Percentiles



Notes: This figure shows the annual trends in patent-related metrics. Panel 1 displays the trends of total patents and citations. Panel 2 presents the yearly trends in patent importance and market value. Panel 3 illustrates the trends for the 80th, 90th, and 95th percentiles of patent value (P80, P90, P95).

A.2 Firm-Level Variables

Table A1: Summary Statistics of Firm-Level Variables

	All							No research firms							Research firms						
	Mean	P50 (Median)	P25	P75	SD	Min	Max	Mean	P50 (Median)	P25	P75	SD	Min	Max	Mean	P50 (Median)	P25	P75	SD	Min	Max
ikc	4.158	0.000	0.000	5.941	9.294	0.000	100.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	7.357	5.336	0.118	8.601	11.372	0.000	100.000
Δlog KC	0.043	0.019	-0.005	0.053	0.114	-0.053	6.398	.	.	.	.	.	.	.	0.043	0.019	-0.005	0.053	0.114	-0.053	6.398
Capx / R&D	11.831	1.187	0.335	4.281	153.445	-28.738	23475.998	.	.	.	.	.	.	.	11.831	1.187	0.335	4.281	153.445	-28.738	23475.998
Physical Capital	719.141	29.924	4.138	203.716	4361.603	0.000	255556.000	836.805	60.359	10.940	328.923	3425.282	0.000	115994.000	628.605	15.258	2.268	122.318	4961.342	0.000	255556.000
cash	0.161	0.079	0.022	0.228	0.197	-0.248	1.000	0.100	0.046	0.014	0.133	0.135	-0.248	1.000	0.208	0.123	0.034	0.317	0.222	-0.070	1.000
Distance to default	5.737	4.687	2.233	8.036	5.037	-4.763	224.585	5.321	4.297	1.973	7.472	4.904	-4.763	224.585	6.075	5.024	2.476	8.475	5.118	-4.251	172.519
Leverage	0.263	0.202	0.040	0.367	0.352	0.000	9.745	0.301	0.260	0.097	0.420	0.305	0.000	8.877	0.234	0.155	0.015	0.319	0.381	0.000	9.745
Age	16.407	13.750	8.000	22.500	11.010	0.750	56.500	16.265	13.750	8.000	22.500	10.792	0.750	56.500	16.516	13.750	8.000	22.750	11.173	0.750	56.500
Total asset	1948.859	143.364	27.513	774.602	9755.559	0.005	444097.000	1775.693	219.155	46.117	960.428	6666.352	0.020	257143.000	2082.101	97.763	19.754	624.626	11582.974	0.005	444097.000
size	5.270	5.245	3.610	6.877	2.318	-5.066	13.002	5.619	5.679	4.151	7.108	2.159	-3.619	12.446	5.001	4.837	3.262	6.647	2.399	-5.066	13.002
Dividend paid	0.105	0.000	0.000	0.000	0.306	0.000	1.000	0.120	0.000	0.000	0.000	0.325	0.000	1.000	0.093	0.000	0.000	0.000	0.291	0.000	1.000
Cashflow	0.005	0.018	-0.003	0.035	0.164	-46.900	40.100	0.016	0.019	0.004	0.035	0.143	-26.225	40.100	-0.004	0.016	-0.012	0.034	0.178	-46.900	4.780
Debt growth	1.598	-0.006	-0.072	0.043	302.432	-1.000	149015.203	2.258	-0.004	-0.060	0.049	430.764	-1.000	149015.203	1.035	-0.009	-0.082	0.037	105.712	-1.000	27779.555
Observations	355764.000							154705.000							201059.000						

Notes: Summary statistics are reported for all firms, non-research firms, and research firms separately. Research firms are defined as those that report R&D expenditures, while non-research firms do not report any R&D expenditures. The sample spans from 1991 to 2016 and includes all firms from Compustat with available data. The variables are defined as follows: *ikc* is the investment rate of knowledge capital, calculated as R&D expenditure divided by the lagged knowledge capital stock; $\Delta \log KC$ is the growth rate of knowledge capital; CAPX/R&D is capital expenditure divided by R&D expenditure; Physical Capital is net property, plant, and equipment; Cash is cash holdings divided by total assets; Distance to Default is a measure of the firm's default risk following [Gilchrist and Zakrajšek \(2012\)](#); Leverage is total debt divided by total assets; Age is the number of years since the firm's first appearance in Compustat; Total Assets is the book value of assets; Size is the natural logarithm of total assets; Dividend Paid is a dummy variable equal to 1 if the firm paid dividends in the fiscal year; Cash Flow is operating income before depreciation scaled by total assets; Debt Growth is the annual growth rate of total debt.

A.3 Alternative Monetary Policy Shock

A natural concern with our baseline results is that the [Ottonello and Winberry \(2020\)](#) shock series, while widely used in the corporate-finance and investment literature, ends in 2007 and may embed a central-bank information effect: part of the high-frequency interest-rate surprise around FOMC announcements can reflect the Federal Reserve's private information about the state of the economy rather than an exogenous change in the policy stance. If firms' innovation and investment decisions respond to this information component rather than to the policy shock itself, our reduced-form estimates would conflate a demand-driven mechanism with a genuine credit-cost channel, and the baseline results would not necessarily speak to the effect of monetary policy on innovation.

To address this concern, we re-estimate every baseline local projection using the [Jarociński and Karadi \(2020\)](#) pure monetary policy shock, which spans 1990–2019. The JK shock is constructed using sign restrictions on the joint response of interest rates and stock prices around FOMC announcements, which allows the authors to separate a conventional "pure" monetary policy shock from a central-bank information shock; only the former is used here. Two features distinguish this exercise from the baseline: first, the identification strategy explicitly purges the information component that may confound our reduced-form estimates; second, the longer sample (through 2019, versus 2007 in the baseline) allows us to check whether the post-crisis, near-zero-lower-bound period materially changes the transmission of monetary policy to firm-level innovation. Apart from the shock series and the sample window, every other element of the specification—firm and time fixed effects, the set of firm-level and macroeconomic controls, and the local projection horizon—is identical to the baseline.

Figure A2: Alternative Monetary Policy Shock: Baseline Outcomes (1/4)

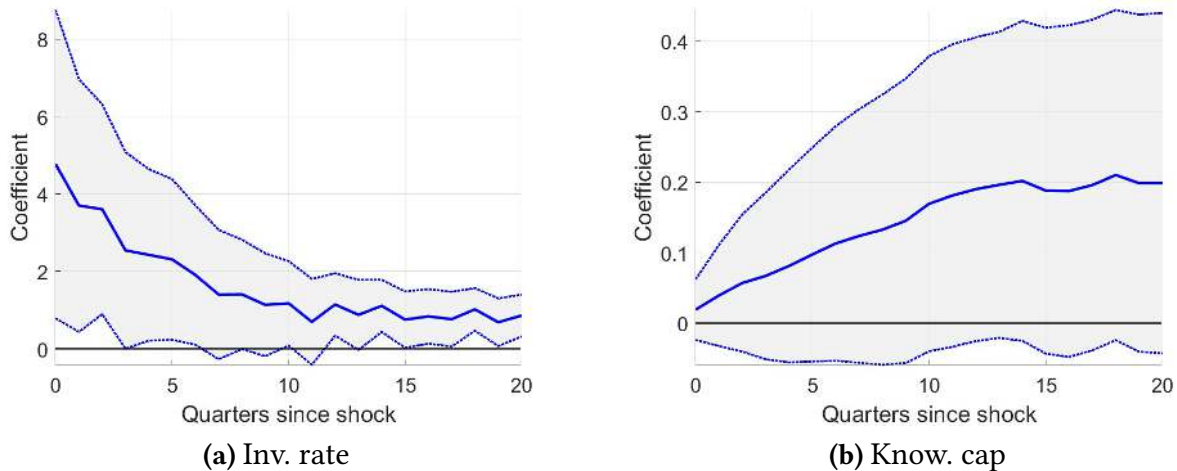
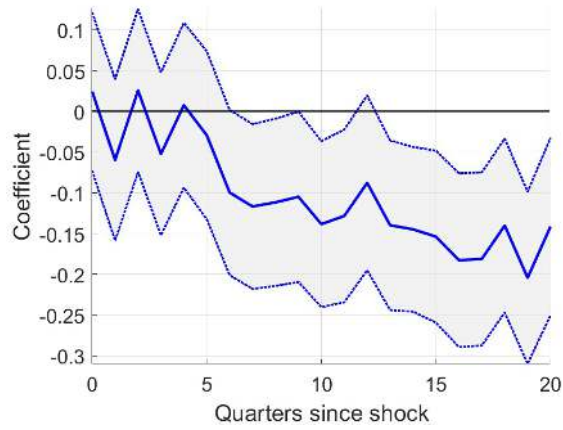
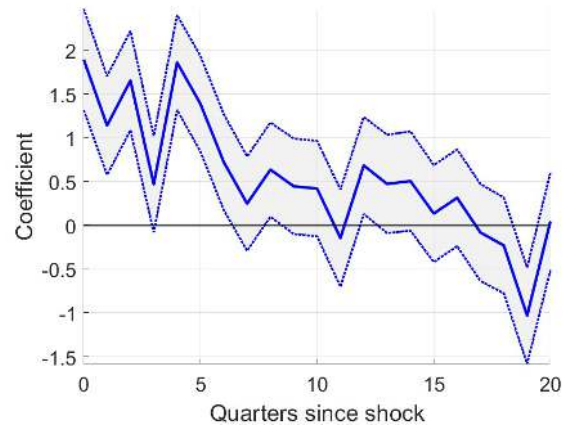


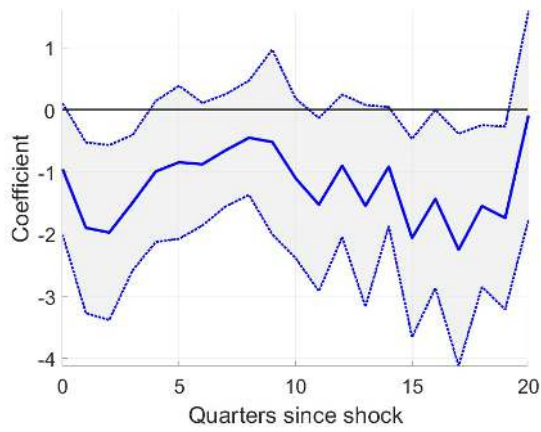
Figure A3: Alternative Monetary Policy Shock: Baseline Outcomes (2/4)



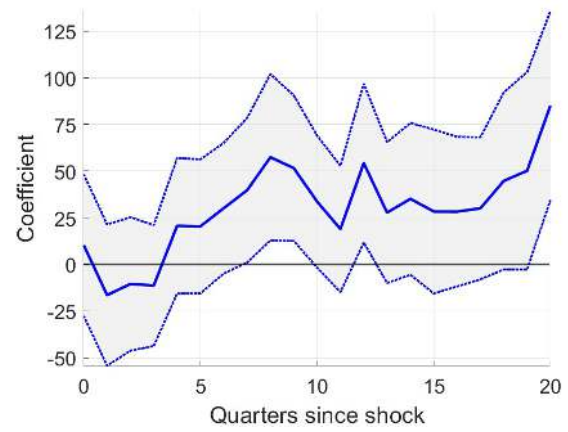
(a) Extensive (Probit)



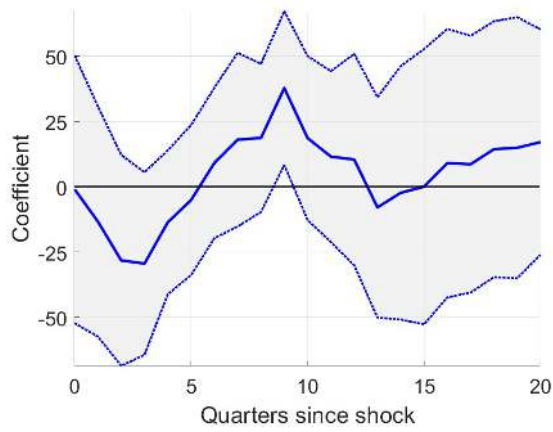
(b) Intensive (Tobit)



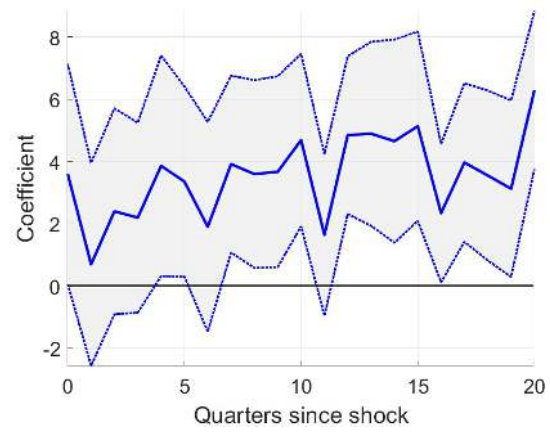
(c) Total patents



(d) Total citations



(e) Total value



(f) Avg. citations

Figure A4: Alternative Monetary Policy Shock: Baseline Outcomes (3/4)

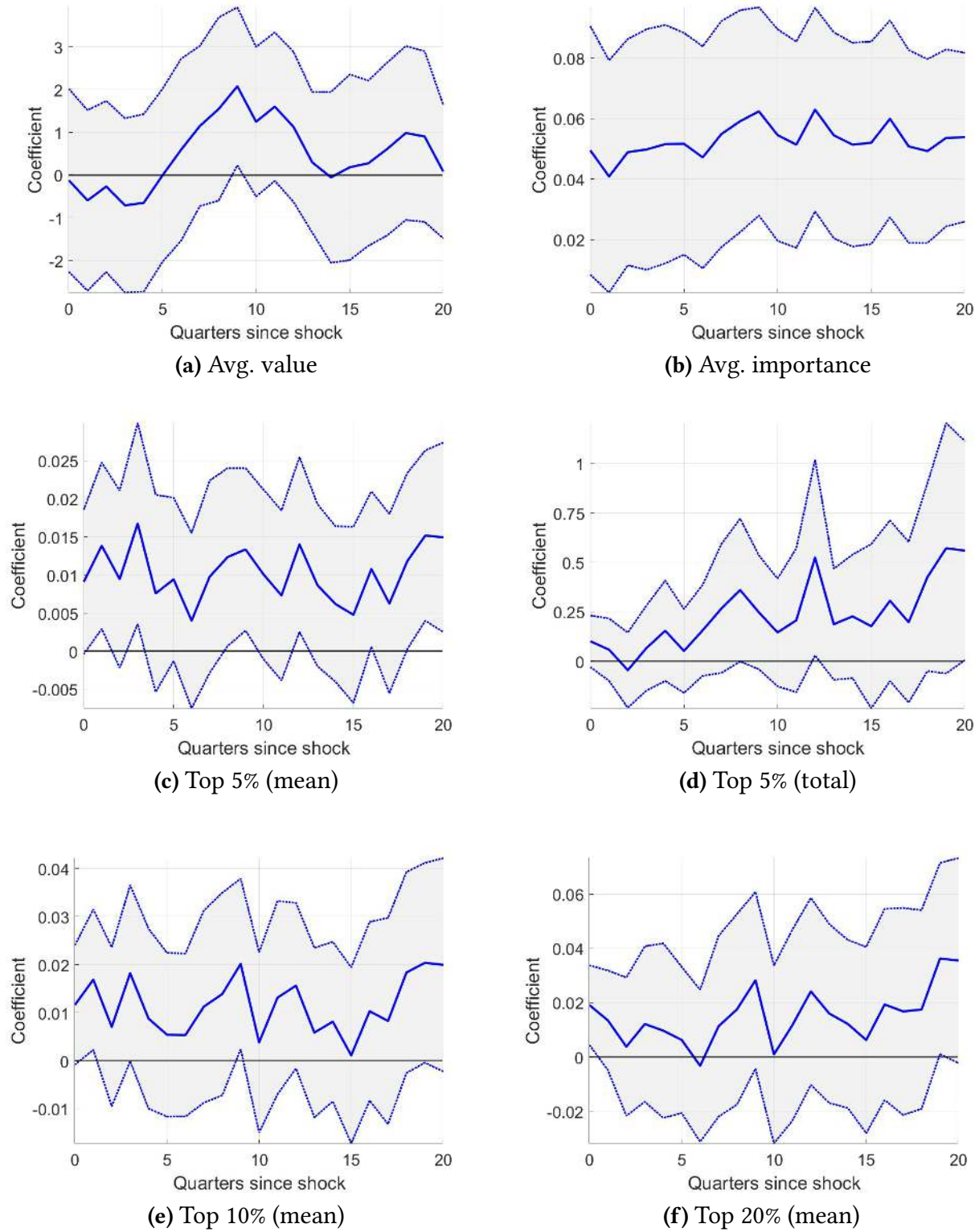
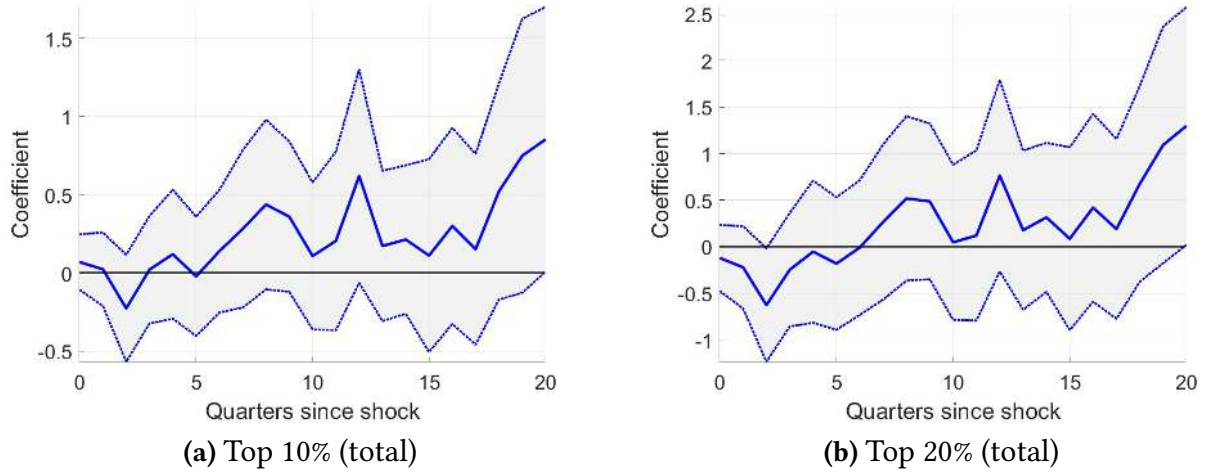


Figure A5: Alternative Monetary Policy Shock: Baseline Outcomes (4/4)



Notes: These figures report the dynamic responses to the [Jarociński and Karadi \(2020\)](#) pure monetary policy shock, 1990–2019, for the full set of baseline outcomes, including the investment rate and growth rate of knowledge capital, the extensive and intensive margins of R&D participation, and the level, citation, value, and importance measures of patenting activity. The solid line is the point estimate at each horizon and the shaded areas represent 90% confidence intervals; the horizontal axis measures quarters since the shock.

Conclusion. The results under the information-robust JK shock closely track the baseline pattern along every dimension we examine. Knowledge-capital growth and the per-patent measures of quality and value rise following an expansionary shock, the extensive margin (captured by the Probit specification) declines modestly, and the intensive margin (captured by the Tobit specification) spikes on impact before gradually reverting toward zero; the raw patent count falls, consistent with firms reallocating effort toward fewer but higher-value projects rather than expanding the sheer volume of filings. This qualitative correspondence with the baseline is reassuring on two counts: it suggests that our results are not an artifact of the central-bank information effect embedded in the Ottonello–Winberry shock, and it shows that the mechanism survives when the sample is extended well past the Great Recession and into the post-crisis low-rate environment.

One nuance is worth flagging. Under the JK shock, the aggregate citation and patent-value responses turn out to be positive over the medium run, whereas the reduced-form baseline shows an initial decline in these aggregate measures before recovering. We interpret this as evidence in favor of the fitted-investment channel developed in the main text: because the JK shock is purged of the information component, it more cleanly identifies the R&D-investment channel through which monetary policy affects patenting, and the resulting aggregate responses line up more closely with the fitted, R&D-driven predictions from our two-stage procedure than with the raw reduced-form estimates. In other words, the discrepancy between the baseline and the JK-shock aggregate responses is itself informative about the mechanism, rather than a sign that the results are fragile.

B Supplements to the Empirical Strategy

B.1 Static Framework for Innovation and Investment

In this section, we present the empirical strategy used to estimate the impact of monetary policy shocks (MPS) on firm-level innovation and investment outcomes. We employ a static panel regression model, incorporating firm and sector-quarter fixed effects, and allowing for heterogeneity in firms' financial conditions. The model specification is as follows:

$$y_{jt} = \alpha_j + \alpha_{st} + \beta(x_{jt-1} - E_j[x_{jt}])\varepsilon_{tm} + \gamma'Z_{jt-1} + e_{jt} \quad (\text{B1})$$

where y_{jt} represents firm-level outcomes such as R&D investment, patenting activity, and firm value. The model includes firm fixed effects (α_j) to capture time-invariant unobserved heterogeneity, as well as sector and quarter-specific fixed effects (α_{st}) to control for sectoral trends and macroeconomic fluctuations. The key independent variable is the monetary policy shock for the quarter (ε_{tm}), while x_{jt} represents the firm's leverage ratio or distance to default, adjusted for the firm's average. The model also incorporates control variables (Z_{jt-1}), including financial position, total assets, sales growth, current asset ratio, and fiscal quarter indicators. Standard errors are doubly clustered by firms and quarters to account for potential correlations in firm- and time-level shocks. Additionally, the model includes interactions between firms' financial conditions and prior quarter GDP growth to capture heterogeneous responses to monetary policy shocks.

To test the robustness of our findings, Table B1 presents robustness checks on the logarithmic change in knowledge capital ($\Delta \log \text{intell_K}$) and the intelligence investment rate. The results confirm that monetary policy shocks significantly impact firm-level innovation and investment decisions.

Table B1: Robustness Checks on $\Delta \log$ KC and Investment Rate of Knowledge Capital

	$\Delta \log$ KC			ikc		
	(1)	(2)	(3)	(1)	(2)	(3)
ffr shock	8.13*** (2.55)	7.53*** (2.40)	7.66*** (2.47)	6.49*** (2.19)	5.96*** (2.03)	6.06*** (2.08)
leverage $\times$ ffr shock		-0.39 (1.01)	-0.55 (1.04)		0.48 (0.59)	0.34 (0.61)
dd $\times$ ffr shock		2.41* (1.24)	2.36* (1.21)		1.65* (0.97)	1.69* (0.97)
size $\times$ ffr shock			-0.01 (1.18)			-0.41 (0.98)
cash flows $\times$ ffr shock			0.29 (0.72)			0.20 (0.50)
$\mathbb{1}\{\text{dividends} > 0\} \times$ ffr shock			-1.82 (2.09)			-1.87 (1.19)
liquidity $\times$ ffr shock			-0.66 (0.81)			-0.83 (0.66)
Observations	104598	70931	70310	113244	76043	75355
R^2	0.167	0.191	0.193	0.205	0.237	0.237
Firm controls	yes	yes	yes	yes	yes	yes
Time sector FE	yes	yes	yes	yes	yes	yes
Time clustering	yes	yes	yes	yes	yes	yes

Notes: This table presents robustness checks on the effects of monetary policy shocks (FFR shock) on firm-level innovation and investment outcomes. The dependent variables are the logarithmic change in knowledge capital ($\Delta \log \text{intell_K}$) and the intelligence investment rate. Standard errors are clustered by firm and quarter. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

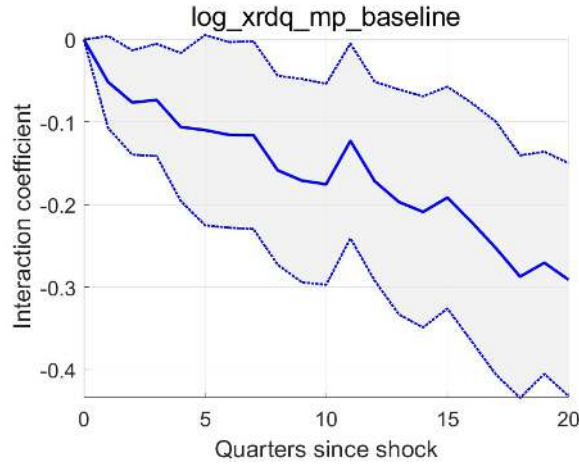
B.2 Dynamic Effects of Monetary Policy on R&D Investment

This section analyzes the impact of monetary policy shocks (MPS) on research and development (R&D) activities. Using a local projection model, we estimate how R&D responses evolve over time in reaction to monetary policy changes. The model controls for firm-level characteristics, past values of R&D, and macroeconomic conditions to isolate the causal effect of monetary policy shocks.

$$\log(\text{R\&D})_{t+h} = \alpha_{j,h} + \beta_h \varepsilon_t + \log(\text{R\&D})_t + \Gamma'_1 Z_{jt-1} + \Gamma'_2 Y_{t-1} + e_{j,t,h}$$

The above equation represents the local projection model, where $\log(\text{R\&D})_{t+h}$ is the logarithmic value of R&D at time $t + h$, capturing the future R&D response; ε_t denotes the monetary policy shock at time t ; $\log(\text{R\&D})_t$ is the lagged logarithmic value of R&D, accounting for persistence in R&D investments; Z_{jt-1} includes firm-level control variables at time $t - 1$ (e.g., size, cash holdings, leverage); Y_{t-1} consists of macro-level control variables at time $t - 1$ (e.g., GDP growth, interest rates); and β_h is the coefficient of interest, capturing the dynamic response of R&D to monetary policy shocks at horizon h . By estimating β_h at various horizons h , we can observe the short- and long-term effects of monetary policy shocks on firms' R&D investments. These results contribute to understanding how external economic factors influence innovation and long-term growth through their impact on R&D.

Figure B1: Local Projection Model for R&D Response



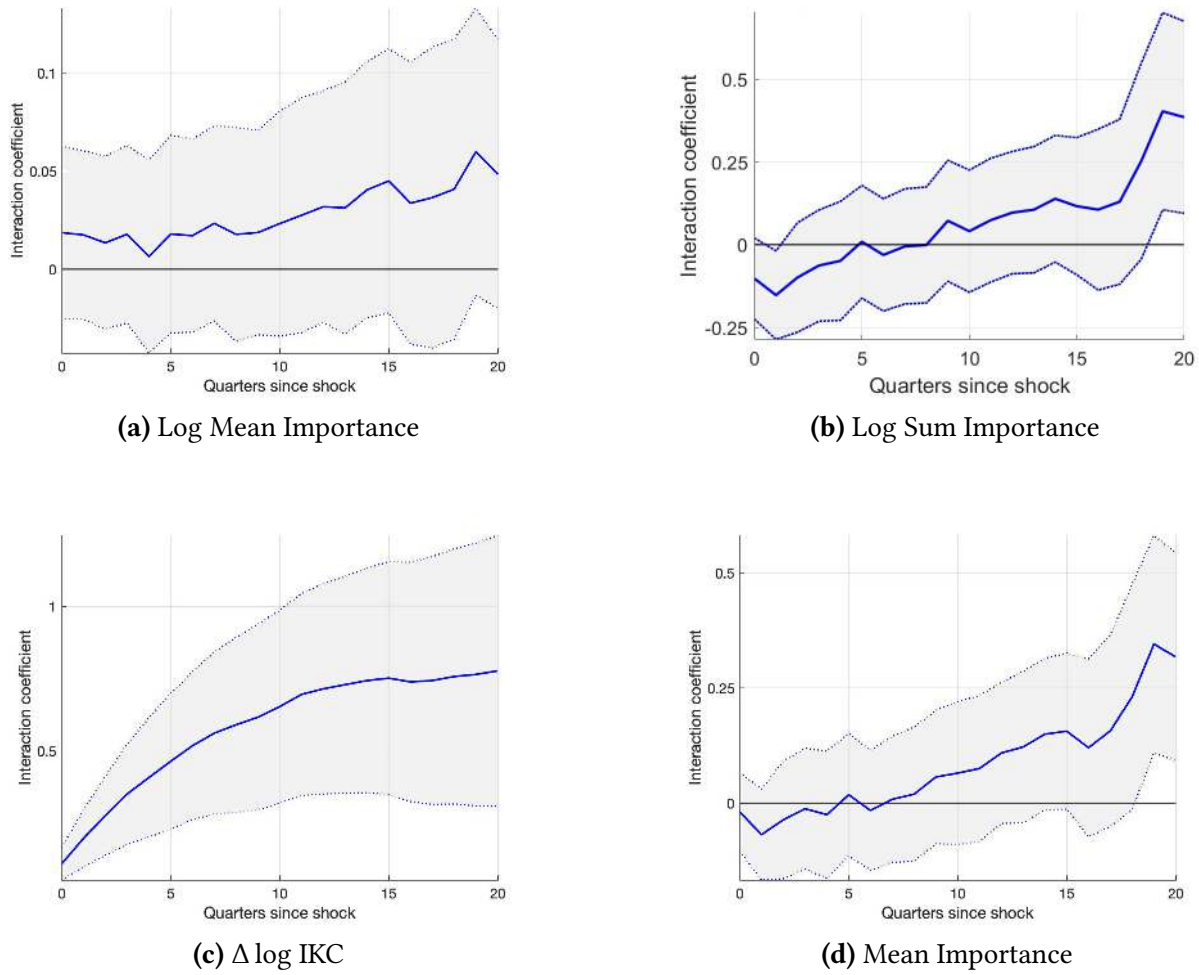
Notes: This figure illustrates the local projection model for R&D responses to monetary policy shocks. The model estimates the effect of shocks (ε_t) on future R&D growth, controlling for past values of R&D and other firm- and macro-level covariates. Specifically, the coefficient β_h measures the dynamic response of R&D at horizon h following a monetary policy shock.

C Supplements to the Baseline Results

C.1 Robustness Analysis of Policy Effects

This part we provides supplementary results using the local projection model to further examine the impact of monetary policy shocks on firms' R&D investment and knowledge capital accumulation.

Figure C1: Probit Model Predictions: Cites and Patent Market Value



Notes: The results capture transitions of firms from inactive to active status, focusing on both the importance and distribution of patents. Panel (a) shows the sum of predicted citations, while panel (b) presents the sum of predicted patent market value. Panel (c) depicts the average of predicted citations, and panel (d) illustrates the average of predicted patent market value. Shaded areas represent 90% confidence intervals, highlighting the uncertainty around predictions.

C.2 Analysis of the Extensive Margin of Investment and Patents

This supplementary analysis complements the main findings by examining the extensive margin of R&D investment and providing additional insights into how monetary policy shapes firms' innovation processes. Specifically, we focus on whether changes in innovation outcomes are driven by firms moving from an inactive state (R&D investment below 1%) to an active state (R&D investment exceeding 1%).

The first stage uses a Probit model to estimate the likelihood of firms transitioning to active R&D investment. This probability is modeled as a function of monetary policy shocks and firm- and aggregate-level controls:

$$P(ikc_{j,t} > 1\%) = \Phi(\alpha_{j,t} + \gamma\varepsilon_t + \Gamma'_1 Z_{j,t-1} + \Gamma'_2 Y_{t-1})$$

Here, $P(ikc_{j,t} > 1\%)$ represents the predicted probability of R&D investment exceeding 1%, derived from the Probit model. The term Φ is the cumulative distribution function of the standard normal distribution, $\alpha_{j,t}$ captures firm-specific fixed effects, and γ measures the impact of monetary policy shocks ε_t . Control variables $\Gamma'_1 Z_{j,t-1}$ and $\Gamma'_2 Y_{t-1}$ represent firm-level and aggregate-level factors. This Probit model focuses on the extensive margin, quantifying how monetary policy shocks influence firms' transitions to active R&D investment.

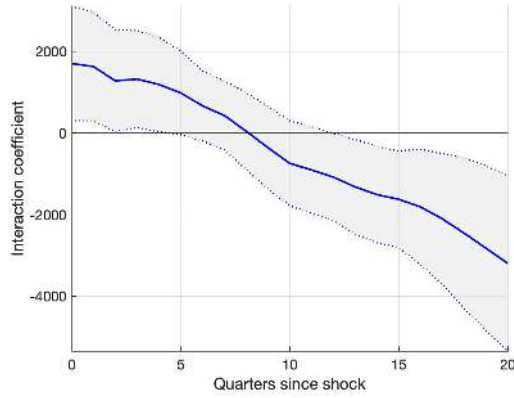
In the second stage, we assess whether the predicted probabilities from the first stage explain changes in innovation outcomes, such as patent numbers, value, and citations:

$$p_{j,t+h} = \alpha_{j,h} + \beta_h \widehat{P(ikc_{j,t} > 1\%)} + \Gamma'_1 Z_{j,t-1} + \Gamma'_2 Y_{t-1} + e_{j,t+h}$$

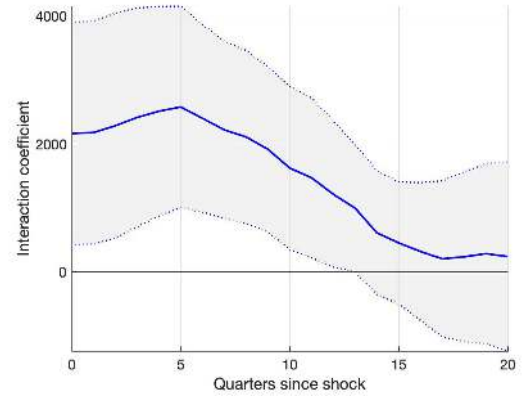
The fitted value $\widehat{P(ikc_{j,t} > 1\%)}$ reflects the likelihood of R&D investment transitions, and β_h quantifies its impact on innovation outcomes. A positive and significant β_h suggests that transitions to active R&D investment due to monetary policy shocks lead to measurable improvements in innovation performance.

This analysis supplements the main findings by focusing on the extensive margin of R&D investment and linking these transitions to observable innovation outcomes, offering additional insights into how monetary policy influences firms' innovation processes.

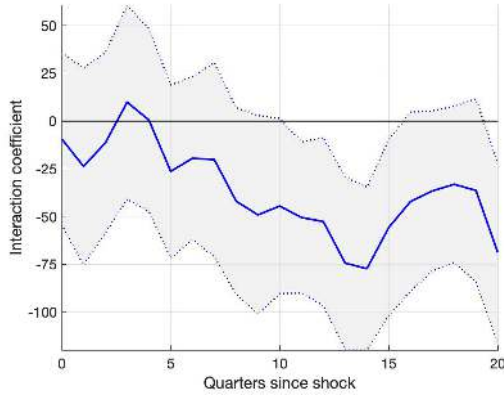
Figure C2: Probit Model Predictions: Cites and Patent Market Value



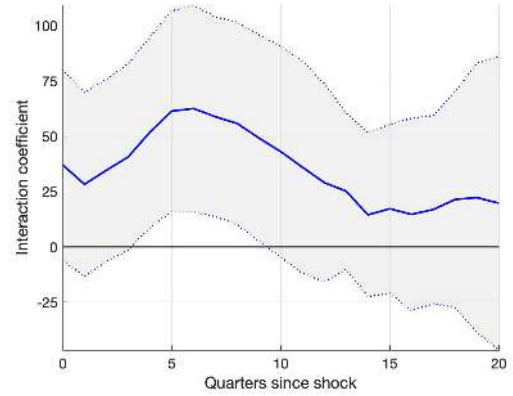
(a) Cites Sum



(b) Patent Market Value Sum



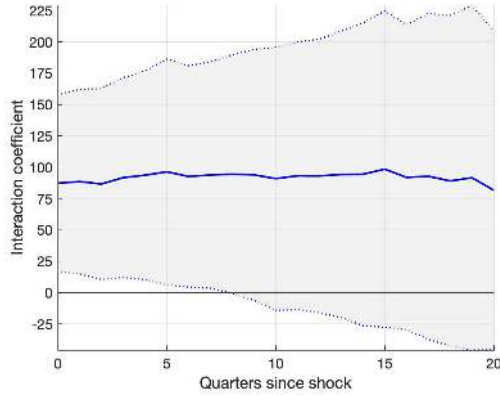
(c) Cites Avg



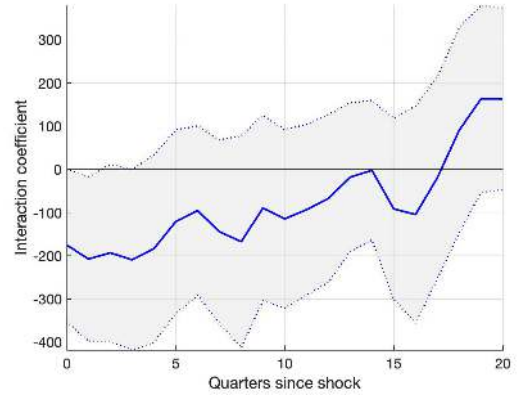
(d) Patent Market Value Avg

Notes: This figure presents predictions from the Probit model focusing on patent metrics. Panel (a) shows the sum of predicted citations, while panel (b) presents the sum of predicted patent market value. Panel (c) depicts the average of predicted citations, and panel (d) illustrates the average of predicted patent market value. Shaded areas represent 90% confidence intervals, highlighting the uncertainty around predictions. The horizontal axis represents transitions of firms from inactive to active status.

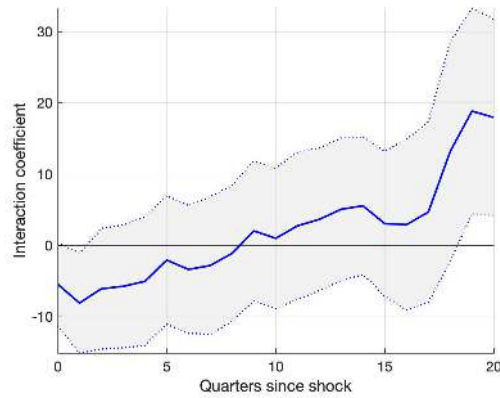
Figure C3: Probit Model Predictions: Importance Metrics



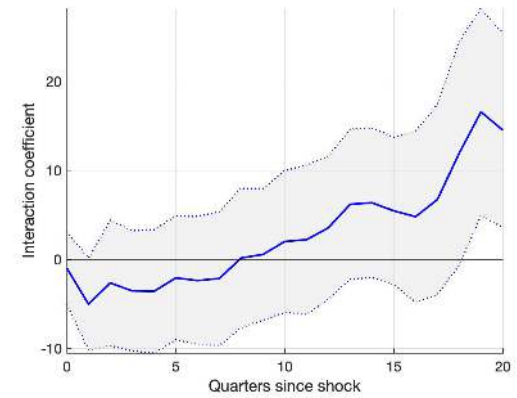
(a) Patent Sum Predict



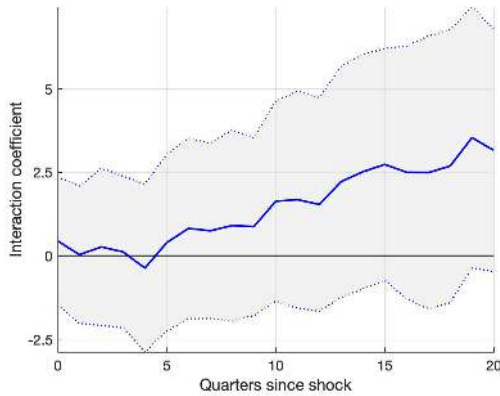
(b) Sum Importance Predict



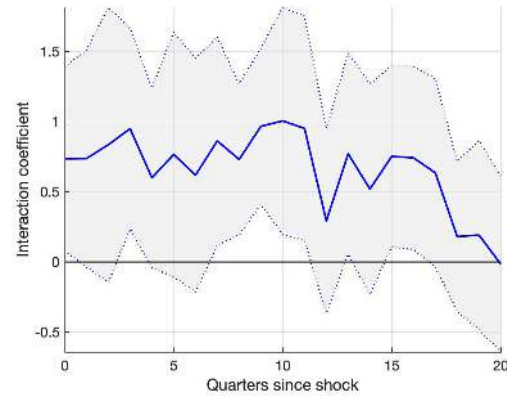
(c) Log Sum Importance



(d) Mean Importance



(e) Log Mean Importance

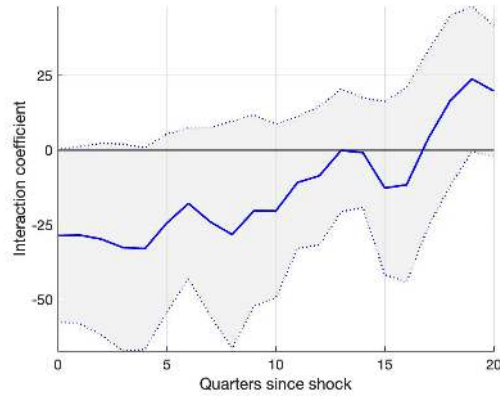


(f) Mean Log Importance

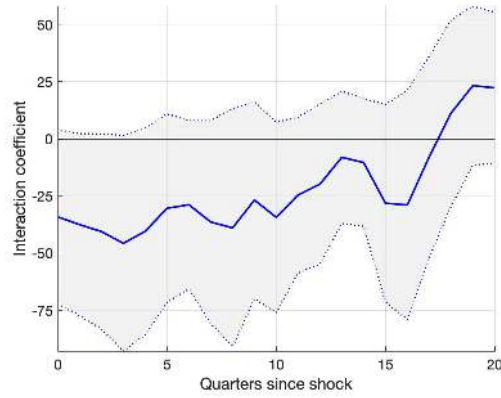
Notes:

This figure presents predictions from the Probit model focusing on importance-related metrics. Panel (a) shows the predicted sum of patents, while panel (b) presents the predicted sum importance. Panel (c) depicts the log of sum importance, and panel (d) shows the mean importance. Panel (e) and panel (f) display the log mean importance and mean log importance, respectively. Shaded areas represent 90% confidence intervals. The horizontal axis denotes the time period or other relevant dimension.

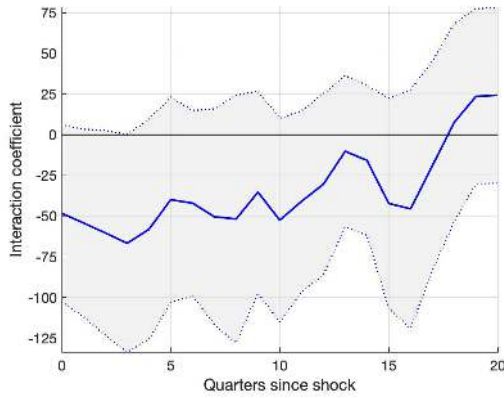
Figure C4: Probit Model Predictions: Percentile-Based Metrics



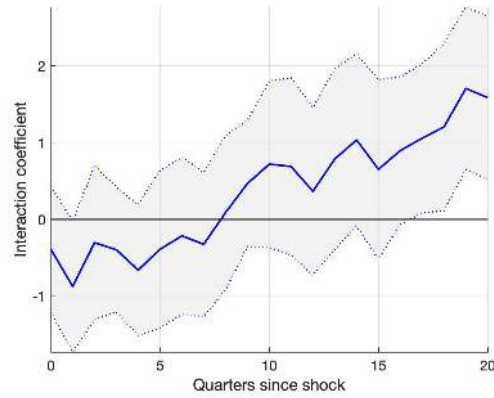
(a) Total P95 Predict



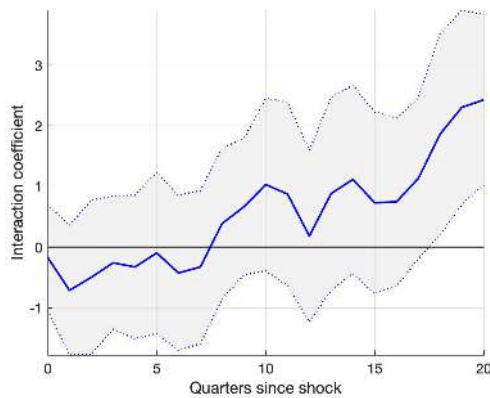
(b) Total P90 Predict



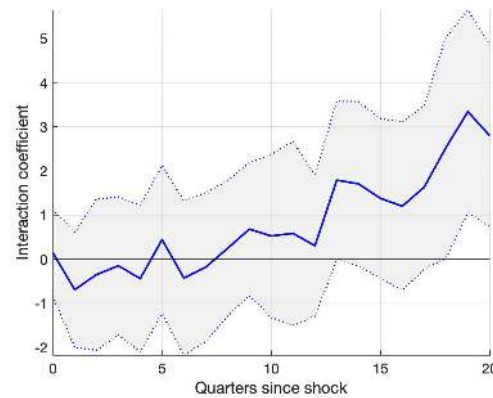
(c) Total P80 Predict



(d) Mean P95 Predict



(e) Mean P90 Predict



(f) Mean P80 Predict

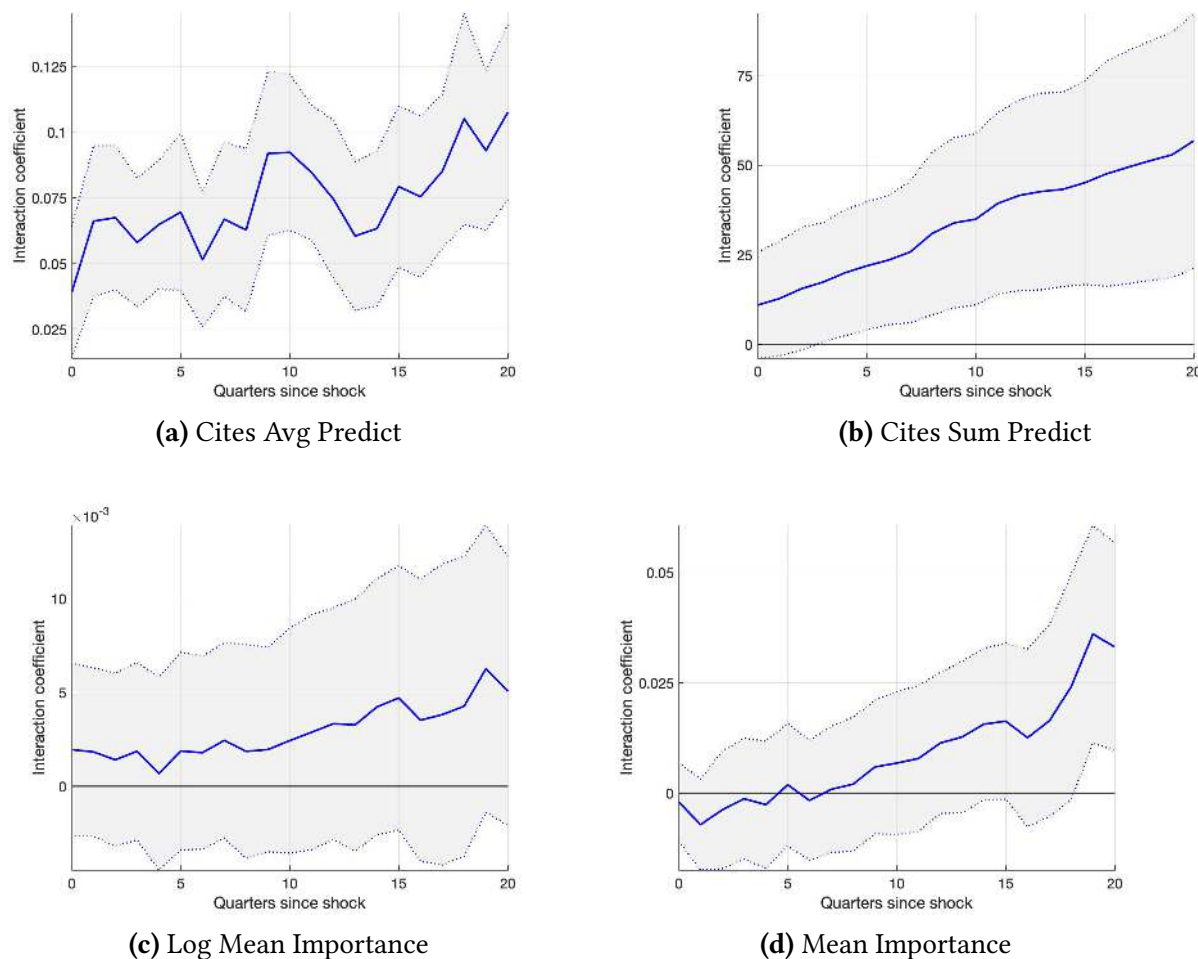
Notes:

This figure presents predictions from the Probit model focusing on percentile-based metrics. Panel (a), (b), and (c) show the total predicted metrics at the 95th, 90th, and 80th percentiles, respectively. Panel (d), (e), and (f) display the mean predicted metrics at the 95th, 90th, and 80th percentiles. Shaded areas represent 90% confidence intervals. The horizontal axis denotes the time period or other relevant dimension.

C.3 Tobit Model Predictions for Innovation Outcomes

This section supplements the intensive margin analysis presented in the main text, where we use the Tobit model to examine how firms that are already engaged in innovation adjust their R&D investment levels in response to monetary policy shocks.

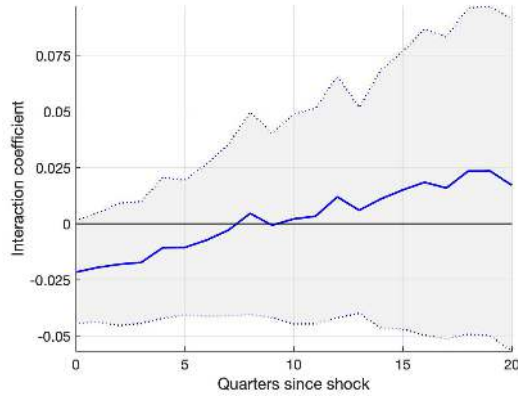
Figure C5: Predictions for the Intensive Margin of Citations and Importance



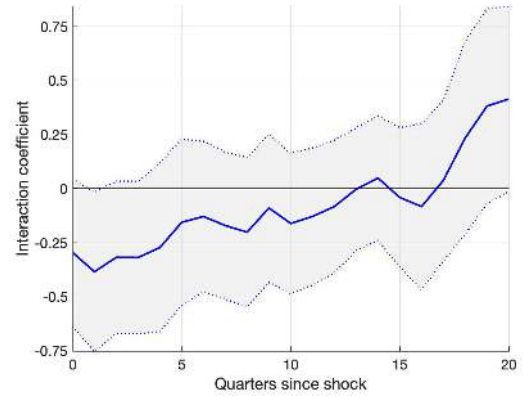
Notes:

This figure presents predictions from the Tobit model focusing on citation and importance metrics. Panel (a) shows the average predicted citations, while panel (b) depicts the sum of predicted citations. Panel (c) presents the log mean importance, and panel (d) illustrates the mean importance of patents. Shaded areas represent 90% confidence intervals. The horizontal axis denotes the time period or other relevant dimension.

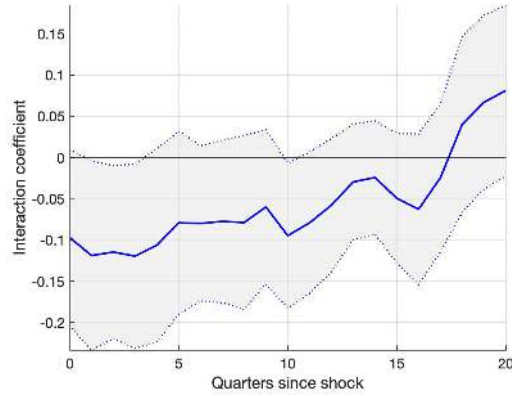
Figure C6: Tobit Model Predictions: Patent and Importance Measures



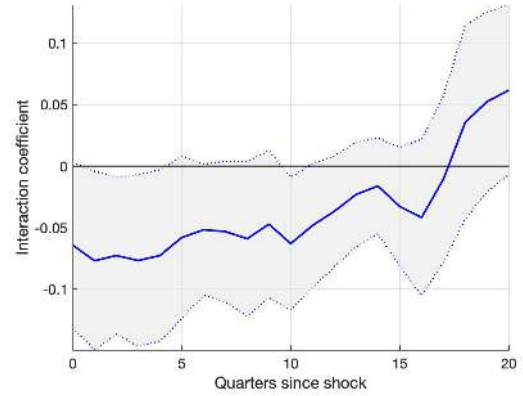
(a) Patent Sum Predict



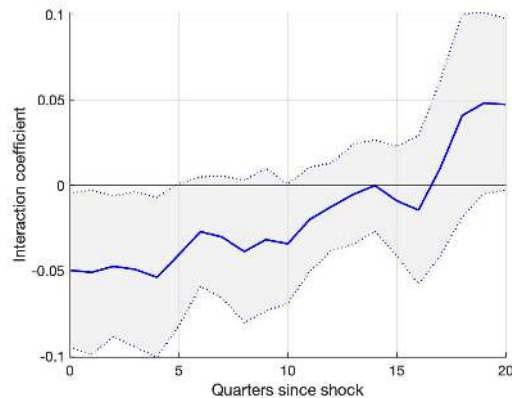
(b) Sum Importance Predict



(c) Total P80 Predict



(d) Total P90 Predict



(e) Total P95 Predict

Notes: This figure presents predictions from the Tobit model focusing on patent and importance measures. Panel (a) shows the predicted sum of patents, while panel (b) illustrates the predicted sum importance. Panels (c), (d), and (e) present total predictions at the 80th, 90th, and 95th percentiles, respectively. Shaded areas represent 90% confidence intervals. The horizontal axis denotes the time period or other relevant dimension.

C.4 Longer Horizon

The baseline local projections trace out responses up to $h = 20$ quarters, or five years, after the shock. While this window is long enough to capture the initial reallocation of R&D effort and the early evolution of patent quantity and quality, it may be too short to reveal the full life cycle of an innovation response: patents can take several years to be filed, examined, and granted, and their citation and market-value trajectories can continue to evolve for a decade or more after the underlying research is conducted. If the true response of innovation to monetary policy is still building at $h = 20$, truncating the horizon there could understate—or, in principle, mischaracterize the sign of—the long-run effect.

To assess this possibility, we extend the projection horizon from $h = 20$ to $h = 30$ (7.5 years) for the full set of inputs and outcomes, estimating

$$y_{i,t+h} = \beta_h s_t + \Gamma' X_{i,t} + \alpha_i + \delta_{q(t)} + \varepsilon_{i,t+h}, \quad h = 0, 1, \dots, 30.$$

This extension is only feasible under the [Jarociński and Karadi \(2020\)](#) shock series used in the previous check, since the Ottonello–Winberry sample ends in 2007 and does not leave enough post-shock quarters to estimate horizons beyond roughly $h = 20$ without running out of data; extending the JK sample through 2019 provides the additional post-shock quarters needed to trace out the response through $h = 30$ while retaining an adequate number of effective observations at each horizon.

Figure C7: Longer Horizon ($h = 30$): Baseline Outcomes (1/3)

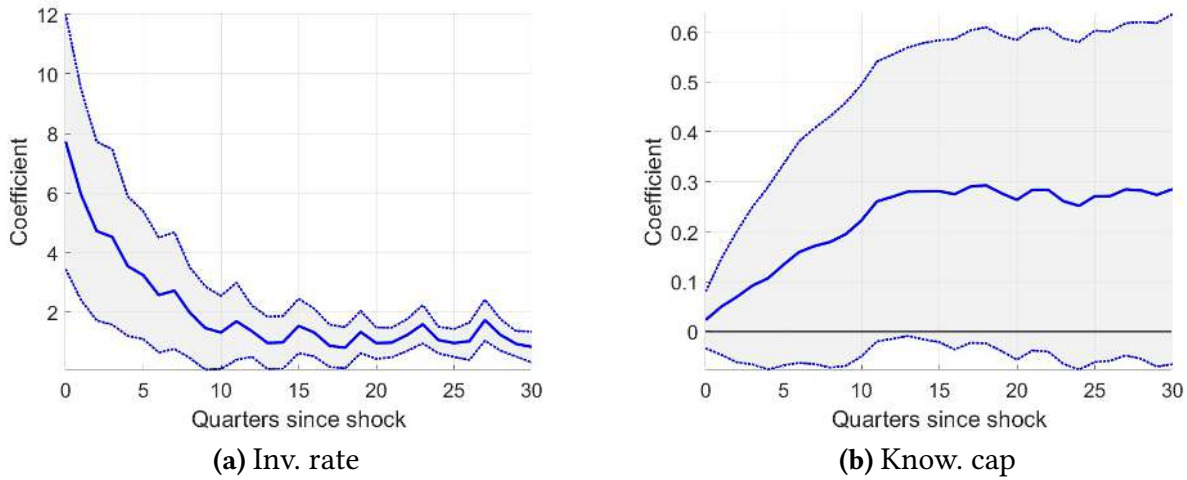


Figure C8: Longer Horizon ($h = 30$): Baseline Outcomes (2/3)

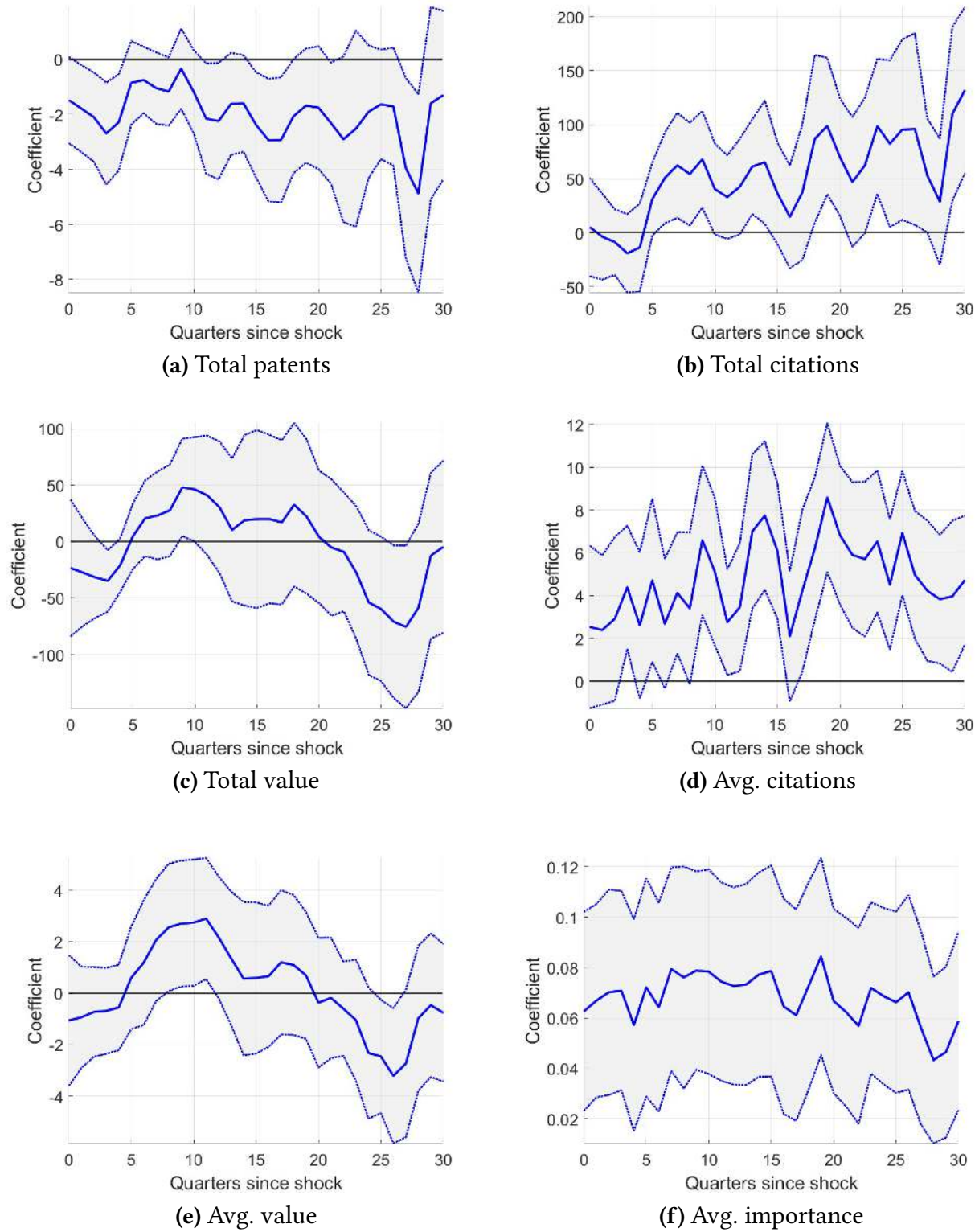
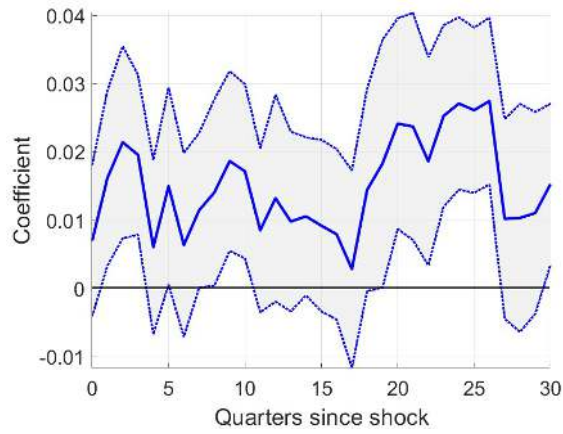
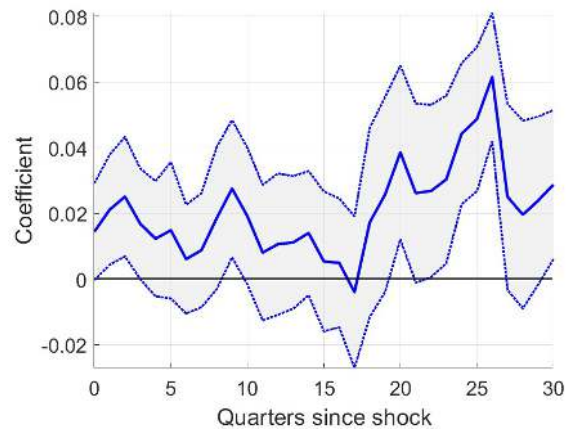


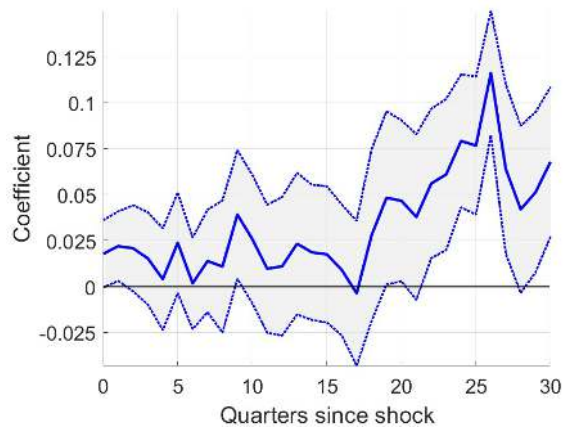
Figure C9: Longer Horizon ($h = 30$): Baseline Outcomes (3/3)



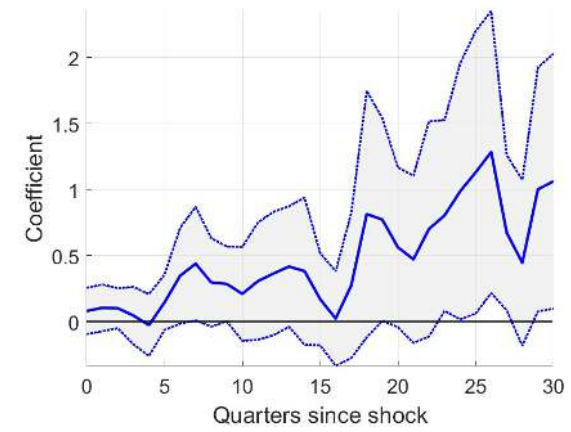
(a) Top 5% (mean)



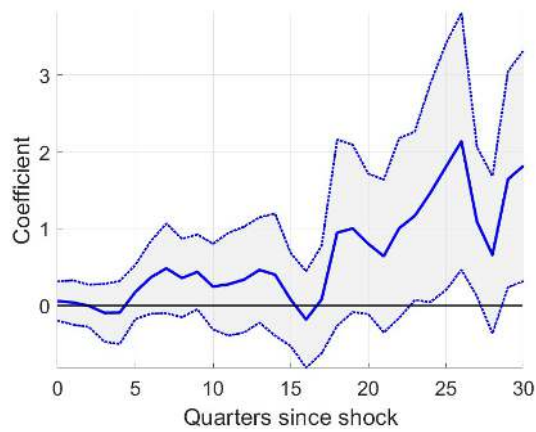
(b) Top 10% (mean)



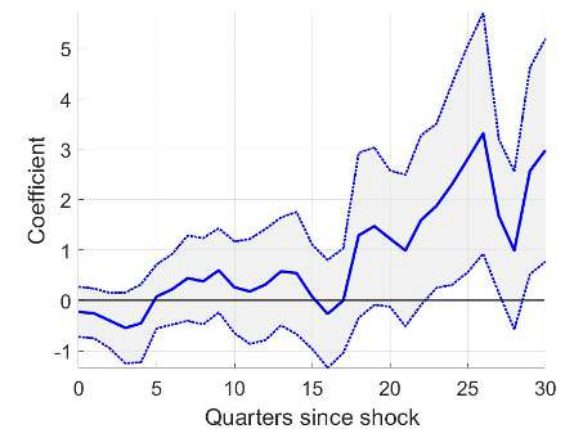
(c) Top 20% (mean)



(d) Top 5% (total)



(e) Top 10% (total)



(f) Top 20% (total)

Notes: These figures extend the baseline local projections to a 30-quarter horizon under the Jarociński and Karadi (2020) shock, for the same set of investment, patenting, and importance outcomes examined in the baseline. Shaded areas represent 90% confidence intervals; note that the bands necessarily widen at longer horizons as the number of post-shock observations available for estimation shrinks.

Conclusion. Extending the horizon confirms that the effects documented in the main text are durable rather than transitory. Knowledge capital plateaus at a permanently higher level rather than reverting to its pre-shock trend, and the totals of breakthrough (top-percentile) patents continue climbing all the way out to $h = 30$, indicating that the reallocation toward higher-quality innovation is not a short-lived response that unwinds once the immediate financing conditions normalize. The value measures—aggregate and average patent value—follow a hump-shaped pattern, peaking somewhere in the middle of the extended horizon before gradually fading, which is consistent with an initial burst of high-value patenting that is only partially sustained as the underlying research projects mature and are eventually superseded by newer innovations. As expected, the confidence bands widen substantially at the longer horizons, reflecting the smaller number of effective post-shock observations available for identification once we move past $h = 20$; we therefore treat the qualitative pattern—persistence in knowledge capital and top-tier patenting, a hump-shaped value response—as the main takeaway from this exercise, rather than the precise point estimates at the furthest horizons.

C.5 Lag Augmentation

Many of the outcomes we study—particularly the investment rate and growth rate of knowledge capital—are highly persistent by construction, since knowledge capital accumulates according to a perpetual-inventory process in which today’s stock mechanically depends on yesterday’s stock plus new R&D spending. If the local projection does not adequately control for this mechanical persistence, part of what looks like a dynamic response to the monetary policy shock could simply reflect the outcome variable’s own autoregressive structure propagating forward, rather than a genuine causal effect of the shock at each horizon.

To separate the shock’s effect from this mechanical persistence, we add four lags of the dependent variable to each local projection:

$$y_{i,t+h} = \beta_h s_t + \sum_{j=1}^p \phi_j y_{i,t-j} + \Gamma' X_{i,t} + \alpha_i + \delta_{q(t)} + \varepsilon_{i,t+h}$$

The baseline specification uses $p = 1$, a single lag of the dependent variable; the lag-augmented specification here sets $p = 4$, absorbing a full year of own-outcome dynamics before estimating the shock response β_h . If the baseline results were an artifact of insufficiently controlling for persistence, we would expect the estimated β_h coefficients to change substantially, and typically to shrink toward zero, once the additional lags are included.

Figure C10: Lag Augmentation ($p = 4$): Baseline Outcomes (1/2)

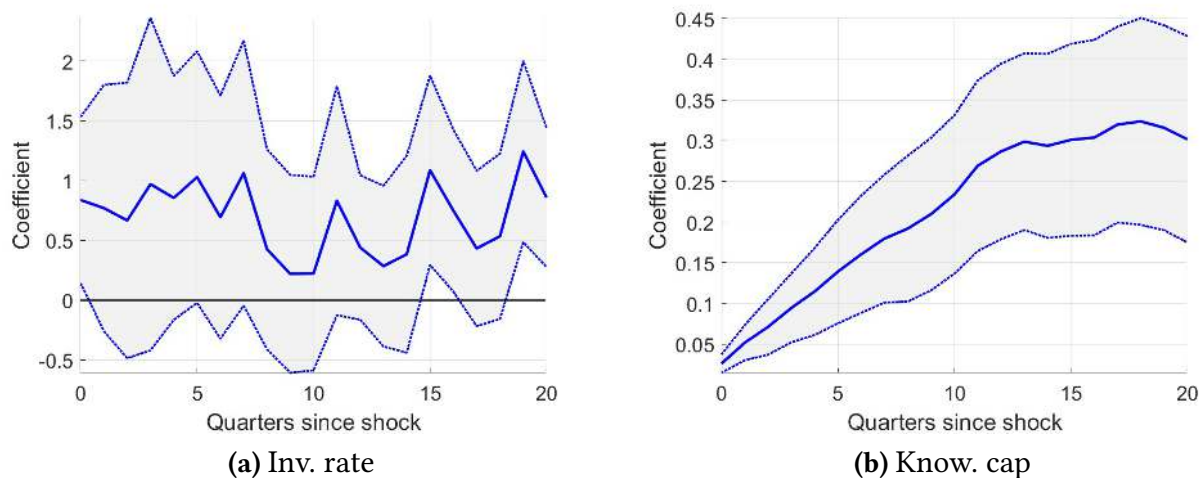
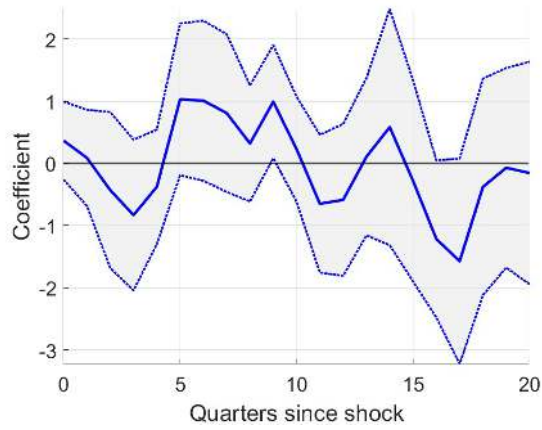
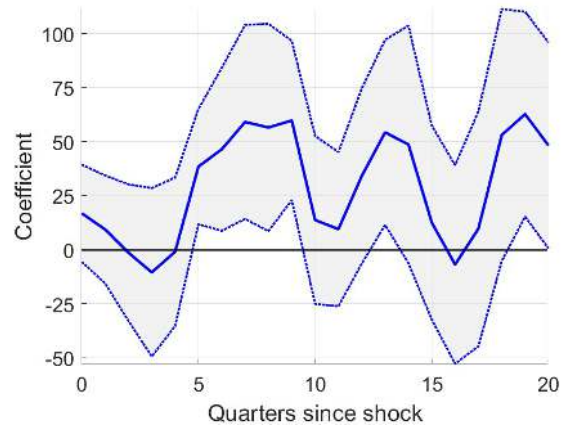


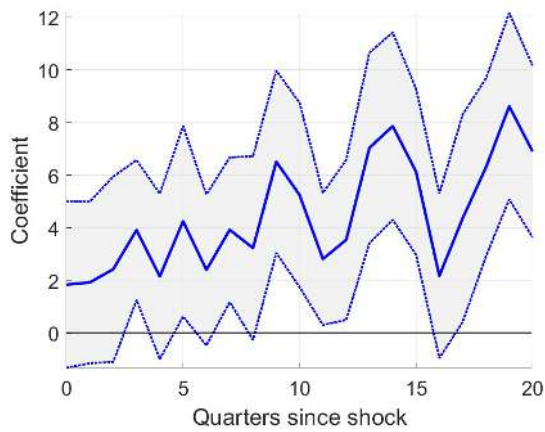
Figure C11: Lag Augmentation ($p = 4$): Baseline Outcomes (2/2)



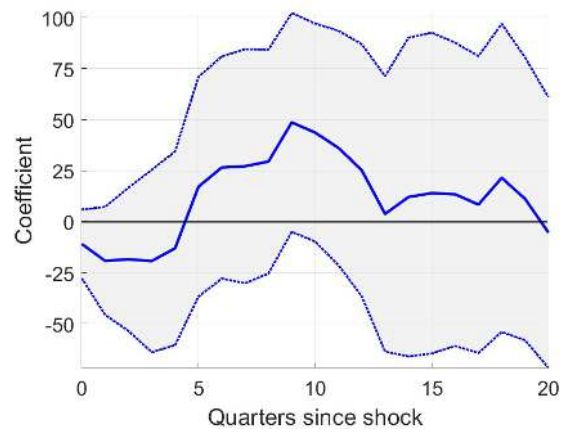
(a) Patents



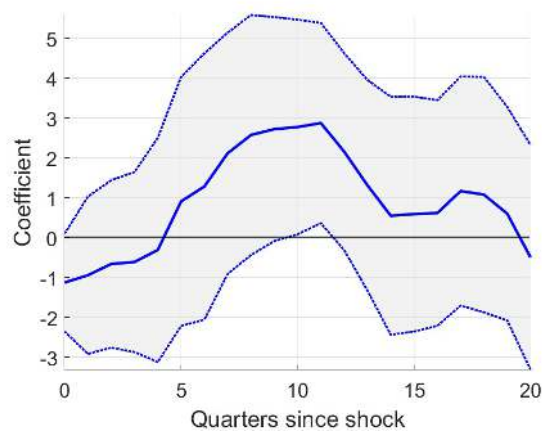
(b) Tot. cites



(c) Avg. cites



(d) Tot. value



(e) Avg. value

Notes: These figures report local projections augmented with four lags of the dependent variable, for the same set of investment, patenting, citation, and value outcomes as in the baseline. Shaded areas represent 90% confidence intervals.

Conclusion. The impulse responses retain the same sign, timing, and broad shape once four lags of the dependent variable are included to absorb own-outcome persistence. This indicates that the dynamic responses documented in the baseline are not a mechanical artifact of the autoregressive structure inherent in the knowledge-capital accumulation process or in patenting activity, but instead reflect a genuine response to the monetary policy shock itself. If persistence had been driving the baseline results, we would have expected the lag-augmented coefficients to attenuate meaningfully relative to the single-lag specification; instead, the estimates are quantitatively close to the baseline throughout the horizon, which strengthens our confidence that the local projection is correctly isolating the causal effect of the shock rather than simply picking up the outcome variable's own dynamics.

D Supplements to Robustness Checks

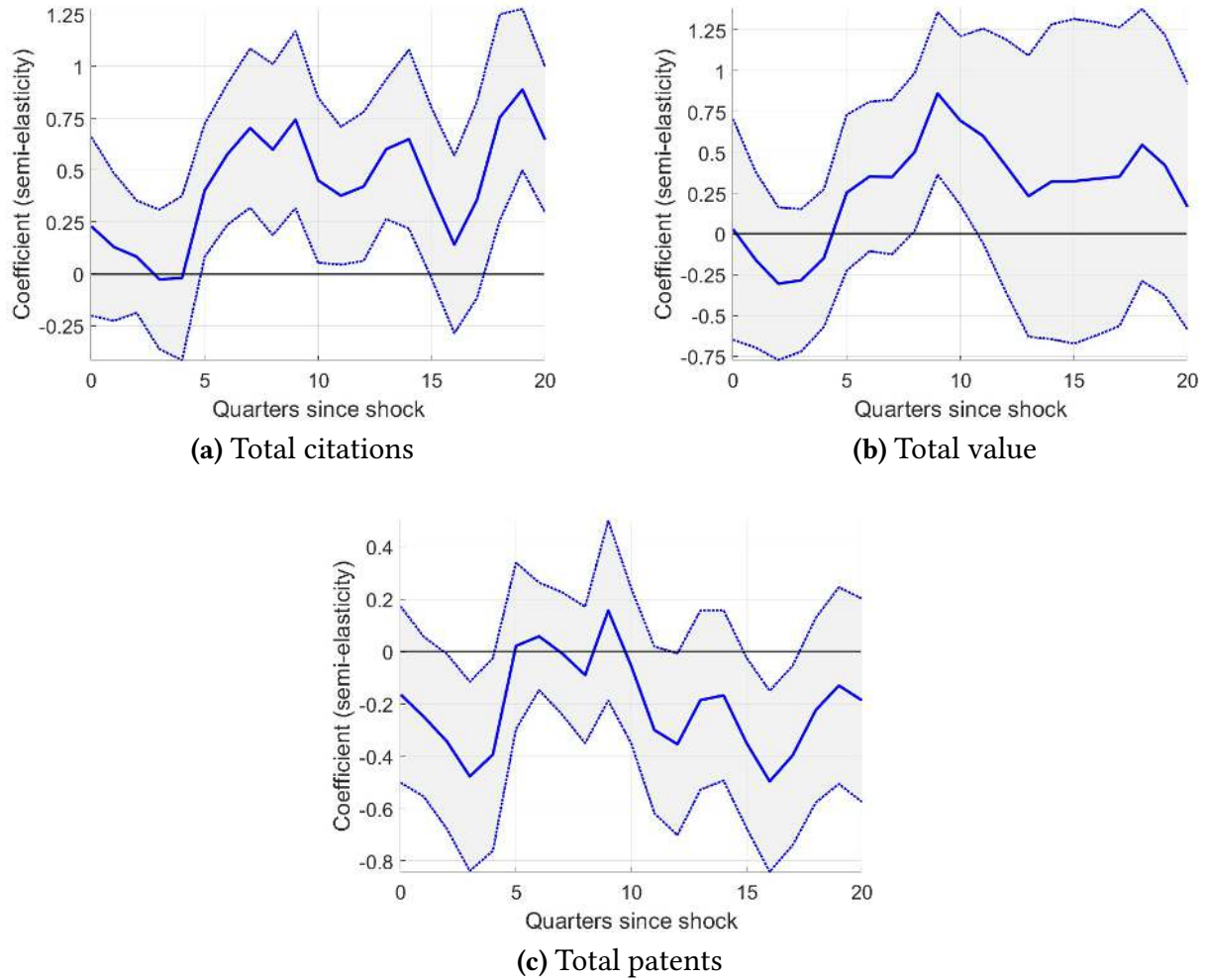
This section reports additional robustness exercises concerning how the baseline results are estimated: alternative functional forms for the count-based outcomes, alternative standard-error clustering, and a range of sample restrictions and firm-level control sets.

D.1 Poisson Estimation (PPML)

Our baseline specification measures innovation output using the logarithm of patent counts and citation counts, a functional form that is convenient because it yields elasticity-like coefficients and helps address the strong right skewness of patent and citation data. However, taking logs is undefined at zero, so any firm-quarter observation with no patents or no citations must be dropped from the estimating sample. Because a substantial share of firm-quarters in our data have zero patents—as documented in the summary statistics in the main text—this could introduce a form of selection: if the probability of a firm reporting a strictly positive patent count is itself correlated with the monetary policy shock, then estimating the log specification only on the subsample with positive counts could bias our estimates of the innovation response.

To address this concern, we re-estimate the count outcomes—total patent citations, total patent value, and total patent counts—using Poisson pseudo-maximum-likelihood (PPML) estimation. PPML models the level of the outcome variable directly rather than its logarithm, so it retains all firm-quarter observations including those with a value of exactly zero, and the resulting coefficients can be interpreted as semi-elasticities (the approximate percentage change in the expected count for a one-unit change in the regressor). This provides a natural robustness check on whether our results are being driven by how the baseline log specification handles the zero-count observations.

Figure D1: Poisson (PPML) Estimation: Citations, Patent Value, and Patent Counts



Notes: These figures report PPML estimates of the response of total citations, total patent value, and total patent counts to the baseline monetary policy shock, estimated on the full sample including firm-quarters with zero patents or zero citations. Shaded areas represent 90% confidence intervals.

Conclusion. Under PPML, total citations and total patent value continue to respond positively and significantly to an expansionary monetary policy shock, mirroring the quality- and value-based findings from the baseline log specification, while the response of the raw patent count remains statistically indistinguishable from zero, just as in the baseline. The fact that the quality-and-value result survives—and, if anything, is estimated with the full sample including firms that report no patents at all—indicates that our central finding is not an artifact of how the baseline log specification handles zero-count observations, nor of any selection induced by dropping those observations. This is a meaningful check because the applied literature on count and gravity-type models has shown that log-linear specifications can produce misleading elasticities when the dependent variable takes on many zero values; the consistency between the PPML and log-based estimates here suggests that concern is not material for our application.

D.2 Alternative Standard Errors

Our baseline inference clusters standard errors two-way by firm and by quarter, which allows for arbitrary within-firm serial correlation and for common shocks affecting all firms in a given quarter. This choice, while standard in the corporate-finance literature, does not explicitly account for correlation across firms within the same industry at a given point in time, nor does it use a Newey–West-type kernel to allow for richer serial dependence in the residuals. If either of these forms of correlation is empirically important and our baseline clustering understates it, our confidence intervals could be too narrow, overstating the statistical significance of the estimated responses.

To evaluate this concern, we recompute standard errors under two alternative approaches for the seven linear specifications in the baseline: Driscoll–Kraay standard errors, which are robust to general forms of cross-sectional and temporal dependence and are particularly well suited to panels with a fixed cross-section and a moderately long time dimension; and two-way clustering by industry $\times$ quarter and by firm, which explicitly allows for correlated shocks within an industry-quarter cell (for example, common technology or demand shocks hitting all firms in an industry at the same time) in addition to firm-level persistence. Because Driscoll–Kraay standard errors require a linear estimator, we omit the Probit and Tobit specifications from this exercise and focus on the seven local projection specifications where the dependent variable enters linearly.

Figure D2: Alternative Standard Errors: Baseline Outcomes (1/3)

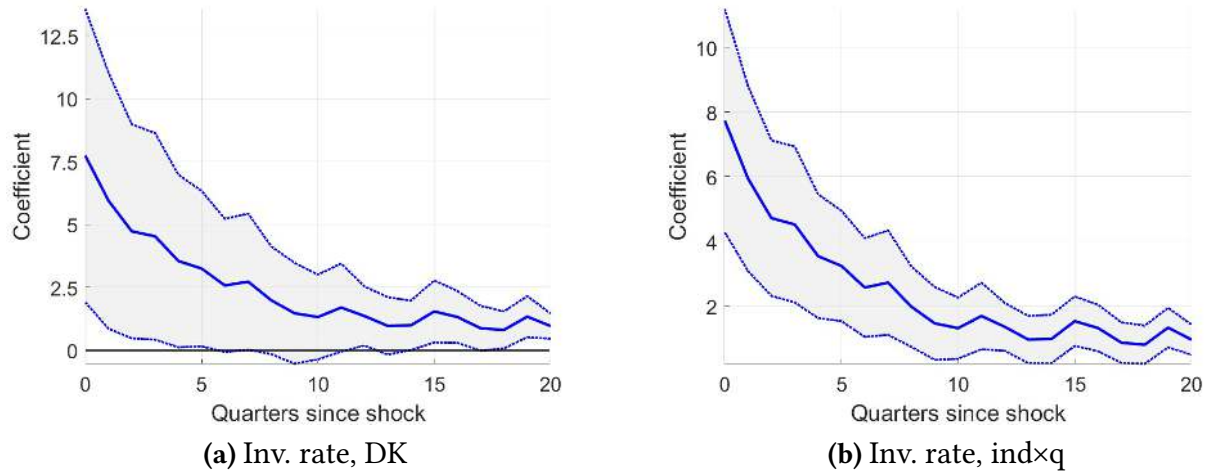
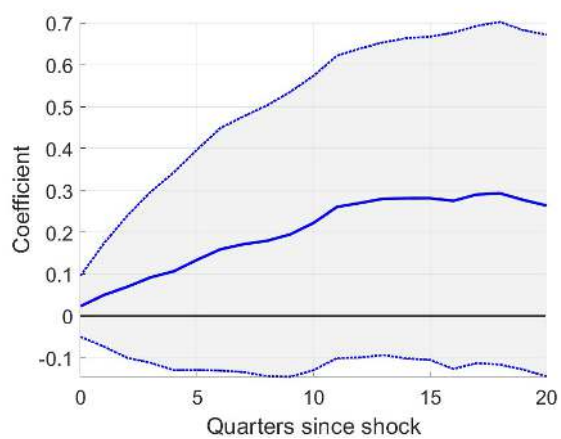
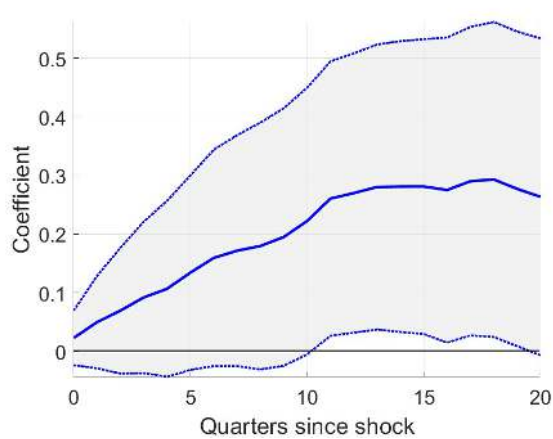


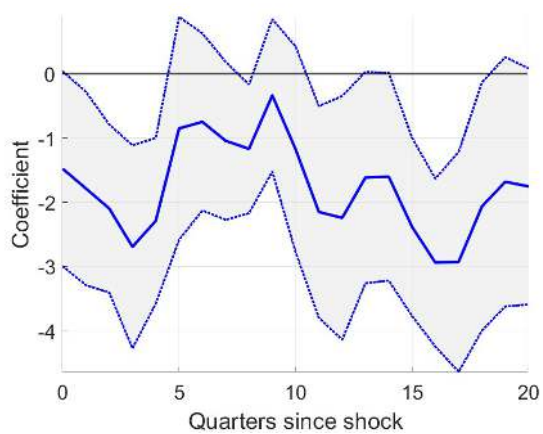
Figure D3: Alternative Standard Errors: Baseline Outcomes (2/3)



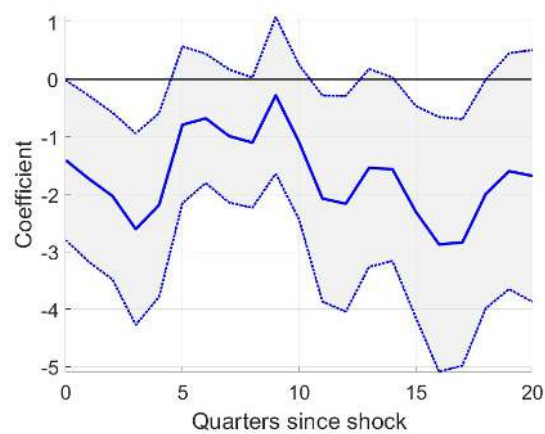
(a) Know. cap, DK



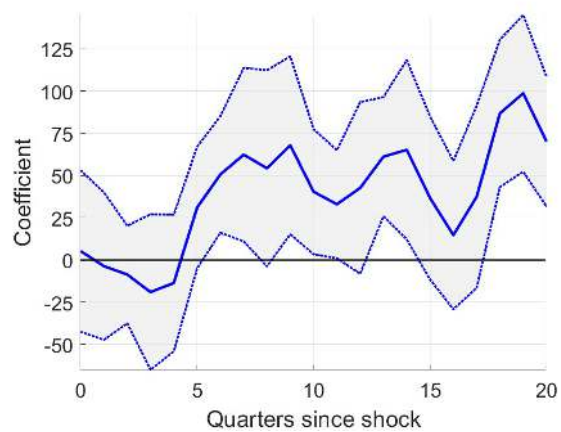
(b) Know. cap, ind×q



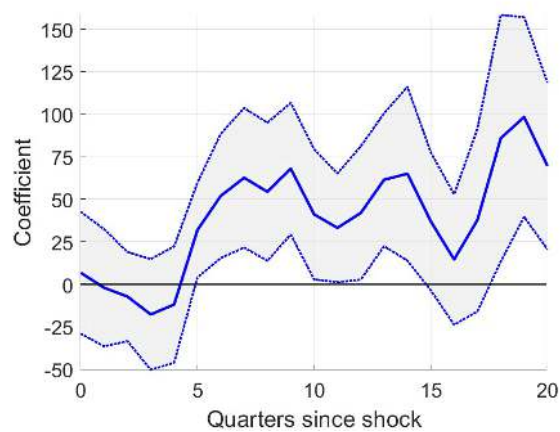
(c) Patents, DK



(d) Patents, ind×q

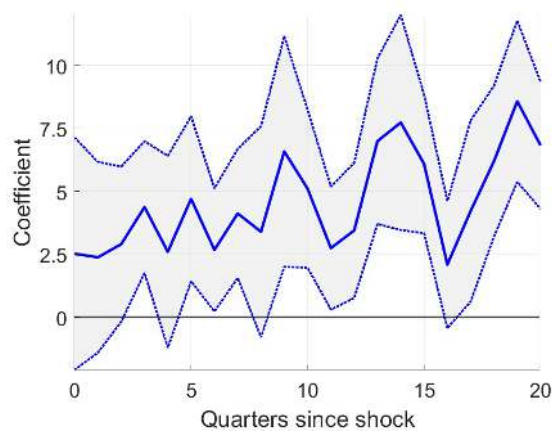


(e) Tot. cites, DK

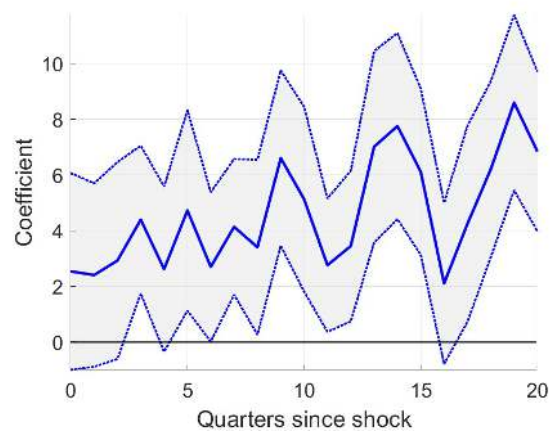


(f) Tot. cites, ind×q

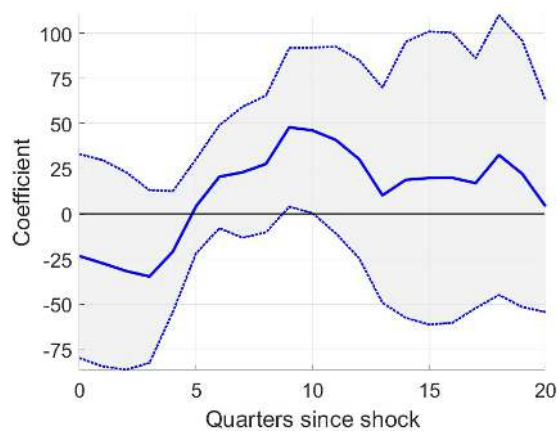
Figure D4: Alternative Standard Errors: Baseline Outcomes (3/3)



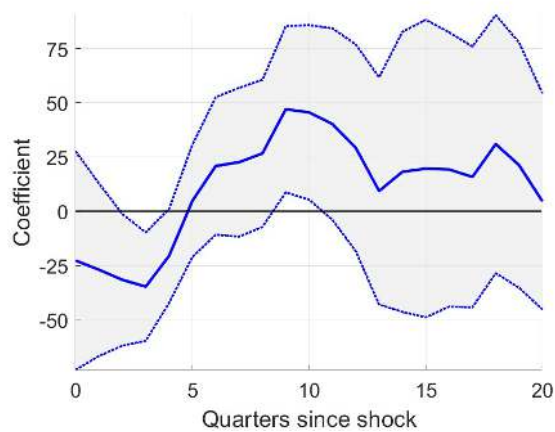
(a) Avg. cites, DK



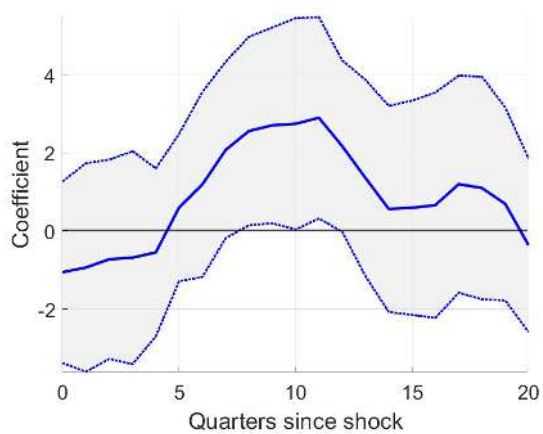
(b) Avg. cites, ind×q



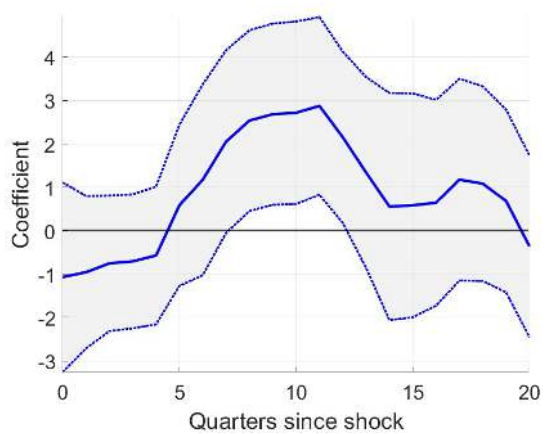
(c) Tot. value, DK



(d) Tot. value, ind×q



(e) Avg. value, DK



(f) Avg. value, ind×q

Notes: DK denotes Driscoll–Kraay standard errors; ind×q denotes two-way clustering by industry×quarter and by firm. Because the point estimates are identical to the baseline by construction (only the standard errors are recomputed), these figures isolate the effect of the clustering choice on the width of the confidence bands. Shaded areas represent 90% confidence intervals.

Conclusion. Since the point estimates are mechanically unchanged under both alternative standard-error procedures, this exercise is entirely about the width of the confidence bands rather than the location of the impulse responses. We find that the bands are nearly identical to the baseline under both Driscoll–Kraay and industry×quarter clustering: responses that are statistically significant in the baseline remain significant, and the null result for the raw patent count remains a null result under both alternative procedures. This suggests that cross-sectional dependence within industry-quarter cells and richer forms of serial correlation are not empirically important sources of additional uncertainty in this setting, and that our baseline two-way firm-and-quarter clustering is not artificially narrowing the confidence intervals or overstating the precision of our estimates. We view this as evidence that the statistical inference underlying our main results is robust to reasonable alternative treatments of the error structure.

D.3 Sample Robustness

A recurring concern in firm-level panel studies of innovation is that aggregate patterns can be driven disproportionately by a small number of outlier firms, by a handful of patent-intensive industries whose R&D behavior may not generalize to the broader economy, or by cyclical dynamics specific to the Great Recession rather than by the transmission of monetary policy per se. We address each of these concerns in turn with three separate sample restrictions.

First, we drop the top 1% of patenters by cumulative patent count, to verify that our results are not mechanically driven by a small number of extremely prolific innovators whose patenting decisions could dominate the aggregate statistics even after weighting. Second, we drop firms in the drug and semiconductor industries, two sectors that are unusually patent-intensive and R&D-dependent relative to the rest of the economy, to check that our findings reflect a broad-based channel rather than a sector-specific pattern in industries where innovation activity is, for institutional and technological reasons, already highly sensitive to financing conditions. Third, we drop all recession quarters (beyond the Great Recession exclusion already imposed in the baseline), to ensure that the estimated responses are not confounded with contemporaneous business-cycle contractions, which independently affect both financing conditions and innovation activity through channels unrelated to monetary policy. The baseline specification, by contrast, uses the full sample and excludes only the Great Recession; these three cuts allow us to test whether the baseline results are driven by the largest patenters, by patent-intensive industries, or by recessions more generally.

Figure D5: Sample Robustness: Baseline Outcomes (1/6)

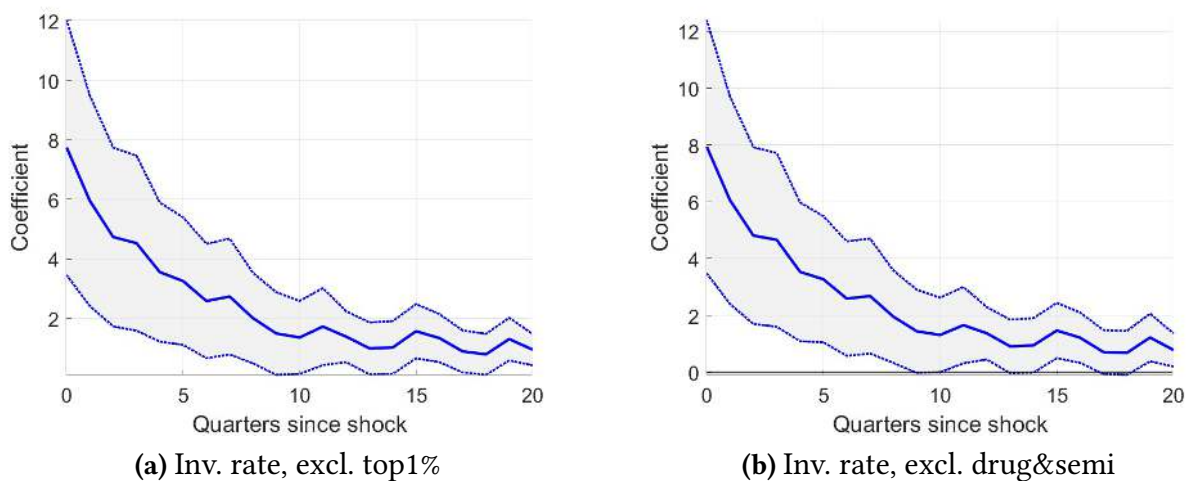
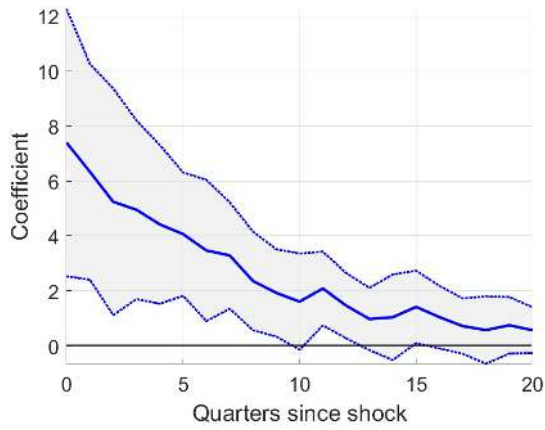
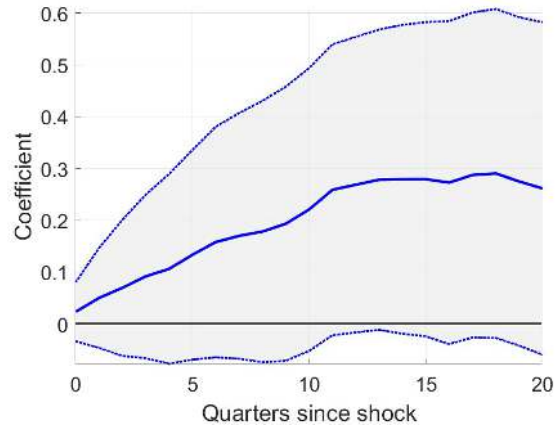


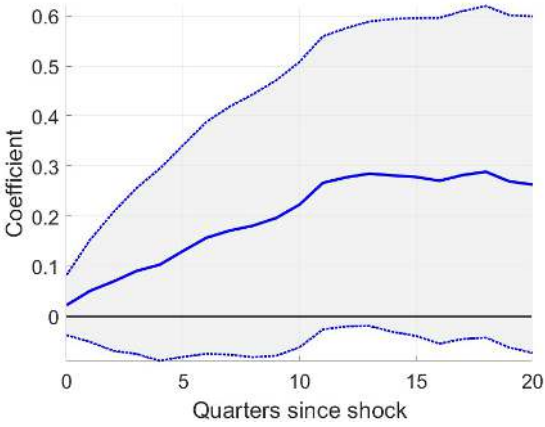
Figure D6: Sample Robustness: Baseline Outcomes (2/6)



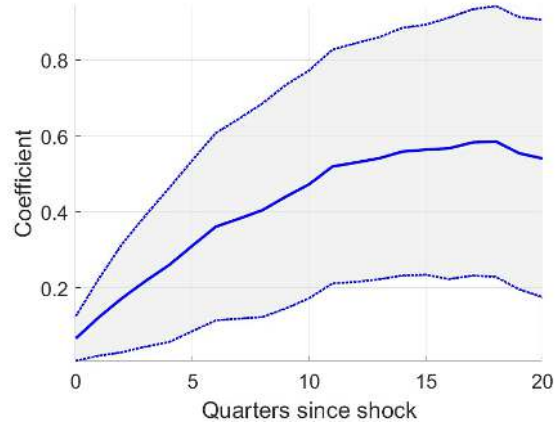
(a) Inv. rate, excl. rec.



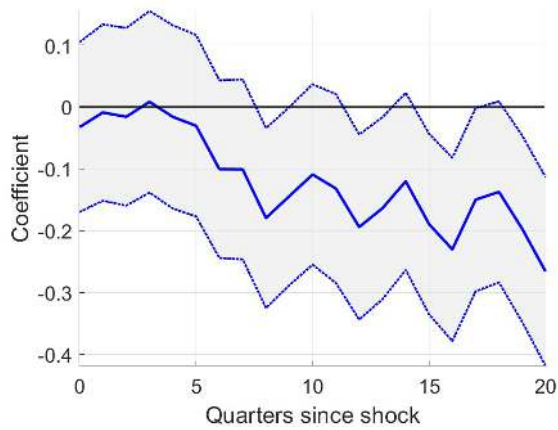
(b) Know. cap, excl. top1%



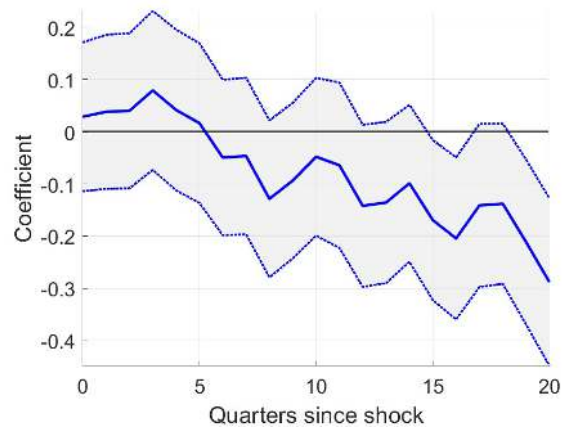
(c) Know. cap, excl. drug&semi



(d) Know. cap, excl. rec.

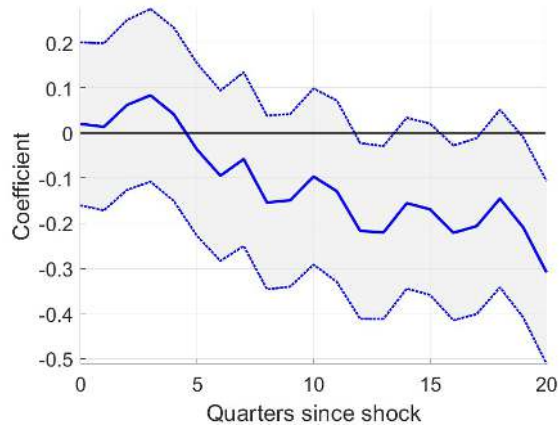


(e) Probit, excl. top1%

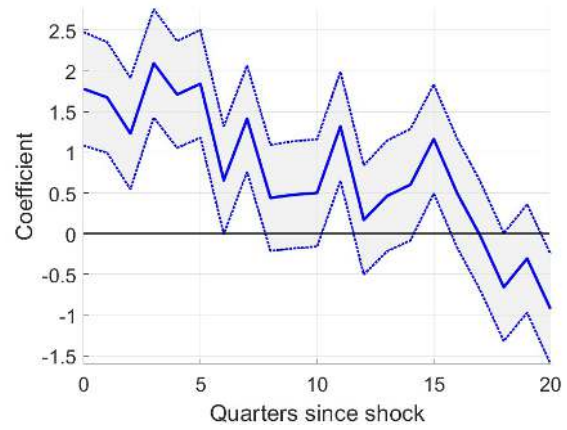


(f) Probit, excl. drug&semi

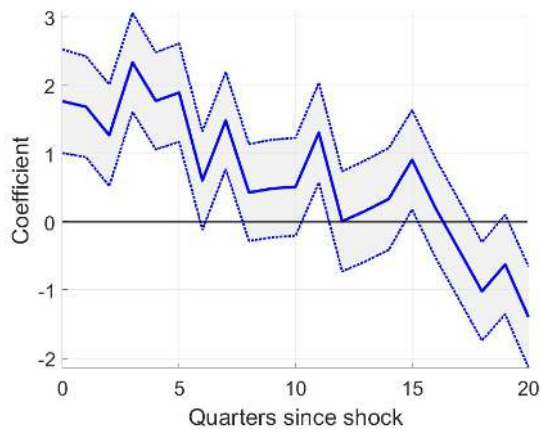
Figure D7: Sample Robustness: Baseline Outcomes (3/6)



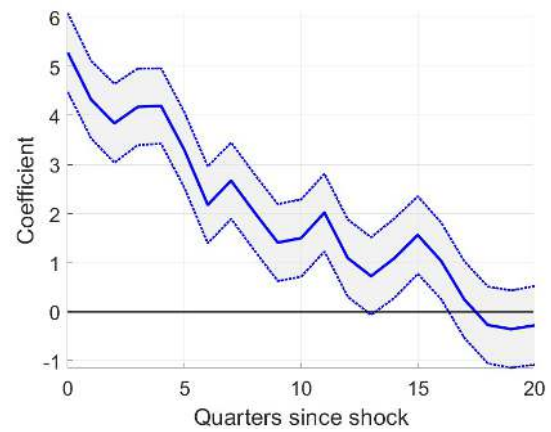
(a) Probit, excl. rec.



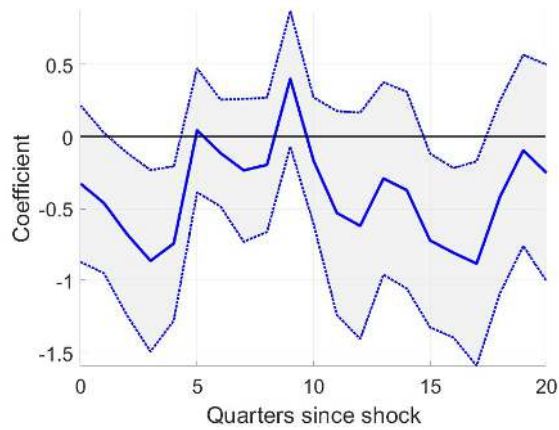
(b) Tobit, excl. top1%



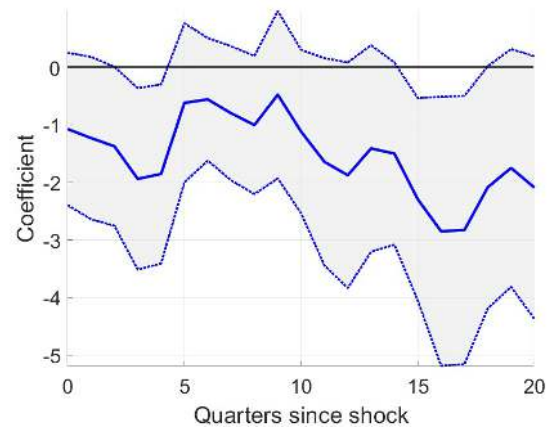
(c) Tobit, excl. drug&semi



(d) Tobit, excl. rec.

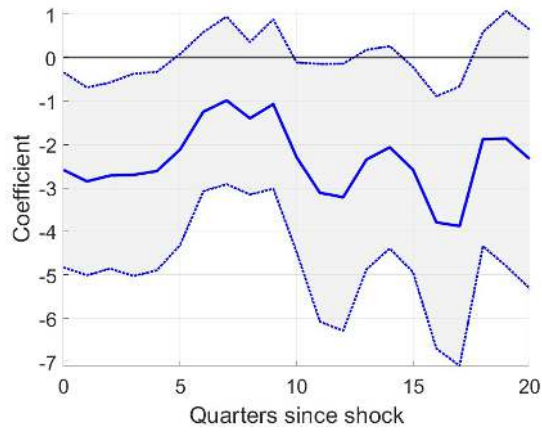


(e) Patents, excl. top1%

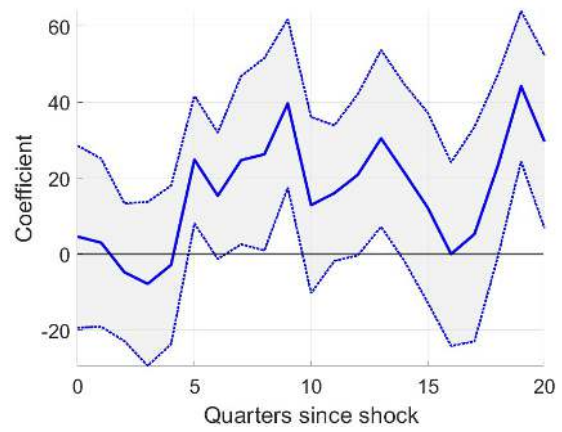


(f) Patents, excl. drug&semi

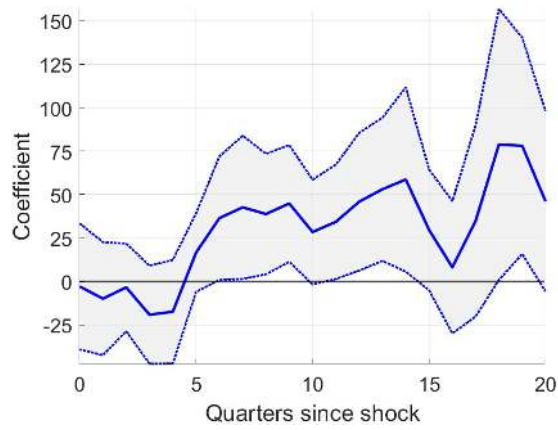
Figure D8: Sample Robustness: Baseline Outcomes (4/6)



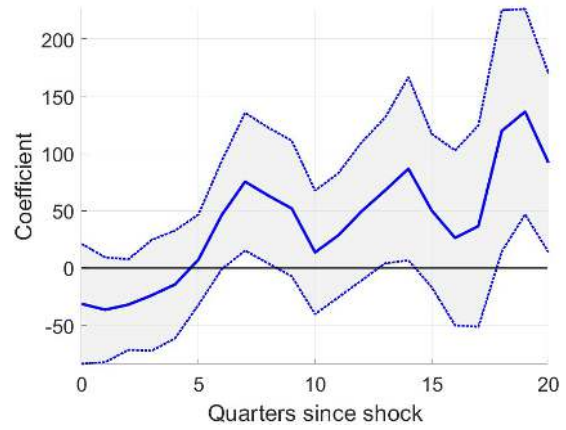
(a) Patents, excl. rec.



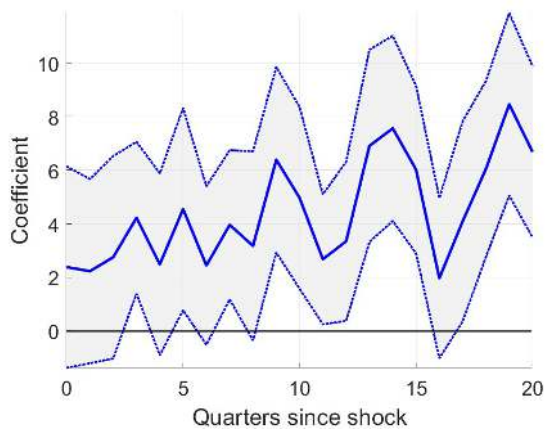
(b) Tot. cites, excl. top1%



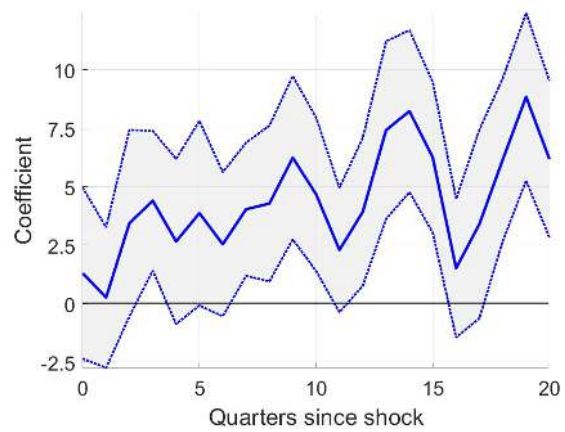
(c) Tot. cites, excl. drug&semi



(d) Tot. cites, excl. rec.

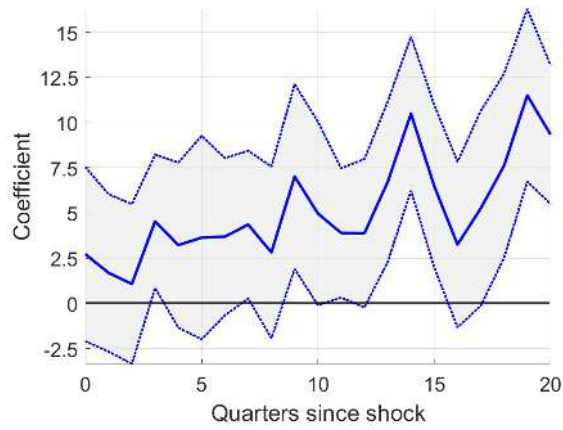


(e) Avg. cites, excl. top1%

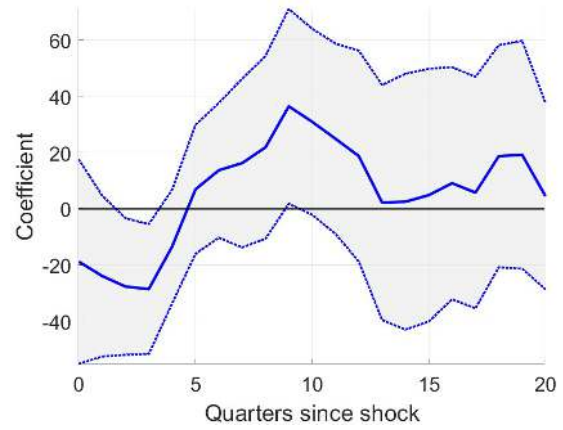


(f) Avg. cites, excl. drug&semi

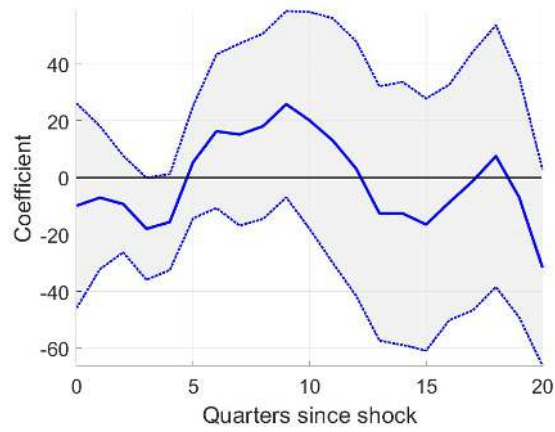
Figure D9: Sample Robustness: Baseline Outcomes (5/6)



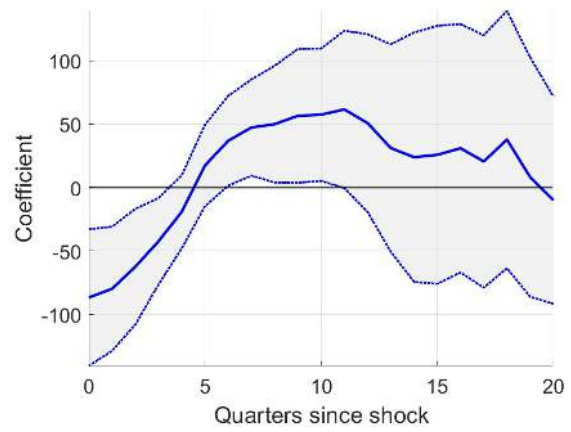
(a) Avg. cites, excl. rec.



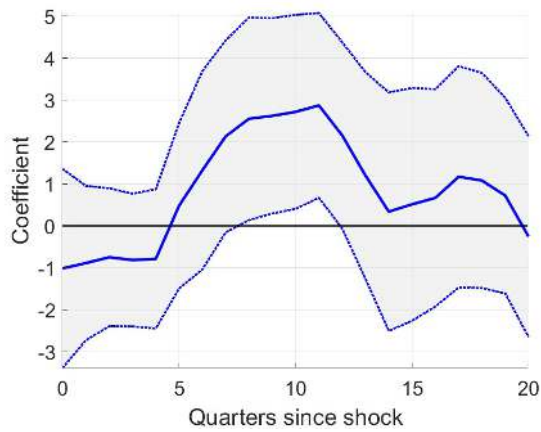
(b) Tot. value, excl. top1%



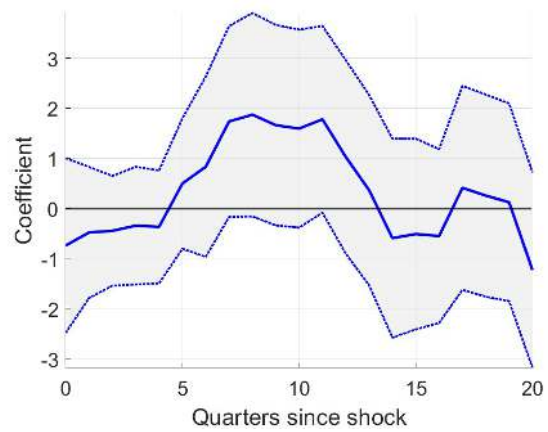
(c) Tot. value, excl. drug&semi



(d) Tot. value, excl. rec.

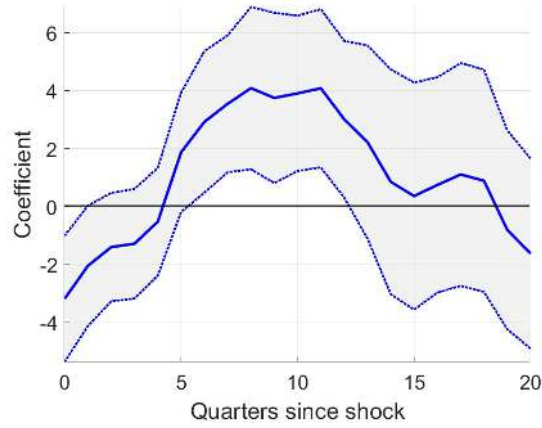


(e) Avg. value, excl. top1%



(f) Avg. value, excl. drug&semi

Figure D10: Sample Robustness: Baseline Outcomes (6/6)



(a) Avg. value, excl. rec.

Notes: Each column reports one sample cut—excluding the top 1% of patenters, excluding the drug and semiconductor industries, and excluding recession quarters—for the same set of investment, extensive- and intensive-margin, patent-count, citation, and value outcomes examined in the baseline. Shaded areas represent 90% confidence intervals.

Conclusion. The sign, timing, and overall shape of the impulse responses are essentially unchanged across all three sample cuts, which is reassuring evidence that the baseline mechanism is not an artifact of a particular subsample. Removing the top 1% of patenters shrinks the magnitude of the count-based outcomes—mechanically so, since these are the firms responsible for the largest share of aggregate patent output—but it does not alter their sign or the timing of the response, indicating that the qualitative pattern documented in the main text is broad-based rather than concentrated in a handful of extremely prolific innovators. Similarly, excluding the drug and semiconductor industries leaves the results intact, suggesting that the innovation-financing channel we document operates across a wide cross-section of industries rather than being specific to the institutional features of these two particularly R&D-intensive sectors. Finally, excluding recession quarters produces a pattern very similar to the baseline, which helps rule out the possibility that our estimates simply reflect the correlation between monetary policy and business-cycle downturns rather than a genuine credit-cost channel operating on firms’ innovation decisions. Taken together, these three cuts support the view that the effect we document is a general feature of how monetary policy transmits to firm-level innovation, rather than a statistical artifact of a specific subset of firms, industries, or macroeconomic episodes.

D.4 Control-Set Sensitivity

The choice of firm-level control variables in a local projection always involves a trade-off. Including too few controls risks omitted-variable bias if some unobserved firm characteristic is correlated both with the monetary policy shock's local effect and with the outcome of interest; including too many controls, on the other hand, risks "bad control" problems if some of the additional variables are themselves outcomes of the monetary policy shock, in which case controlling for them would absorb part of the very effect we are trying to estimate. Our baseline specification controls for sales growth, firm size, and the current-asset share, a parsimonious set intended to capture the main dimensions of firm heterogeneity relevant to financing constraints without over-controlling.

To assess how sensitive our results are to this particular choice, we re-estimate the local projections using two alternative control sets. The expanded set adds cash holdings, cash flow, and leverage growth to the baseline controls, allowing for a richer characterization of each firm's financial position and its capacity to self-finance R&D independently of external credit conditions. The minimal set, at the other extreme, includes only firm size, stripping out all of the financial controls and testing whether our results survive with essentially no controls beyond a basic measure of firm scale. Formally, we estimate

$$y_{i,t+h} = \beta_h s_t + \Gamma' \mathbf{X}_{i,t} + \alpha_i + \delta_{q(t)} + \varepsilon_{i,t+h}$$

where $\mathbf{X}_{i,t}$ denotes the firm control vector, varied across the expanded and minimal specifications described above; all other elements of the specification, including the fixed effects and the shock series, are unchanged from the baseline.

Figure D11: Control-Set Sensitivity: Baseline Outcomes (1/3)

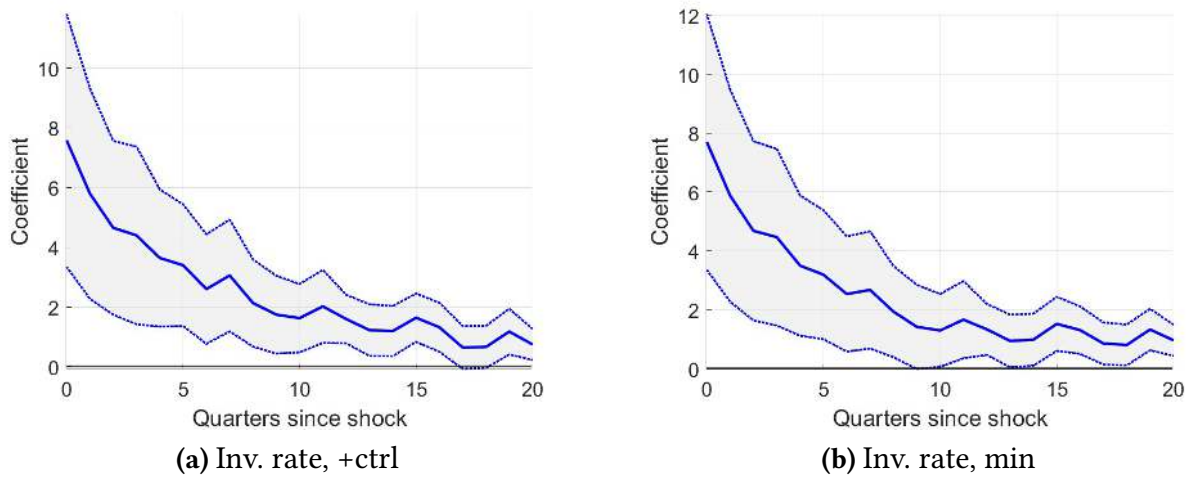
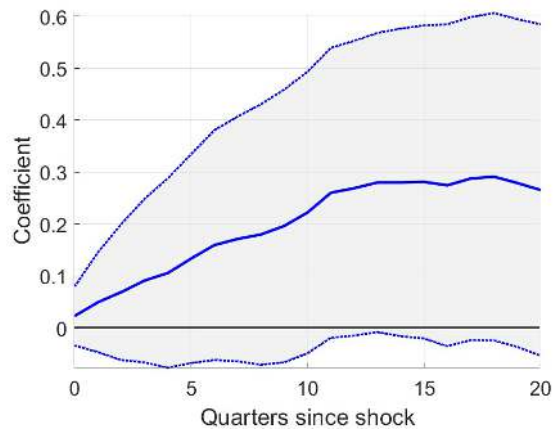
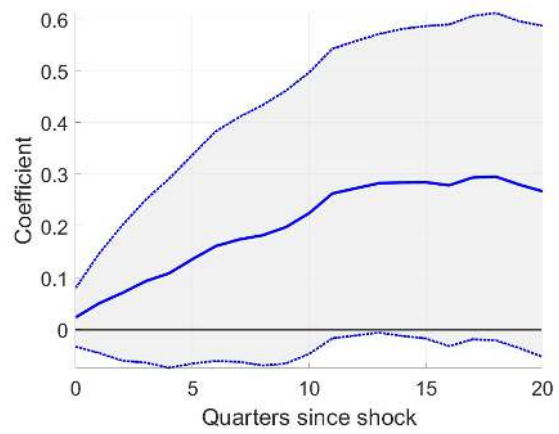


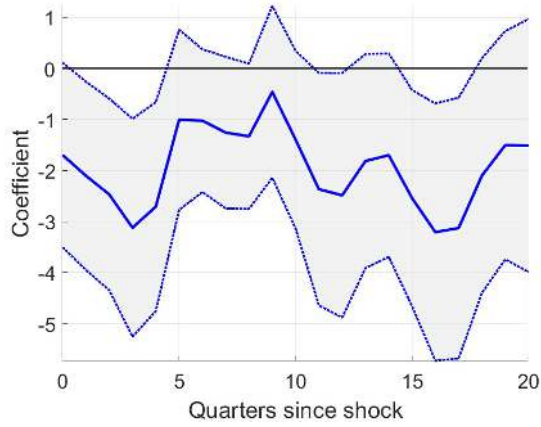
Figure D12: Control-Set Sensitivity: Baseline Outcomes (2/3)



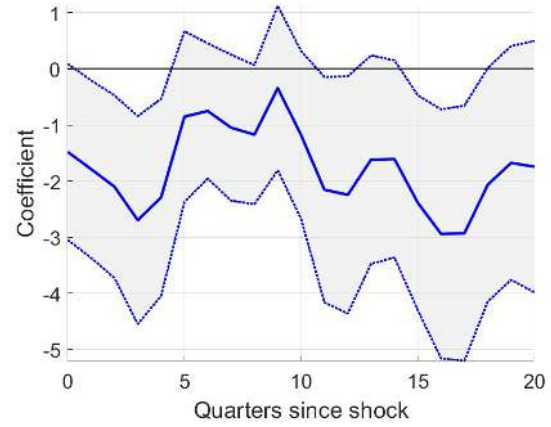
(a) Know. cap, +ctrl



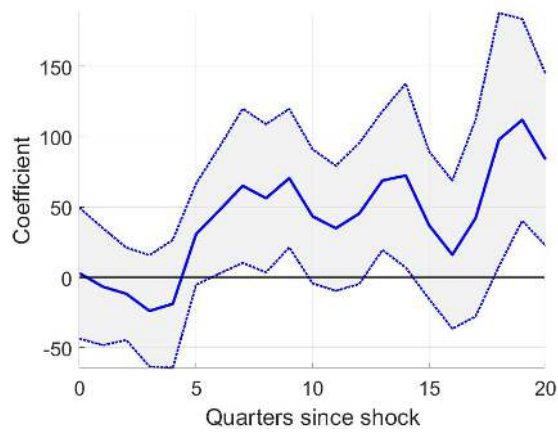
(b) Know. cap, min



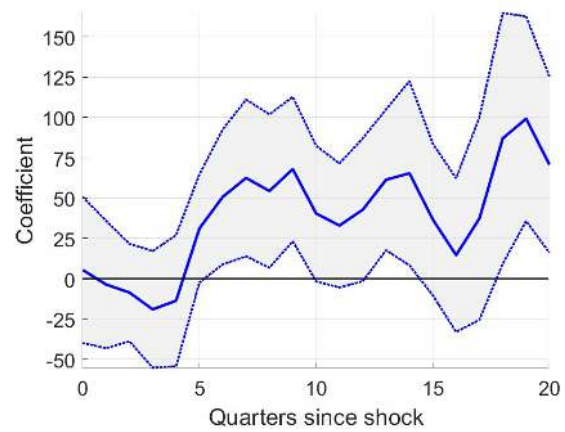
(c) Patents, +ctrl



(d) Patents, min

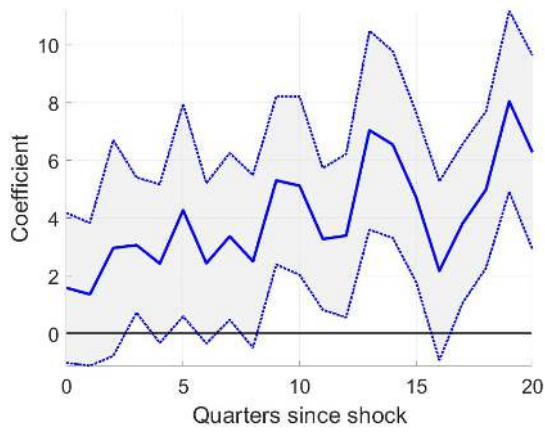


(e) Tot. cites, +ctrl

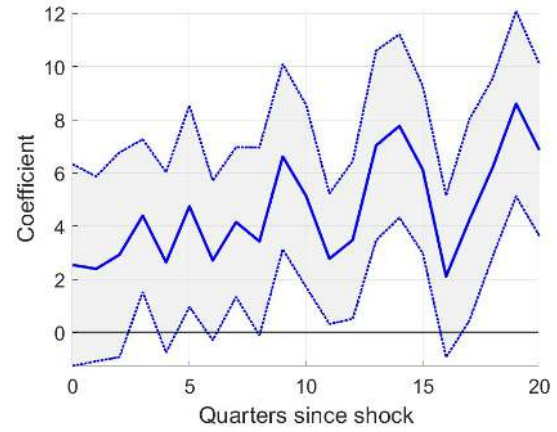


(f) Tot. cites, min

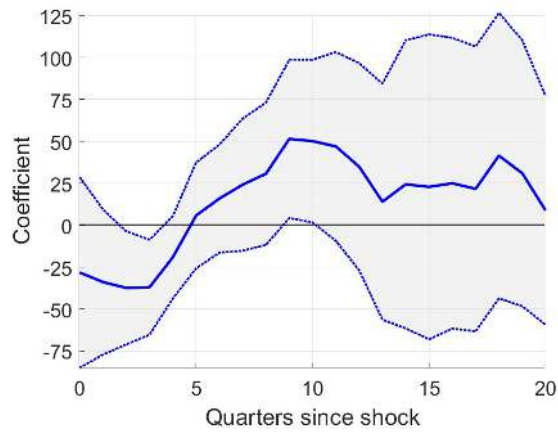
Figure D13: Control-Set Sensitivity: Baseline Outcomes (3/3)



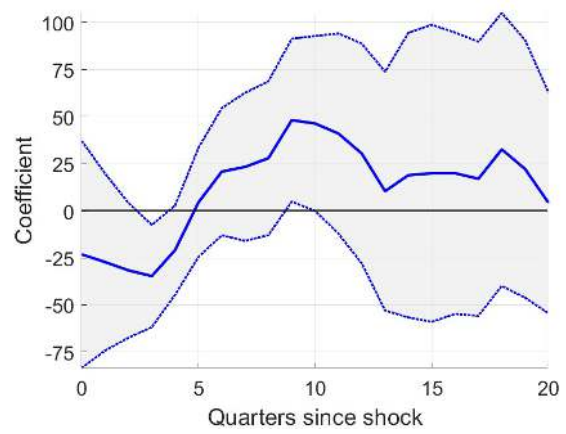
(a) Avg. cites, +ctrl



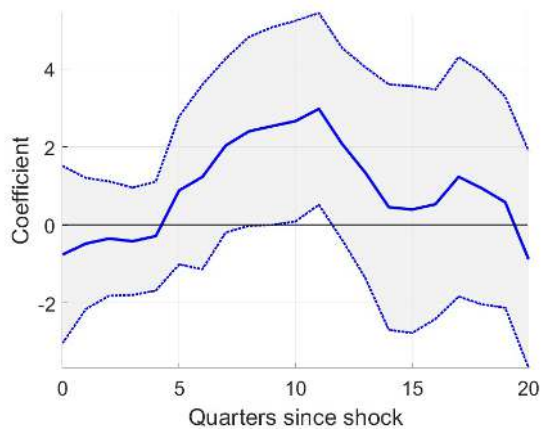
(b) Avg. cites, min



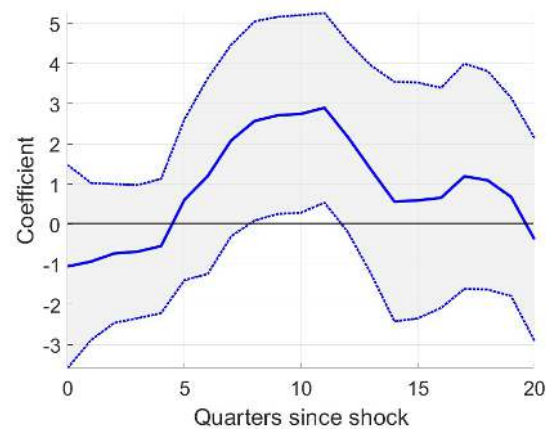
(c) Tot. value, +ctrl



(d) Tot. value, min



(e) Avg. value, +ctrl



(f) Avg. value, min

Notes: +ctrl denotes the expanded control set (baseline controls plus cash holdings, cash flow, and leverage growth); min denotes the minimal control set (firm size only). Shaded areas represent 90% confidence intervals.

Conclusion. The impulse responses under the expanded and minimal control sets are nearly indistinguishable from one another and from the baseline, both in sign and in magnitude, across the full set of investment, patenting, citation, and value outcomes. This is informative in both directions: the fact that adding richer financial controls (cash holdings, cash flow, leverage growth) does not meaningfully change the estimated responses suggests that our baseline controls are not omitting an important confounder along these dimensions; and the fact that stripping down to a minimal control set of firm size alone also leaves the results largely intact suggests that our findings are not an artifact of a particular, possibly fragile, combination of controls, nor are we inadvertently absorbing part of the treatment effect through an over-specified control vector. Taken together, this exercise indicates that the responses we document are driven by the monetary policy shock itself rather than by the specific choice of conditioning variables, which strengthens the case that our baseline specification is a reasonable and robust characterization of the underlying relationship between monetary policy and firm-level innovation.